

REVIEW

A Review of TRiSM Frameworks in Artificial Intelligence Systems: Fundamentals, Taxonomy, Use Cases, Key Challenges and Future Directions

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ABSTRACT

The rapid expansion of generative AI—particularly large language models (LLMs)—into mission-critical domains has underscored the urgent need for unified frameworks that embed trust, risk and security management (TRiSM) throughout the AI lifecycle. In this work, we present a comprehensive review and synthesis of AI TRiSM, uniting five foundational pillars: explainability with real-time drift monitoring, ModelOps governance, application-level security, data protection and privacy, and adversarial resilience. We introduce three aligned taxonomies for trust dimensions (e.g., fairness, transparency, accountability, inclusiveness, ethical alignment), risk categories (e.g., model, data, legal, operational, societal, cognitive, emergent, third-party) and security controls (e.g., access management, infrastructure hardening, runtime enforcement, privacy-enhancing techniques). Building on these, we develop a detailed toxicity taxonomy for generative AI—covering hate, violence, self-harm, misinformation, bias, jailbreak attacks, multimodal harms, and more—each mapped to specific TRiSM safeguards. Through cross-domain case studies in finance, healthcare, autonomous vehicles, public sector, cybersecurity, and beyond, we illustrate practical integration patterns and governance workflows. We also identify key adoption challenges—fragmented tooling, late-stage governance, scalability constraints, evolving threats and regulations—and chart a forward-looking roadmap toward adaptive, AI-driven policy engines, causal explainability, privacy-by-design pipelines, continuous real-time assurance, federated governance, quantum-safe architectures, and sustainable “green AI” practices. This article aims to guide researchers and practitioners in designing, evaluating and scaling resilient, ethical, and compliant AI systems at enterprise scale.

1 | Introduction

The rapid proliferation of AI across mission-critical domains—ranging from algorithmic trading (Dakalbab et al. 2024; Han et al. 2023) platforms in finance and personalised treatment planning in healthcare to perception stacks in autonomous vehicles (Sharma and Rana 2024; Thakur and Mishra 2024) and citizen-facing chatbots (Alsubayhay and Abdalla 2024; Saheb et al. 2025) in the public sector—has dramatically elevated both the promise and the peril of intelligent systems (Verdecchia et al. 2023; Ng et al. 2023; Shaygan et al. 2022;

Minh et al. 2022). As organisations entrust AI with decisions that directly impact human safety (Arunthavanathan et al. 2024; Perez-Cerrolaza et al. 2024), financial stability (Ozili 2024), and public welfare (Murko et al. 2024; Nayak and Tiyyadi 2024), questions around explainability (Chander et al. 2025; Hassija et al. 2024; Panagoulas et al. 2024; Carloni et al. 2025), bias (Hanna et al. 2025; Kumar et al. 2024; Koçak et al. 2025), privacy (Solove 2025; Yanamala et al. 2024), and adversarial resilience (Malali and Praveen Madugula 2025; Brunner et al. 2024) have moved from theoretical debates to boardroom imperatives. High-profile incidents—such as

loan-application (Ratan et al. 2024) denials attributed to spurious correlates, misdiagnoses driven by unbalanced medical imaging datasets, or safety overrides triggered by adversarial perturbations (Pan et al. 2024; Apruzzese et al. 2024)—have underscored the fragile boundary between AI’s transformative potential and its capacity for harm. In this context, the concept of AI TRiSM has crystallised as a unifying paradigm designed to embed comprehensive governance, robust risk controls, and end-to-end transparency into every phase of the AI lifecycle. By bringing together explainability mechanisms, continuous operational monitoring, disciplined ModelOps practices, hardened application-level security, rigorous data protection, and adversarial-defence techniques, AI TRiSM (Habbal et al. 2024) frameworks strive to ensure that AI systems not only perform at scale but also do so reliably, ethically, and in alignment with evolving regulatory requirements.

Despite significant advances in specialised toolkits for fairness auditing (Li and Goel 2025; Murikah et al. 2024), interpretability (Talaat et al. 2024), privacy preservation (Alkaeed et al. 2024), and adversarial defence, many organisations still grapple with fragmented point solutions that operate in isolation. Siloed teams maintain separate workflows for data governance, security reviews, and model validation, often resorting to “after-the-fact” compliance checks just prior to deployment. Such late-stage interventions lead to extensive rework, inconsistent policy enforcement across model portfolios, and protracted release cycles. Key challenges include obtaining holistic, traceable visibility into how data moves through pipelines and how features influence model outcomes; reconciling diverse risk vocabularies across data science, security, and legal functions; scaling continuous assurance for hundreds or thousands of models in production; and rapidly adapting governance controls to counter emerging threats—whether newly disclosed vulnerabilities in third-party components or shifting regulatory mandates. Without a coherent, integrated approach, organisations risk undetected drift in fairness or performance metrics, blind spots in their security posture, and ultimately, erosion of stakeholder trust and potential regulatory penalties.

This article provides a systematic, end-to-end review of AI TRiSM frameworks, taxonomies, toolkits, real-world use cases, challenges and future direction, with the dual aim of equipping researchers with a unified conceptual foundation and enabling practitioners to design, evaluate, and operationalise holistic trust, risk, and security controls. We first distill the foundational pillars of AI TRiSM—explainability and model monitoring, ModelOps governance, application-level security, data protection and privacy, and adversarial resilience—and survey leading methodologies in each area. Next, we introduce consolidated taxonomies for trust dimensions, risk categories, and security controls, illustrating how these taxonomies guide both tool selection and process design. We then examine concrete deployments across finance, healthcare, autonomous systems, the public sector, and other domains, highlighting patterns, trade-offs, and lessons learned. Finally, we identify critical gaps and chart a forward-looking research agenda—spanning adaptive AI-powered governance, causal explainability, scalable privacy-enhancing technologies, and quantum-safe security—that will shape the next generation of trustworthy and secure AI

ecosystems. In particular, we seek to: (i) clarify core AI TRiSM pillars—explainability, ModelOps, AppSec, data protection, adversarial resilience, (ii) develop unified taxonomies for trust, risk, and security controls, (iii) illustrate real-world AI TRiSM deployments across finance, healthcare, autonomous systems, public sector, and cybersecurity, and (iv) chart future research avenues—adaptive governance, causal explainability, PETs, and quantum-safe architectures. Table 1 presents key abbreviations and their full forms used in this article.

The key contributions of this work are:

- A comprehensive consolidation of AI TRiSM foundations—explainability, ModelOps governance, application security, data protection, and adversarial resilience—into a single cohesive framework.
- The development of unified trust, risk, and security taxonomies that map core dimensions, risk categories, and defence controls across the AI lifecycle
- A detailed generative-AI toxicity taxonomy for LLMs—spanning hate and harassment, violent content, self-harm, misinformation, privacy leaks, bias, jailbreak attacks, multimodal harms, and more—aligned with corresponding TRiSM safeguards, illustrating how each toxicity vector is identified, measured, and mitigated in practice.
- Rich, cross-domain case studies demonstrating how TRiSM principles are operationalised to deliver transparent, resilient, and compliant AI.
- A systematic identification and analysis of key obstacles—fragmented tooling, siloed processes, late-stage governance, scalability constraints, and evolving threats/regulations—and strategies to overcome them.
- A forward-looking roadmap outlining emerging opportunities—adaptive AI-driven governance, causal explainability, privacy-enhancing technologies, quantum-safe architectures, and sustainable “green AI” practices—to guide next-generation TRiSM solutions.

The remainder of this article is organised as follows: Section 2 presents related works. Section 3 presents the core AI TRiSM fundamentals. Section 4 develops unified taxonomies for trust, risk, and security management. Section 5 shows taxonomy for toxicity of generative AI-based LLMs. Section 6 explores representative use cases across multiple industries. Section 7 analyzes key adoption challenges. Section 8 outlines prospective research directions. Section 9 presents a detailed aspects and contextualization of whole article in terms of discussions of various matter of interest. Finally, Section 10 summarises our findings and suggests pathways for advancing resilient, compliant, and transparent AI systems.

2 | Related Works

The AI Trust, Risk, and Security Management (AI TRiSM) paradigm has attracted significant scholarly attention, with several studies exploring its theoretical foundations, practical architectures, and domain-specific applications. Habbal et al. (2024) provide a comprehensive survey of AI TRiSM,

TABLE 1 | Glossary of acronyms and full forms.

Acronym	Full form
ABAC	Attribute-Based Access Control
AES	Advanced Encryption Standard
AI	Artificial Intelligence
AI TRiSM	AI Trust, Risk, and Security Management
API	Application Programming Interface
AppSec	Application Security
ART	Adversarial Robustness Toolbox
AUC	Area Under the Curve
AWS	Amazon Web Services
BYOAI	Bring Your Own AI
CCPA	California Consumer Privacy Act
CI/CD	Continuous Integration/ Continuous Deployment
CLI	Command-Line Interface
CNN(s)	Convolutional Neural Network(s)
CSP	Cloud Service Provider(s)
CTR	Click-Through Rate
CVE	Common Vulnerabilities and Exposures
DAST	Dynamic Application Security Testing
DLP	Data Loss Prevention
DP-SGD	Differentially Private Stochastic Gradient Descent
DSAR	Data Subject Access Request
DT	Decision Tree
DVC	Data Version Control
EDR	Endpoint Detection and Response
ELT	Extract, Load, Transform
ELK	Elasticsearch, Logstash, Kibana
FGSM	Fast Gradient Sign Method
FL	Federated Learning
FLOPs	Floating-Point Operations per Second
GDPR	General Data Protection Regulation
Grad-CAM	Gradient-weighted Class Activation Mapping
HE	Homomorphic Encryption
HSM	Hardware Security Module
IAM	Identity and Access Management
IDS/IPS	Intrusion Detection/Prevention System
JSON	JavaScript Object Notation

(Continues)

TABLE 1 | (Continued)

Acronym	Full form
JWT	JSON Web Token
KG	Knowledge Graph
KPI	Key Performance Indicator
LIME	Local Interpretable Model- agnostic Explanations
LLM	Large Language Model
MFA	Multi-Factor Authentication
MLOps	Machine Learning Operations
ModelOps	Model Operations
mTLS	Mutual TLS
MRA	Model Risk Assessment
NER	Named Entity Recognition
NIST	National Institute of Standards and Technology
NIST RMF	NIST Risk Management Framework
NLI	Natural Language Inference
OCI	Open Container Initiative
OPA	Open Policy Agent
PET	Privacy Enhancing Technologies
PGD	Projected Gradient Descent
PHI	Protected Health Information
PII	Personally Identifiable Information
PSI	Population Stability Index
RAG	Retrieval-Augmented Generation
RASP	Runtime Application Self-Protection
RBAC	Role-Based Access Control
RLHF	Reinforcement Learning from Human Feedback
SAST	Static Application Security Testing
SCAP	Security Content Automation Protocol
SDK	Software Development Kit
SHAP	SHapley Additive exPlanations
SIEM	Security Information and Event Management
SLI/SLO	Service Level Indicator/Objective
SLA	Service Level Agreement
SMPC	Secure Multi-Party Computation
SOC	Security Operations Center
SOAR	Security Orchestration, Automation, and Response

(Continues)

TABLE 1 | (Continued)

Acronym	Full form
TLS	Transport Layer Security
WAAP	Web Application and API Protection
XAI	Explainable AI
ZKP	Zero-Knowledge Proof

synthesising its evolution across smart cities, healthcare, manufacturing, and virtual worlds. They identify core challenges—including adversarial attacks, regulatory volatility, and skill shortages—and outline future research directions emphasising stakeholder collaboration and continuous adaptation. Mathew (2024) complements this overview by presenting a detailed methodology for integrating trust-building, risk assessment, and security enforcement into cyber-physical systems, illustrating block diagrams and algorithmic flows that underpin proactive AI governance.

In the realm of digital asset management, Tasatanattakool et al. (2024, 2025) propose AI TRiSM-driven architectures to streamline transparency and trust within organisational repositories. Their paper (Tasatanattakool et al. 2024) reports quantitative improvements in efficiency (mean = 4.91, SD = 0.14) when AI TRiSM principles guide metadata cataloguing and usage auditing. The follow-up journal article (Tasatanattakool et al. 2025) elaborates a nine-expert-validated system architecture, highlighting modular components for secure data ingestion, access control, and compliance reporting (mean = 4.53, SD = 0.49).

Healthcare emerges as a critical domain for AI TRiSM adoption. Sumathi and Devilakshmi (2025) develop a specialised framework to ensure patient-centric AI applications adhere to confidentiality, ethical, and legal standards. Their work integrates privacy-by-design, transparency mechanisms, and interdisciplinary governance to maintain data integrity while supporting diagnostic and prognostic AI tools. The study underscores the necessity for real-world case studies to validate AI TRiSM efficacy under HIPAA and GDPR constraints.

Education has also benefitted from AI TRiSM insights. El Khatib et al. (2023) investigate how AI TRiSM influences decision-making for educational executives, demonstrating that robust governance protocols enhance trust and reduce cognitive load in e-learning environments. They employ mixed-methods interviews to reveal that personalization engines powered by AI require clear accountability, bias audits, and data-retention schedules to support both administrative and pedagogical goals.

Generative AI risk management in digital economies is addressed by Chang, Draper, and Yu (2024), who analyse content personalization frameworks within healthcare procurement. Their qualitative study evaluates ethical and operational hazards—such as hallucinations and IP infringement—and recommends layered defences combining network security, data validation, and ethical oversight. This work illustrates the fine balance between leveraging LLMs for efficiency and guarding against misinformation or unauthorised data exposure. The

formation of trust in AI among end users is further explored by Wong et al. (2024), who apply partial least squares–structural equation modelling to travel and tourism contexts. They differentiate between institutional factors (e.g., regulatory protection) and self-related perceptions (e.g., threat concerns), finding that regulatory safeguards and individual trust propensity significantly influence AI adoption. Their framework offers a nuanced perspective on how policy and personal attitudes coalesce to shape AI acceptance. Recent tourism AI studies have used a modified total interpretive structural modelling approach to dissect competitiveness drivers in tourism firms (Sharma et al. 2022), examined the ethical and technological challenges of AI adoption in tourism contexts (Ferhataj and Memaj 2024), and modelled the deployment and positioning of AI-based Tourism Relationship Management systems (Hojjatitalemy 2021).

From an industry analyst standpoint, Litan (2024) delivers practical guidance on AI governance strategies in a Gartner research article. He prescribes a five-step governance framework—comprising policy definition, application inventory, data classification, AI TRiSM technology implementation, and continuous monitoring—to mitigate data breaches, hallucinations, and compliance failures. This prescriptive model has informed many enterprise AI initiatives, underscoring the importance of organisational alignment between technical and business units. IBM's white paper by Gomstyn and Jonker (2025) offers an integrative overview of AI TRiSM's five pillars: governance, trustworthiness, fairness, reliability, and data protection. They emphasise explainability techniques (e.g., SHAP, LIME), runtime inspection, and adversarial resilience, presenting real-world use cases in healthcare and finance. Their unified approach serves as a blueprint for organisations seeking to harmonise disparate AI governance frameworks into a coherent strategy.

Despite the breadth of existing work, several gaps remain. First, there is a scarcity of longitudinal studies measuring the sustained impact of AI TRiSM interventions on organisational risk posture and ROI. Second, cross-domain transferability of AI TRiSM architectures—particularly from regulated sectors like healthcare to emerging areas such as the Metaverse—requires empirical validation. Third, the interplay between AI TRiSM governance and cultural or regional regulatory differences (e.g., GDPR vs. CCPA) has not been systematically investigated. Finally, automated tooling for policy-as-code enforcement and real-time compliance auditing remains underdeveloped. Table 2 presents a comparison of key related works on AI TRiSM.

Key lessons from the literature include the necessity of embedding AI TRiSM principles early in the development lifecycle; the efficacy of interdisciplinary governance bodies combining technical, legal, and ethical expertise; and the value of explainability and uncertainty quantification for stakeholder trust. Furthermore, modular, policy-driven architectures show promise for scaling AI TRiSM across both cloud and edge environments, while federated and privacy-preserving learning approaches mitigate data-sharing risks in multi-party settings. These insights collectively inform a roadmap for advancing AI TRiSM research and practice.

TABLE 2 | Comparison of key AI TRISM publications.

Article (Author et al.)	Year	Type of article	Technology	Contribution	Relevance and correctness
Habbal et al. 2024	2024	Review article	Conceptual framework	Systematic review of AI TRISM principles, applications across domains, and future research agenda	High relevance; comprehensive, technically accurate synthesis
Mathew 2024	2024	Methodology paper	ModelOps and governance controls	Methodology for embedding trust, risk, and security in cyber-physical AI systems	Highly relevant; correct in outlining proactive governance steps
Tasatanattakool et al. 2024	2024	Conference paper	AI-driven metadata cataloguing and auditing	Empirical study showing efficiency gains in digital asset management using AI TRISM	Relevant; quantitative results clearly reported and valid
Tasatanattakool et al. 2025	2025	Journal article	Modular ingestion, access control, compliance monitoring	Expert-validated secure architecture for asset management under AI TRISM	Very relevant; methodology and evaluation are technically sound
Sumathi and Devilakshmi 2025	2025	Book chapter	Privacy, explainability, interdisciplinary workflows	Healthcare-specific AI TRISM integrating privacy-by-design and governance	High relevance; correctly addresses HIPAA/GDPR constraints
El Khatib et al. 2023	2023	Empirical research	Adaptive learning governance, bias audits	Impact assessment of AI TRISM in education through primary interviews	Relevant; findings based on sound mixed-methods design
Chang, Draper, and Yu 2024	2024	Conference paper	GenAI governance, network security, data integrity	Qualitative analysis of generative AI risks in healthcare procurement	Moderately relevant; correct best practices in qualitative insights
Wong et al. 2024	2024	Journal article	PLS-SEM modelling of trust factors	Framework for trust formation in AI in travel tech context	Highly relevant; statistically rigorous and valid conclusions
Litan 2024	2024	Industry research note	TRISM lifecycle with policy, monitoring, inventory	Five-step enterprise AI governance strategy	Very relevant; practical and accurate for enterprise settings
Gomstyn and Jonker 2025	2025	White paper	Explainability, runtime inspection, adversarial defences	Overview of five AI TRISM pillars with real-world cases	Highly relevant; cohesive synthesis, though less empirical

Despite the breadth of existing work, our review distinguishes itself by synthesising AI TRiSM frameworks into a unified, end-to-end lifecycle architecture that bridges theory and practice across domains. Unlike prior surveys that focus on isolated pillars (e.g., explainability or privacy) or domain-specific implementations, this article offers (i) a coherent “technology sandwich” model integrating built, BYOAI and embedded AI under a central TRiSM governance core; (ii) consolidated taxonomies mapping trust dimensions, risk categories and security controls to concrete phases of ModelOps pipelines; and (iii) cross-industry case studies that demonstrate how AI TRiSM principles can be operationalised in finance, healthcare, autonomous vehicles, smart cities and beyond. By articulating both the technological components and the organisational processes required for continuous, adaptive governance, our review provides the first comprehensive roadmap for designing, deploying and scaling resilient, ethical and compliant AI systems under a single, cohesive TRiSM umbrella.

3 | AI TRiSM Fundamentals

AI TRiSM Fundamentals lay the groundwork for embedding trust, risk management, and security directly into every phase of the AI lifecycle (Gomstyn and Jonker 2025; Litan 2024). It begins by defining the core pillars—explainability with real-time monitoring, robust ModelOps practices, application-level security, data protection, and adversarial resilience—that collectively ensure transparent, resilient AI (Gartner Glossary 2025; Das 2025). Through explainability and drift detection, teams gain visibility into model decisions and can promptly address performance degradation or bias (National Institute of Standards and Technology 2025; Litan 2025). ModelOps transforms experimentation into production-grade engineering by automating versioning, testing, deployment, and retirement under strict governance (Wu 2024; Siddiqui 2024). AI application security extends traditional AppSec with defences against prompt injection, data poisoning, and model theft, safeguarding AI services from novel threat vectors (Checkpoint 2025; Takyar and Exploring 2025). Data protection and privacy controls—from anonymization to federated learning and differential privacy—preserve confidentiality without stifling innovation (Ivanov and Ampilogova 2024; Consultancy.eu 2024; Ramos 2024). Finally, adversarial defences harden models against evasion and poisoning attacks, closing the loop on a truly end-to-end AI TRiSM framework (Aisera 2025; HP Online Store 2025). We will revisit in Section 4 where AI TRiSM pillars shall be used again.

3.1 | AI TRiSM Pillar

Figure 1 presents the TRiSM pillar. The diagram illustrates the AI TRiSM framework as a sturdy architectural structure supported by five essential pillars. At its foundation and roof sits the “AI TRiSM Framework” itself, symbolising the overarching governance that holds everything together. Rising between these beams are five vertical columns, each representing a core domain: “Explainability and Model Monitoring,” “Model Operations (ModelOps),” “AI Application Security (AppSec),” “Data Protection and Privacy,” and “Adversarial Attacks Resistance.” In the remainder of this section, we introduce

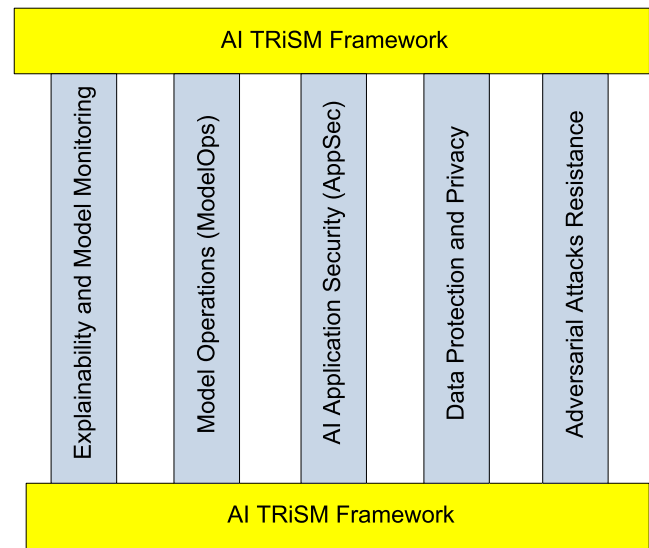


FIGURE 1 | Overview of the AI TRiSM pillars.

the five foundational pillars of the AI TRiSM framework—(1) Explainability and Model Monitoring, (2) Model Operations, (3) AI Application Security, (4) Data Protection and Privacy, and (5) Adversarial Attacks Resistance—defining their key objectives, methods, and technical controls. Here, our focus is on establishing the conceptual underpinnings and core capabilities of each domain. In contrast, Section 3.4 (“Adoption of TRiSM for Secure and Responsible AI”) will revisit these same pillars from a practical standpoint, illustrating how organisations operationalise them through end-to-end workflows, governance processes, and tooling. By separating the “what” (this section) from the “how” (the Adoption subsection), readers will gain both a clear theoretical framework and concrete guidance for embedding trust, risk, and security into their AI lifecycles.

3.1.1 | Explainability and Model Monitoring

Within the AI TRiSM framework, explainability and model monitoring provide essential transparency and operational oversight for all machine learning systems (Simister A 2024; Stellarix 2024). Explainability employs eXplainable AI (XAI) methods (Yeo et al. 2025; Carloni et al. 2025; Herrera 2025)—such as SHapley Additive exPlanations (SHAP) (Salih et al. 2025), Local Interpretable Model-agnostic Explanations (LIME) (Donmez et al. 2025), feature attribution heatmaps, and attention-based visualisation—to deconstruct “black box” algorithms and reveal how individual input features drive predictions (Ukwuoma et al. 2025; De-Giorgio et al. 2025). For example, SHAP computes per-feature contribution scores that sum to the model output, enabling practitioners to compare the impact of each variable across thousands of inferences. Saliency maps for convolutional neural networks highlight image regions that most influence classification, while decision-tree surrogate models approximate complex ensembles with human-readable rules (Müller 2025; Schwenke and Atzmueller 2025). Concurrently, model monitoring tracks performance metrics in real time—accuracy, precision, recall, F1 score, ROC-AUC, and calibration error—against established baselines. Statistical drift detection techniques (e.g., Population Stability Index,¹

Kolmogorov–Smirnov,² Jensen–Shannon divergence; Gholami and Hodsani (2025) continuously compare incoming feature distributions with training data to surface covariate drift, while concept drift detectors flag shifts in the relationship between features and targets. All inference metadata—including input fingerprints, prediction confidences, execution latency, and resource consumption—is logged to centralised observability platforms (e.g., Prometheus/Grafana,³ ELK stack⁴). Automated alerting triggers retraining workflows or human review when performance degradation or data anomalies exceed predefined thresholds. Integrating XAI checks and drift tests into CI/CD pipelines ensures that every code or data change reruns explainability analyses and drift assessments before deployment, firmly aligning transparency and reliability with evolving production conditions.

3.1.1.1 | Model Operations. Model Operations or ModelOps Within the AI TRISM Framework Codifies End-To-End Governance Over the Entire ML Lifecycle—From Initial Development Through Deployment, Maintenance, and Retirement (Glossary (2025); Sharma (2023)). In the Development Phase, Robust Version Control (e.g., Git,⁵ DVC⁶) Tracks Code, Data Schemas, Feature-Engineering Scripts, and Hyperparameter Configurations. Automated Unit Tests Validate Data Transformations and Expected Model Outputs, While Integration Tests Ensure Compatibility Between Preprocessing Pipelines and Model Inference Code. Containerization Technologies (e.g., Docker,⁷ OCI⁸) Encapsulate Reproducible Runtime Environments, and Infrastructure as Code (e.g., Terraform,⁹ CloudFormation¹⁰) Defines Compliant Compute, Storage, and Networking Resources. Upon Passing Quality Gates, CI/CD Orchestrators (e.g., Jenkins,¹¹ GitLab CI/CD,¹² Argo CD¹³) Automatically Build Model Artefacts, Conduct Security Scans (SAST, Dependency Vulnerability Checks), and Deploy Microservices to Staging Clusters. Canary and Blue-Green Deployment Strategies Route a Fraction of Live Traffic to New Model Versions, Measuring Key Indicators—Latency Percentiles, Success/Error Rates, Resource Utilisation—To Detect Regressions. In Production, Continuous Observability Integrates With Prometheus, Grafana, or Datadog AI¹⁴ Ops to Monitor Both Infrastructure Metrics (e.g., CPU/GPU Load, Memory Usage) and Model-Specific KPIs. Scheduled Retraining Pipelines Leverage Drift-Detected Triggers or Performance Declines to Refresh Models With Updated Data, Ensuring Relevance and Accuracy. Policy-As-Code (i.e., Open Policy Agent; Open Policy Agent (2025)) Enforces That Only Models Meeting Security, Fairness, and Performance Thresholds Progress Along Deployment Stages. Finally, Model Retirement Archives Artefacts, Metadata, and Retraining Logs in a Centralised Registry, Preserving Traceability and Compliance With Governance and Data Retention Policies.

3.1.1.2 | AI Application Security. AI Application Security (AppSec) Extends Traditional Software Security Practices to the Unique Vulnerabilities of AI Services (Robust Intelligence (2025); Tooley (2025)). To Prevent Prompt-Injection and Adversarial Input Attacks, Input Validation Layers Enforce Strict schema Checks, Length Limits, and Sanitization Routines That Strip or Encode Malicious Tokens. Training Data Pipelines Incorporate Provenance Verification, Anomaly-Based Sanitizers, and Outlier Detection (e.g., Isolation Forests (Chugh et al. (2025); Gao et al. (2025); Wu, Sheng, et al. (2025)), Autoencoders

(Bergmann and Stryker (2023)) to Mitigate Data Poisoning and Hidden Backdoors. During Development and CI, Static Application Security Testing (SAST) (Chembakottu and Robillard (2025)) and Software Composition Analysis (SCA) (Nocera et al. (2025)) Tools Uncover Code-Level Flaws and Vulnerable Dependencies, While Dynamic Application Security Testing (DAST) (Hieu et al. (2025)) Simulates Runtime Behaviour to Detect Injection or Authorization Flaws. Containers Are Hardened Through Minimal Base Images (i.e., Distrosless), seccomp Profiles, Capability Dropping, and Read-Only Root Filesystems. Inference APIs Reside Behind API Gateways Enforcing OAuth2 or JWT for Authentication, Rate Limiting, and Attribute-Based Access Control (ABAC) (Bello et al. (2025)) to Ensure That Only Authorised Actors Invoke Model Endpoints. Runtime Application Self-Protection (RASP) (Seth et al. (2025)) Agents Instrument AI Services to Detect Anomalous Call Patterns, Unusual Input Distributions, or Exploitation Attempts in Real Time, Triggering Automated Isolation or Rollback to Safe Model Versions. Regular Red-Teaming Exercises Simulate Both Evasion and Poisoning Strategies, Validating Detection Signatures and Refining Defence Rules. Incident Response Playbooks Define Containment Procedures—Like Rotating API Keys via Secrets Managers (Vault¹⁵), quarantining Compromised Containers, and Reverting to Known-Good Artefacts—Ensuring Rapid Recovery and Minimal Damage.

3.1.1.3 | Data Protection and Privacy. Data Protection and Privacy Controls Are Foundational to AI TRISM, Safeguarding Sensitive Information From Collection Through Final Disposal. Data Minimization Policies Ensure Only Attributes Strictly Necessary for Model Objectives Are Ingested. Anonymization Techniques— k -Anonymity, ℓ -Diversity, t -Closeness—Reduce Re-Identification Risk in Shared Datasets, While Differential Privacy (DP) Mechanisms (Gu et al. (2025); Demelius et al. (2025); Mooradian et al. (2025)) Inject Calibrated Noise to Bound an Individual Record's Influence Under (ϵ, δ) Privacy Budgets. Federated Learning (FL) Architectures Orchestrate Decentralised Training Across Edge Nodes or Data Silos, Transmitting Only Encrypted Model Updates to a Central Aggregator and Preserving Raw Data Locality. Privacy-Enhancing Technologies (PETs) Such as Secure Multi-Party Computation (SMPC) (Gamiz et al. (2025); Salako et al. (2025)) and Homomorphic Encryption (HE) (Singhi et al. (2025)) Allow Computation on Ciphertexts, Enabling Collaborative Analytics Without Exposing Plaintext. Comprehensive Encryption-At-Rest (AES-256) and In-Transit (TLS 1.3) Protect Persisted Stores and Network Channels. Consent Management Systems Capture, Store, and Enforce User Preferences, Automatically Filtering or Revoking Data Usage When Consent Expires. Automated Privacy Audits Embedded in CI/CD Pipelines Verify DP Budget Consumption, FL Configuration Integrity, and Compliance With General Data Protection Regulation (GDPR), California Consumer Privacy Act (CCPA), or Health Insurance Portability and Accountability Act (HIPAA). End-To-End Data Lineage Tracking Logs Every Transformation—From Ingestion Through Model Training—To Maintain Accountability, Support Forensic Investigations, and Fulfil Audit or Regulatory Requests.

3.1.1.4 | Adversarial Attacks Resistance. Adversarial Attacks Resistance Fortifies AI Systems Against Input Manipulation and Model-Targeted Exploits. At Training Time,

Adversarial Training Injects Perturbations—Using Fast Gradient Sign Method (FGSM) (Naseem 2024) or Projected Gradient Descent (PGD) (Chaddad et al. 2025) —To Expose Models to Crafted Adversarial Examples, Improving Robustness. Defensive Distillation Trains a Secondary “Student” Model on Soft Labels From a “Teacher” Network, Smoothing Decision Boundaries and Reducing Susceptibility to Small Input Perturbations. Model Ensembling Aggregates Predictions Across Diverse Architectures, Diluting Individual Model Vulnerabilities. Feature Squeezing Reduces Attack Surface by Compressing Inputs (e.g., Bit-Depth Reduction, Spatial Smoothing), limiting Adversary Manipulation Vectors. Runtime Detectors Monitor Gradient Norms and Activation Patterns to Identify Potential Adversarial Inputs, Flagging Suspects for Fallback to Safe-Mode Predictions. In Continuous Integration, Defence Efficacy Is Validated via Red-Team Toolkits (e.g., CleverHans,¹⁶ Foolbox¹⁷) That Launch Automated Adversarial Attacks to Evaluate Resilience. Incident Response Integrates Model Rollback Procedures and Patch Deployment Pipelines to Rapidly Remediate Discovered Weaknesses, Ensuring That AI Installations Maintain Both Accuracy and Integrity in Adversarial Environments.

3.2 | AI TRiSM Technology Sandwich

Figure 2 presents the layered AI TRiSM-Rich sandwich. At the foundation, Centralised Data consolidates enterprise datasets. Directly above this, the Built AI layer represents in-house model development that leverages these consolidated datasets. The core AI TRiSM layer enforces trust, risk, and security controls across all AI-related activities. Above this, the BYOAI tier integrates third-party AI services within the same governance framework. The Embedded AI layer refers to embedding inference capabilities directly within applications. At the top, Data Everywhere signifies the pervasive availability of data, supporting continuous model improvement and refinement. The “TRiSM-Rich AI Technology Sandwich” presents a layered enterprise architecture in which data and AI components coexist under a robust governance core (Litan 2024; Mesaglio and LeHong 2024). At its foundation sits Centralised Data, the unified repository that aggregates enterprise information from disparate operational systems. This layer comprises data lakes and warehouses engineered to ingest structured and unstructured records via ETL or ELT pipelines. Here, stringent data quality controls—schema validation, referential integrity checks, and duplicate

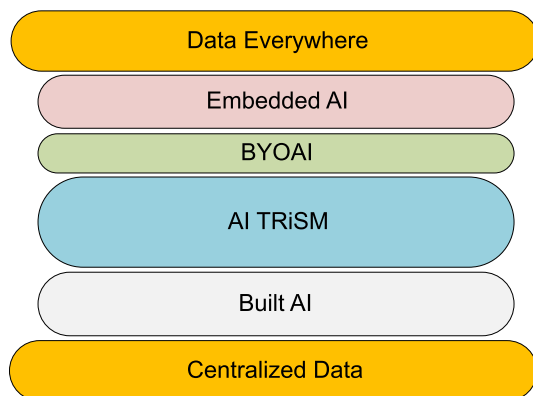


FIGURE 2 | Overview of AI TRiSM-Rich sandwich architecture.

detection—ensure that all downstream processes consume reliable inputs. The RBAC and ABAC enforce least-privilege principles, while encryption at rest (e.g., AES-256) and in transit (e.g., TLS 1.3) protect sensitive fields. Metadata management tools catalogue data lineage, enabling auditors and data stewards to trace each record's origin, transformations, and usage, thereby facilitating compliance with regulations.

Directly above centralised data, the Built AI layer embodies in-house developed machine learning and deep learning solutions. Data scientists leverage frameworks like TensorFlow, PyTorch, and scikit-learn to train models on the enterprise data foundation. Reproducibility is guaranteed by versioning both code and data within Git repositories, while containerization technologies standardise runtime environments across development, staging, and production. The CI/CD pipelines automatically execute unit tests, integration tests, and performance benchmarks—measuring accuracy, precision, recall, and Receiver Operating Characteristic Area Under the Curve (ROC-AUC)—to validate model readiness. Automated canary and blue-green deployment strategies minimise user impact during rollouts. Resource monitoring integrates with observability stacks to detect regressions and trigger retraining workflows when model performance degrades or drift thresholds are exceeded.

At the sandwich's center sits AI TRiSM, the governance engine that permeates every tier. Policy-as-code frameworks—such as Open Policy Agent—codify organisational rules mandating bias detection, adversarial robustness tests, and security vulnerability scans before any model or application can progress. The XAI methodologies provide transparency by attributing prediction outcomes to individual features. Drift detection algorithms monitor covariate and concept shifts via statistical tests and anomaly-based detectors. Privacy-enhancing techniques—Differential Privacy and federated learning—preserve confidentiality during training. Compliance with standards and the EU AI Act is enforced through continuous auditing and third-party assessments, ensuring that AI initiatives operate within defined risk tolerances and regulatory boundaries.

Stacked above TRiSM is Bring Your Own AI (BYOAI) (van der Meulen et al. 2024), the integration point for external AI services and pre-trained models. Enterprises consume software-as-a-service (SaaS) APIs offering LLMs, computer vision, and natural language processing capabilities without building from scratch. Secure API gateways—enforcing OAuth 2.0 or JWT authentication, rate limiting, and request schema validation—protect these endpoints from misuse and prompt injection attacks. Input sanitization routines pre-empt malicious payloads, while usage telemetry tracks invocation patterns to detect anomalous spikes. Although BYOAI accelerates innovation and reduces time-to-market, TRiSM governance ensures each imported model undergoes a rigorous vetting process—covering vendor security posture, bias evaluations, and ongoing performance monitoring—to align with the organisation's ethical and risk management standards.

Above BYOAI, Embedded AI refers to AI-powered features seamlessly woven into existing business applications. Whether embedded in customer relationship management (CRM) platforms to auto-classify support tickets or integrated into supply

chain management software to forecast demand, these micro-services encapsulate inference logic, decoupling AI workloads from core application code. Low-latency feature stores deliver real-time input vectors, while service meshes (Istio¹⁸) enforce mutual TLS (mTLS) and micro-segmentation across containerized services. Event streaming architectures supply continuous data feeds, enabling embedded models to update predictions dynamically. DevOps teams instrument these services with distributed tracing (Jaeger¹⁹) and metrics collection, ensuring that AI augmentation remains performant, observable, and secured against unauthorised access.

Atop the entire stack sits Data Everywhere, signifying the democratisation of information throughout the enterprise. Edge computing nodes preprocess and filter real-time telemetry—Internet of Things (IoT) sensor readings, user interaction logs, and clickstreams—reducing latency and bandwidth consumption. These event streams flow into the centralised data layer and directly fuel both built and BYOAI models, creating an adaptive feedback loop for continuous learning. Real-time ingestion frameworks (Flink,²⁰ Spark Streaming²¹) power analytical dashboards and alerting systems, while lightweight agents capture user-level signals for personalization engines. By embracing pervasive data collection and leveraging high-velocity pipelines, organisations deliver contextual insights wherever users engage, ensuring that AI-driven decision-making remains timely, relevant, and intimately tied to evolving business needs.

3.3 | AI TRiSM Technology Functions

Figure 3 presents the AI technology function architecture. It depicts a hierarchical, five-layer AI TRiSM technology function architecture, progressing from foundational cybersecurity controls to strategic governance. At its base, Traditional Technology Protection secures networks, endpoints, and applications through established firewalls, IDS/IPS, and EDR. Above this, the Infrastructure and Stack layer enforces hardened compute environments and container orchestration. The central Information Governance tier governs data protection, classification, and asset management. AI Runtime Inspection and Enforcement then monitors live model interactions, enforcing policies and detecting anomalies. Finally, the apex AI Governance layer defines risk frameworks, ethical guidelines, and oversight structures that steer all underlying functions. Within the AI TRiSM framework, the foundation is “Traditional Technology Protection” or TTP, comprising time-tested

cybersecurity measures that secure an organisation's entire IT estate. This layer leverages established controls—firewalls, intrusion detection/prevention systems (IDS/IPS), anti-malware, and endpoint detection and response (EDR)—to defend against commonplace threats. The RBAC and multi-factor authentication (MFA) guard administrative and user privileges, while Security Information and Event Management (SIEM) platforms aggregate logs for real-time correlation and alerting. Network segmentation and micro-segmentation isolate sensitive systems, reducing lateral movement opportunities for attackers. Data loss prevention (DLP) tools detect and block exfiltration of confidential files. Although these conventional protections are not AI-specific, they form the first line of defence, ensuring that all downstream AI services operate upon a hardened and monitored infrastructure, free from basic compromise vectors that could undermine higher-order AI governance efforts (Joynt 2025).

Sitting just above this base, Infrastructure and Stack addresses the hardware, virtualization, and cloud platforms that host AI workloads. This tier involves provisioning secure compute clusters—bare-metal servers, GPU/TPU accelerators, and Kubernetes-orchestrated containers—configured with hardened operating systems and minimal attack surface images. Secure credentials management integrates secrets vaults to store API keys and certificates, while Infrastructure as Code (IaC) (Bessghaier et al. 2025; Bykov and Shiboldenkov 2025) templates define network topologies, security group rules, and resource lifecycles in version-controlled repositories. Observability tools continuously monitor CPU/GPU utilisation, memory usage, and I/O operations, enabling capacity planning and anomaly detection. Immutable infrastructure patterns ensure that each deployment is reproducible and can be rolled back instantly in case of misconfiguration or emerging vulnerabilities, thus safeguarding AI components against infrastructure-level risks.

The middle layer, Information Governance: Data Protection, Classification, and Asset Management, ensures that the raw materials fueling AI—data—are managed securely and compliantly. This function encompasses data classification schemes that tag records by sensitivity (e.g., public, internal, confidential, regulated), enforced via data discovery tools that scan repositories for personally identifiable information or intellectual property. Encryption frameworks apply AES-256 at rest and TLS 1.3 in transit, while tokenization replaces sensitive fields in non-production environments. Data catalogues and metadata management systems record lineage, ownership, and transformation logs, enabling traceability from ingestion through model training to inference. Automated privacy compliance checks validate adherence to existing regulatory standards or industry-specific mandates, flagging datasets that require consent renewal or anonymization. Together, these controls guarantee that data governance practices support both traditional business intelligence (BI) needs and AI's heightened privacy requirements.

One level above, AI Runtime Inspection and Enforcement governs real-time interactions with deployed AI models, applications, and agents. At inference time, runtime agents intercept input payloads to perform content anomaly detection—looking for out-of-distribution requests or prompt injections—that could trigger hallucinations or malicious outputs. Policy enforcement

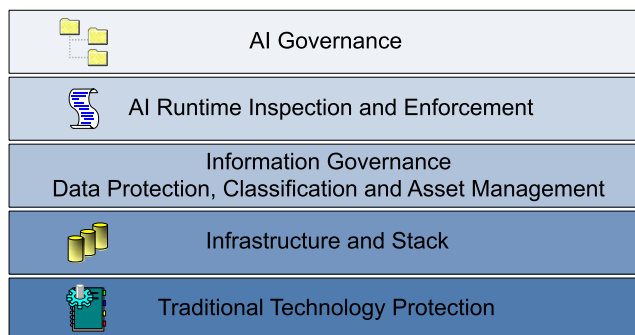


FIGURE 3 | Overview of AI TRiSM technology function architecture.

points validate each request against allow-lists, block-lists, or dynamic context-based rules and can apply redaction or transformation to sensitive inputs before model processing. Continuous monitoring captures inference logs: input features, model confidences, execution latency, and output summaries. Automated alerting pipelines feed into Security Orchestration, Automation, and Response (SOAR) platforms (Gulati et al. 2024; Nguyen et al. 2024; Niyomdi and Oluoch 2025), enabling rapid containment of suspicious behaviour. Runtime sandboxing of models within secure enclaves (e.g., Intel SGX,²² AWS Nitro Enclaves²³) prevents data leakage while feedback loops flag performance degradation or adversarial evasion, triggering governance workflows that quarantine affected endpoints until mitigation is complete.

At the apex of the pyramid, AI Governance defines the strategic controls, policies, and organisational structures that guide responsible AI adoption throughout the enterprise. This function incorporates risk assessment frameworks—such as the NIST AI Risk Management Framework (RMF)²⁴ or ISO/IEC 42001²⁵—to classify AI initiatives by impact and maturity, then assign appropriate oversight levels. Governance councils and ethics boards review model use cases, approving or rejecting deployments based on fairness, accountability, and transparency criteria. Model documentation (e.g., model cards, data-sheets) summarises intended purpose, performance metrics, known limitations, and mitigation strategies for bias or inequity. Governance workflows integrate with ticketing systems (Jira,²⁶ ServiceNow²⁷) to track policy exceptions, audit findings, and compliance attestations. Training programs certify data scientists and developers on AI ethics, legal requirements, and secure coding practices. By embedding governance at the highest level, organisations align AI solutions to corporate values, stakeholder expectations, and evolving regulatory landscapes, ensuring that every model—whether built in-house or consumed through BYOAI—operates within a trusted, risk-managed environment.

Figure 4 presents an end-to-end AI TRiSM technology stack²⁸ (Litan 2025) arranged both vertically—by domain of control—and horizontally—across the AI lifecycle phases from “Initial Steps” through “Run.” At its foundation lie Traditional Technology Protections such as Software Composition Analysis and Application Security Testing, reinforced at runtime by Web

Application and API Protection (WAAP), endpoint security agents, API gateways, and Identity and Access Management (IAM). Above this, the Infrastructure and Stack layer hardens the environment via Trusted Execution Environments (e.g., confidential computing enclaves), specialised AI chipsets, and container-orchestration safeguards. The Information Governance tier relies on Privacy-Enhancing Technologies (PETs) and Data Security Posture Management (DSPM)—including centralised data governance and automated classification—to ensure that both training and inference datasets remain protected and auditable. During deployment and execution, the AI Runtime Inspection and Enforcement layer employs real-time defence mechanisms, dynamic data labeling, contextual access controls, and an AI Gateway to intercept anomalous behaviour or policy violations. Finally, overarching the entire pipeline is AI Governance, encompassing continuous posture management, systematic adversarial testing, formal AI validation processes, and third-party risk oversight, all aligned to the “Develop and Update,” “Deploy,” and “Run” phases of ModelOps. This stacked architecture ensures that every stage of an AI system’s lifecycle is both securely managed and rigorously governed.

3.4 | Adoption of TRiSM for Secure and Responsible AI

In today’s digital economy, embedding trust, risk, and security management into the very fabric of AI delivery is no longer optional—it is imperative. The AI TRiSM framework provides a structured approach to transform unmanaged AI risks into managed, resilient capabilities. At its core, AI TRiSM integrates four interdependent disciplines—explainability/model monitoring, privacy, ModelOps, and AI application security—across the entire AI lifecycle, from data ingestion through model deployment and ongoing operations. Figure 5 illustrates the staged adoption of the AI TRiSM framework as an evolution from unmanaged to managed risk domains. Along the horizontal axis, “Unmanaged Risks” are transformed by the central AI TRiSM intervention point into “Managed Risks.” Below this axis, four core capability areas—Explainability & Model Monitoring, Privacy, ModelOps, and AI Application Security—are depicted with distinct icons. Together, these pillars constitute the operational components

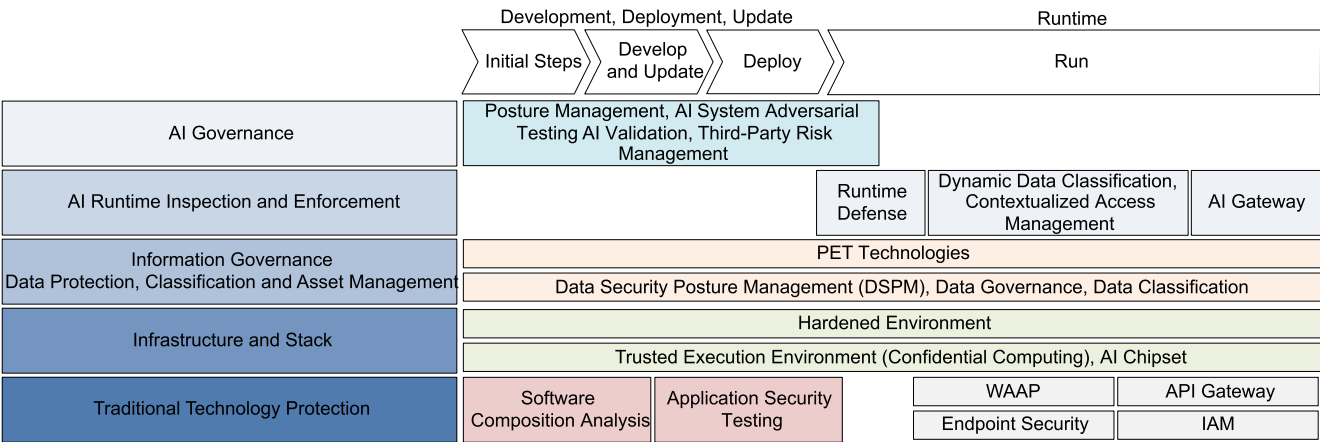


FIGURE 4 | Sample of AI TRiSM technologies.

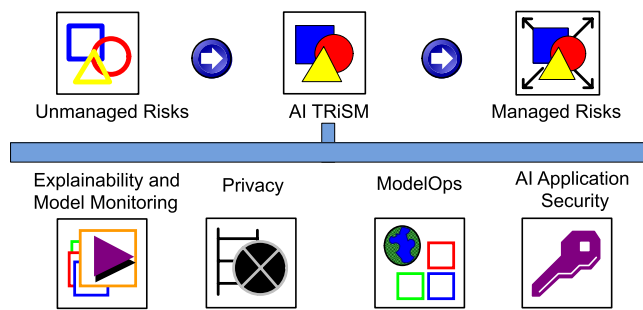


FIGURE 5 | Overview of AI TRiSM adoption strategy.

of the AI TRiSM adoption strategy, ensuring that each phase of risk management is addressed in a coordinated and systematic manner.

This subsection closely mirrors the core pillars of the AI TRiSM framework—Explainability & Model Monitoring, Privacy, ModelOps, and AI Application Security—by presenting them as four interdependent disciplines rather than isolated capabilities. However, whereas the pillar descriptions provide a static taxonomy of capabilities, this adoption narrative reframes them as a dynamic, phased journey: it illustrates how each discipline is sequentially orchestrated to guide an organisation from “unmanaged” to “managed” AI risk. In this way, the adoption perspective transforms the theoretical pillar taxonomy into an actionable roadmap for embedding trust, risk, and security throughout the AI lifecycle.

3.4.1 | Explainability and Model Monitoring

To operationalise explainability, organisations must weave interpretability tools directly into their AI development lifecycle. At training time, SHAP values quantify each feature’s marginal contribution to individual predictions; these values are logged alongside raw inputs and outputs in a feature store. Concurrently, LIME constructs linear surrogates around each inference, enabling rapid human review of decision boundaries without exposing full model internals (Mersha et al. 2024; Akman and Schuller 2024; Bennetot et al. 2024). Complementing these local techniques, global surrogate models—such as shallow decision trees trained on model inputs and outputs—surface overarching patterns, for instance flagging that “loan denial” often correlates with low credit utilisation. Model monitoring extends beyond basic performance metrics. Real-time telemetry captures latency, error rates, and confidence distributions, but it also registers higher-order indicators: Population Stability Index for input drift, Jensen–Shannon divergence for output distribution changes, and subgroup performance metrics to detect fairness regressions. Automated pipelines compare current metrics to historical baselines, triggering retraining workflows or human alerts when defined thresholds—for example, a 3% drop in AUC—or new data profiles emerge. These monitoring systems feed centralised dashboards that visualise trends across multiple models, business units, and geographies, enabling governance committees to pinpoint emerging risks before they materialise. Rich metadata—model cards, data cards, and “What-If” analysis outputs—are versioned within a model registry. This

registry tracks lineage: which data snapshot, hyperparameters, code commit, and explainability artefact produced each model iteration. Audit hooks ensure that any change to feature engineering, labeling rules, or reference data spawns a new version with its own card, preserving reproducibility. For regulated environments, this comprehensive logging fulfils compliance mandates, while steering continuous improvement: stakeholders can review why a scoring threshold was adjusted and how that change affected false-positive rates across demographic groups.

3.4.2 | Privacy

Embedding privacy into AI TRiSM requires a holistic, data-centric strategy—from initial ingestion to long-term storage and inference. At the data-acquisition layer, classification engines tag every field according to sensitivity—public, internal, confidential, or regulated—populating a metadata catalogue that drives downstream controls. Data subject consent flags are stored in a consent management system and propagated through the feature store; unauthorised records are automatically excluded from training, ensuring respect for user preferences. To protect against re-identification, data is transformed using advanced anonymization techniques. k -anonymity groups records so that each quasi-identifier appears in at least k records, while ℓ -diversity guarantees that sensitive attributes within each group display sufficient variation. For high-risk scenarios, differential privacy injects calibrated noise into training gradients or query results, mathematically bounding any individual record’s influence on model outputs. The privacy budget (ϵ) is monitored over time to prevent budget exhaustion, preserving long-term privacy guarantees. Infrastructure security underpins these algorithms: (i) data at rest is encrypted with AES-256, and (ii) in transit with TLS 1.3. Hardware Security Modules (HSMs) manage encryption keys under strict access policies, rotated regularly to limit exposure. The RBAC and attribute-based policies enforce “least privilege” across data lakes, feature stores, and model serving endpoints. Automated policy-as-code (PAC) systems (Ray 2024)—written in languages like Rego²⁹ for Open Policy Agent—validate every data pipeline change against privacy rules, rejecting code that would expose unauthorised fields. Finally, for compliance with regulations, automated workflows handle data-subject requests: “right to erasure” jobs scrub personal records from databases and trigger model retraining to expunge their influence. Immutable audit logs capture every request and action—username, timestamp, action type—creating a verifiable record for internal audits and external regulators.

3.4.3 | ModelOps

ModelOps transforms AI development into a disciplined, production-grade engineering practice by codifying every stage of the model lifecycle. In source control, Git repositories house not only code but also data schemas, preprocessing pipelines, and hyperparameter configurations. Automated CI jobs run static code analysis (e.g., Bandit³⁰ for Python security checks) and unit tests for data transformations, ensuring that

schema changes or library upgrades do not introduce vulnerabilities or break data contracts. Upon merge, CD workflows build container images—bundling model artefacts, dependencies, and inference code—tagged by semantic version numbers. These images pass through bias detection gates: specialised tests compute fairness metrics (e.g., demographic parity, equal opportunity difference) on synthetic and hold-out datasets. Simultaneously, vulnerability scanning tools (e.g., Clair³¹ or Trivy³²) inspect container layers for outdated packages, reporting critical CVEs. Only when both bias and security gates pass does the pipeline promote the image to staging clusters. In staging, canary deployments route a small percentage of live traffic to the new model, while shadow deployments mirror production traffic to parallel inference pods. Real-time monitoring collects business KPIs—transaction volumes, conversion rates—and technical metrics—latency, resource utilisation—to detect regressions. Runbooks automate rollback (Sneha and Goedtel 2024) to the previous stable version if error rates exceed configurable thresholds. Observability platforms visualise metrics alongside tagging by model version, simplifying root-cause analysis when anomalies occur. When stability is confirmed, blue-green or rolling updates move the canary to full production. The IaC frameworks ensure that model serving clusters (Feitosa et al. 2024; Konala et al. 2025), autoscaling policies, and network rules are consistently provisioned across environments. Governance plugins enforce policy-as-code: every production model must be accompanied by signed artefacts, preventing any “shadow” model from escaping oversight.

3.4.4 | AI Application Security

AI AppSec addresses novel threat vectors introduced by machine-learning components, layering specialised defences across the model infrastructure. At the edge, input validation enforces strict schemas and rate limits to prevent adversarial injection attacks—maliciously crafted inputs designed to manipulate model behaviour. Techniques such as feature bounding and input sanitization remove anomalies that lie outside the training distribution, rejecting suspicious requests before they reach the

inference engine. During training and serving, workloads execute in container sandboxes governed by Kubernetes Network Policies, isolating pods into micro-segmented namespaces. Strict egress controls prevent models from exfiltrating data unexpectedly, while ingress rules ensure that only authenticated, authorised clients may invoke inference APIs. The RBAC and mTLS authenticate service-to-service calls, preventing lateral movement within the cluster. To resist model extraction and theft, organisations embed digital watermarks—unique perturbations in model parameters that remain imperceptible to legitimate use but reveal stolen copies. Monitoring systems track inference patterns: elevated request volumes, unusually high confidence scores, or off-peak traffic surges trigger alerts for potential scraping or probing campaigns. The SIEM systems ingest these AI-specific indicators of compromise and correlate them with infrastructure logs—SSH sessions (Vielberth 2025; Tendikov et al. 2024), container restarts—to rapidly identify incidents. Adversarial training further hardens models by exposing them to crafted examples—generated via FGSM or PGD—during training, enhancing resilience against perturbation attacks. Continuous fuzz testing channels random noise and targeted adversarial inputs through serving endpoints, uncovering blind spots. Runtime Application Self-Protection (RASP) (Gartner 2025) agents instrument inference binaries, detecting anomalous input patterns and dynamically enforcing containment policies (e.g., throttling, quarantine).

3.5 | AI TRiSM Enterprise Integration Architecture

Figure 6 presents an AI TRiSM enterprise integration architecture³³ (Litan 2025). At the perimeter of the enterprise network sits a hardened ingress layer composed of a network proxy and configuration-change router. Incoming requests—from browser widgets, cloud functions, or embedded device agents—are first subject to TLS/mTLS enforcement and dynamic routing rules based on versioned deployment metadata. This ensures that only compliant traffic reaches the core AI services and that any configuration drifts are caught and rolled back before they can impact production workloads.

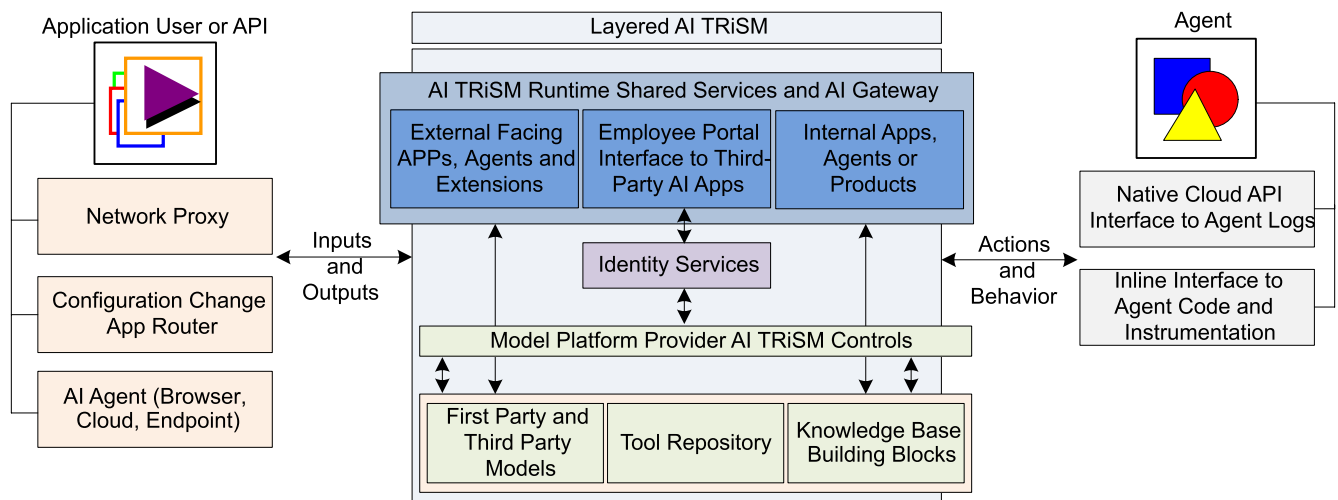


FIGURE 6 | AI TRiSM enterprise architecture.

Beyond the ingress, all flows converge on the AI TRiSM Gateway, which exposes three distinct service planes:

- External-Facing Extensions: Relates to hosting customer-facing APIs and partner integrations
- Employee-Portal Interfaces: Connected to powering internal dashboards and third-party SaaS connectors
- Internal Apps and Agents: Aimed for servicing back-office microservices and on-device agents

Each plane is fronted by a unified Identity Services mesh implementing single-sign-on, OAuth/OIDC token brokering, and ABAC. This identity fabric ensures that every request carries cryptographically verifiable context and can be authoritatively mapped to a user, service, or agent identity. Authenticated requests then enter the Model Platform Provider's TRiSM Control Plane. Here, a policy engine references registries of first- and third-party models, taps into a centralised tool repository (for data preprocessing, RAG pipelines, or stylistic transforms), and consults a structured knowledge base of ontologies and vetted prompt templates. Real-time policy checks—content-safety gates, privacy-budget calculators, fairness thresholds—are applied to each inference invocation, and every decision is risk-scored and immutably logged before execution. On the agent side, runtime nodes are dual-instrumented for maximum observability and control. Native Cloud-API Hooks stream audit logs, anomaly alerts, and performance telemetry back to the control plane via secure message buses, while Inline SDK Instrumentation weaves policy enforcement and detailed event capture—prompt modifications, user overrides, drift signals—directly into the agent's code path.

3.6 | Significance of Integration of AI TRiSM Framework

Figure 7 synthesises the core value propositions of the AI TRiSM framework by arranging seven thematic layers around a central “AI TRiSM Significance” axis. At the top, the framework emphasises enhancing understandability and transparency, hardening models against adversarial attacks, and embedding

continuous monitoring and adaptation. Beneath the significance core, it highlights the importance of mitigating risks associated with third-party or generative AI providers, securing data privacy when utilising external AI services, and ensuring end-to-end regulatory compliance.

3.6.1 | Enhancing Understandability and Transparency

Embedding explainability and transparency into AI workflows is essential for clarifying complex model behaviours and building stakeholder confidence. Within the AI TRiSM framework, XAI techniques are integrated into both training and inference pipelines to illuminate how feature values contribute to each prediction. By generating feature importance scores, saliency maps over image inputs, or local surrogate decision trees for tabular models, XAI transforms inscrutable “black-box” algorithms into transparent, auditable systems. Complementary to algorithm-level insights, comprehensive Model Cards catalogue metadata including intended use cases, training data distributions, evaluation metrics (e.g., accuracy, ROC-AUC, calibration), and known limitations, providing end-to-end documentation. Interactive dashboards present drift indicators, counterfactual explanations, and bias heatmaps, enabling data scientists to pinpoint spurious correlations and business users to validate outputs against domain knowledge. Regulatory compliance is also streamlined: transparency requirements under laws like the EU AI Act (Eu Artificial Intelligence Act 2025) or the UK's Data Protection and Digital Information Bill³⁴ mandate clear explanations of automated decisions, which XAI artefacts and Model Cards readily satisfy. Ultimately, robust explainability and transparency initiatives under AI TRiSM empower cross-functional teams to detect hidden biases, improve model quality, and ensure that AI-driven recommendations align with ethical and legal standards.

3.6.2 | Protecting Against Adversarial Attacks

Adversarial attacks pose a severe threat to AI systems by subtly perturbing inputs or poisoning training data to provoke erroneous outcomes or disclose sensitive information. Within AI TRiSM's security pillar, a multi-layered defence strategy is prescribed. Adversarial training augments the training set with examples crafted by FGSM or PGD, hardening models against evasion tactics. Defensive distillation leverages softened teacher-student network paradigms to smooth decision boundaries, reducing local sensitivity to crafted perturbations. At runtime, anomaly detectors continuously monitor gradients, confidence scores, and feature activations for deviations from baseline distributions, flagging suspicious inference requests in real time. In parallel, comprehensive red teaming exercises—using open-source libraries like CleverHans or Foolbox—simulate both evasion and backdoor scenarios to validate detection algorithms and refine response protocols. When an attack is detected, automated incident response workflows isolate affected model endpoints, rotate compromised keys via secrets management solutions, and revert traffic to a previously validated model snapshot stored in a centralised registry. Detailed forensic logs, including input hashes and perturbation vectors, support post-incident analyses and compliance reporting.

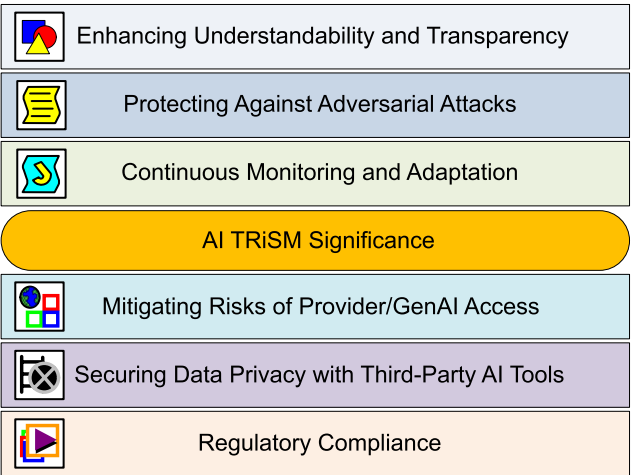


FIGURE 7 | Overview of AI TRiSM technology significance.

3.6.3 | Continuous Monitoring and Adaptation

AI models in production inevitably encounter data distributions that shift over time, a phenomenon known as drift, which can degrade performance and erode trust. The AI TRiSM framework mandates continuous observability and adaptive retraining mechanisms to maintain reliability. Observability stacks—built on tools like Prometheus for metric collection, Grafana for visualisation, and Datadog AI Ops for anomaly detection—track key performance indicators (KPIs) such as accuracy, precision, recall, and ROC-AUC in near real-time. Statistical drift detectors calculate metrics like PSI and Jensen–Shannon divergence to compare incoming feature distributions against training baselines. Upon detecting drift beyond predefined thresholds, automated ModelOps pipelines trigger retraining routines: fresh data is ingested, cleaned, and split; models are retrained and validated against hold-out datasets; and new artefacts are versioned in an immutable registry. Deployment to production employs safe rollout strategies—such as canary or blue-green deployments—to minimise user impact, with real-time comparison of latency, throughput, and error rates. Immutable audit logs record every training run, data snapshot, and deployment event, facilitating traceability and regulatory audits (Antony et al. 2024; Strielkowski et al. 2025; Enemosah 2025). This continuous monitoring and adaptation loop closes the gap between static model development and dynamic production environments, ensuring that AI systems remain effective, fair, and aligned with evolving real-world conditions.

3.6.4 | Mitigating Risks of Provider

The proliferation of BYOAI and shadow generative AI deployments introduces significant governance challenges, as enterprise users may access unvetted external models outside official IT oversight (Vysek 2025). AI TRiSM extends corporate guardrails to every AI interaction by channelling all third-party API calls through secure gateways. These gateways enforce data classification policies, ensure content moderation, implement rate limiting, and generate comprehensive request/response logs for auditability. Before whitelisting a vendor's GenAI endpoint, organisations conduct thorough risk assessments that evaluate model provenance, documented training data governance, security certifications, and compliance commitments. The ABAC policies define fine-grained permissions, restricting sensitive data flows based on user role, data sensitivity label, and contextual risk factors. Token-based authentication (e.g., OAuth 2.0 or JWT) ensures that only authorised applications and personnel can access external AI resources. Moreover, metadata tagging and policy-as-code rules dynamically enforce usage constraints—such as disallowing the transmission of personally identifiable information or regulated data segments—to prevent inadvertent exposures. Centralised monitoring dashboards correlate internal and external AI usage metrics, alerting governance teams to shadow AI spikes or abnormal data flows.

3.6.5 | Securing Data Privacy with Third-Party AI Tools

When integrating external AI platforms, organisations often must share substantial volumes of proprietary or sensitive

information, elevating privacy and confidentiality concerns. AI TRiSM addresses these through a suite of data protection controls. Data minimization dictates that only attributes strictly necessary for a model's objective are collected, while de-identification techniques—such as k -anonymity and ℓ -diversity—obfuscate direct identifiers. Differential Privacy (DP) (Nawshin et al. 2024; Zhang, Li, et al. 2024) adds mathematically calibrated noise to aggregated training statistics or query outputs under a defined (ϵ, δ) privacy budget, bounding any single record's influence. Encryption at rest employs AES-256 keys managed by hardware security modules, and encryption in transit leverages TLS 1.3 to protect network channels. Consent management frameworks record individual user preferences and dynamically filter or revoke data usage in compliance with existing regulations. Privacy-Enhancing Technologies (PETs) (d'Aliberti et al. 2024; Dritsas et al. 2024)—including Secure Multi-Party Computation (SMPC) and Homomorphic Encryption (HE)—facilitate joint model training or inference on encrypted data, enabling collaboration with external partners without exposing plaintext inputs. Automated privacy audits embedded in CI/CD pipelines verify DP budget consumption, inspect federated learning configurations, and ensure that no sensitive fields slip through transformation steps. Comprehensive data lineage tracking, spanning source ingestion through feature engineering and model training, maintains an auditable chain of custody, thereby reinforcing accountability and streamlining regulatory reporting.

3.6.6 | Regulatory Compliance

Global regulatory regimes—ranging from the EU AI Act's risk-based classification to the CCPA and industry-specific mandates like HIPAA—require demonstrable adherence to principles of transparency, fairness, and data protection. AI TRiSM integrates policy-as-code engines directly into CI/CD workflows, automating checks for bias thresholds, privacy guardrail enforcement, and completeness of documentation prior to any production release. Centralised registries store Model Cards, Data Sheets, audit trails, and versioned artefacts that capture model lineage, hyperparameters, evaluation results, and deployment metadata. These registries support on-demand reporting to external auditors or regulators, significantly reducing manual audit overhead. Role-based dashboards provide real-time compliance status, highlighting any deviations from approved policies or drift in fairness metrics. Automated alerts flag upcoming privacy budget exhaustion under DP schemes or policy violations triggered by new data ingestion (Demelius et al. 2025; Vadisetty and Polamarasetti 2024). Thus, by codifying regulatory requirements into automated gates and preserving immutable logs, AI TRiSM ensures that organisations can rapidly demonstrate compliance, avoid costly penalties, and maintain the flexibility to adapt to evolving legal frameworks without disrupting AI innovation.

4 | AI Trust, Risk and Security Management Taxonomy

This section presents a unified taxonomy for AI Trust, Risk, and Security Management, offering a clear framework to navigate the intertwined challenges of building and governing

responsible AI. We map out key trust dimensions—such as fairness, explainability, and transparency—alongside a comprehensive risk catalogue and security controls that span the model lifecycle.

4.1 | Trust Taxonomy

Figure 8 presents the taxonomy of AI trust. In the section below, we organise the core dimensions of AI trust into a structured taxonomy, illustrating how systems can be designed, evaluated, and governed to meet rigorous expectations for fairness, explainability, transparency, accountability, ethical alignment, inclusiveness, human oversight, trust calibration, and sustainability.

4.1.1 | Fairness

Ensuring fairness in AI systems involves actively mitigating biases that may emerge from dataset imbalances or skewed feature representations (Luo et al. 2025; Chinta et al. 2025; Nepomuceno and Petrillo 2025). In practice, fairness requires deploying statistical measures—such as demographic parity, equalised odds, or disparate impact ratio—to quantify and correct discriminatory outcomes across protected groups (e.g., race, gender, age). Machine learning pipelines must incorporate pre-processing techniques (like resampling or reweighting), in-processing interventions (e.g., adversarial debiasing), and post-processing adjustments (e.g., output thresholding) to uphold equitable decision boundaries. Embedding fairness constraints within the optimization objective harmonises utility and equity, while fairness-aware feature selection prevents selection of proxies that encode sensitive attributes (Parraga et al. 2025; Cao et al. 2025). Moreover, continuous monitoring for concept drift ensures that evolving data distributions do not erode fairness guarantees over time. Integrating fairness governance into ModelOps workflows entails documenting provenance of training data, versioning bias mitigation models, and maintaining an audit trail for regulatory compliance—particularly under statutes such as the EU AI Act or the U.S. Algorithmic Justice and Online Platform Transparency (AJOPT)³⁵ proposals. Cross-functional teams comprising data

scientists, domain experts, and ethicists collaboratively define fairness metrics tied to business goals, ensuring that algorithmic outcomes align with organisational values. To enhance stakeholder trust, fairness reports and model cards disclose performance disparities and corrective measures, fostering transparency in AI-driven decisions. Thus, a robust fairness framework transforms AI systems from opaque “black boxes” into accountable, equitable solutions that reflect societal norms and legal mandates, strengthening public confidence in automated decision-making (Pant et al. 2025; Pulivarthy and Whig 2025).

4.1.2 | Explainability

Explainability in AI—often referred to as XAI—focuses on elucidating how complex models derive specific outputs, thereby enabling stakeholders to interpret and validate algorithmic behaviour (Baron 2025; Mumuni and Mumuni 2025; Bárcena et al. 2025). Techniques such as LIME, SHAP, and counterfactual reasoning convert high-dimensional feature interactions into human-understandable narratives or visualisations. For neural networks, saliency maps and layer-wise relevance propagation highlight input regions that most influence model predictions, while rule extraction methods translate decision boundaries into logical if-then constructs. Embedding explainability into the design phase involves selecting inherently interpretable architectures—such as decision trees, generalised additive models (GAMs), or linear models—for high-stakes applications like credit scoring or medical diagnosis. During ModelOps, explainability dashboards provide real-time insights, enabling operators to trace anomalies and detect model degradation. Rigorous documentation, including model cards and datasheets for datasets, captures design choices, training parameters, and known limitations, facilitating reproducibility and compliance with emerging governance frameworks (e.g., ISO/IEC 42001). Furthermore, explainability enhances model validation by revealing hidden correlations and unintended biases, guiding data refinement and feature engineering (Pavlidis 2024; Galli et al. 2024). Collaborative evaluation sessions with end-users and domain experts ensure that explanations align with business semantics and regulatory expectations, reducing the risk of adversarial exploitation or stakeholder mistrust.

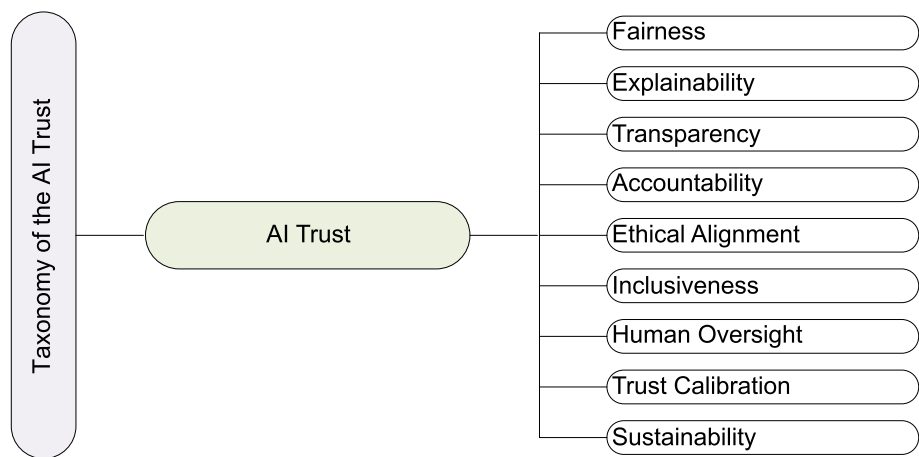


FIGURE 8 | Taxonomy of AI trust.

4.1.3 | Transparency

Transparency in AI underscores the obligation to make every facet of model development, deployment, and operation open for inspection by relevant stakeholders (Cheong 2024; Akinrinola et al. 2024). This encompasses disclosing the provenance and lineage of datasets (e.g., data versioning, source documentation), clarifying preprocessing transformations (e.g., feature encoding, normalisation), and detailing model architecture choices (e.g., hyperparameters, training epochs, regularisation techniques). Transparency mandates that organisations maintain immutable audit logs—capturing model training runs, parameter changes, and deployment events—often facilitated by MLOps platforms (e.g., MLflow,³⁶ Kubeflow³⁷). These logs not only support compliance requirements (e.g., GDPR's accountability principle) but also enable root-cause analysis when adverse incidents occur. Transparency further extends to contractual agreements with third-party model providers, stipulating data usage rights, IP protections, and privacy safeguards (Akinrinola et al. 2024; Fehr et al. 2024). Publishing model cards at API endpoints communicates performance metrics across demographic slices, risk factors, intended use cases, and known limitations, empowering integrators and end users to make informed judgements. Additionally, transparent governance frameworks establish clear roles and responsibilities—defining who owns model approvals, who monitors runtime metrics, and who escalates potential failures. Interfacing with interactive XAI tools, transparency initiatives facilitate stakeholder feedback loops, where domain experts can annotate explanation outputs and suggest corrective refinements (Tang, Li, et al. 2024).

4.1.4 | Accountability

Accountability in AI demands that organisations clearly delineate responsibility for each stage of the AI lifecycle, from data collection and feature engineering to model selection, deployment, and ongoing maintenance (Ojewale et al. 2025; Schmidt, Bartsch, Adam, and Benlian 2025; Schmidt, Bartsch, Zweidinger, and Benlian 2025). Establishing accountability frameworks involves defining governance policies—codified in internal AI ethics charters or External AI Accountability Schemes—that assign roles such as Data Steward, Model Owner, and Compliance Officer. These roles carry specific obligations: Data Stewards manage data quality and privacy compliance (e.g., ensuring consent under GDPR), Model Owners oversee validation tests and performance benchmarks, and Compliance Officers audit adherence to legal and ethical standards. Implementing formal approval gates within ModelOps pipelines ensures that no model progresses to production without sign-off by multidisciplinary review boards comprising legal, security, and ethics experts. Automated policy enforcement tools—integrated via PAC frameworks—validate that each artefact (e.g., data schema, training script, model binary) satisfies organisational rules before deployment. Traceability mechanisms, such as model lineage visualisations, facilitate forensic investigations in the event of adverse outcomes (e.g., discriminatory lending decisions), enabling root-cause analysis and targeted remediation. Accountability also extends to external stakeholders through

transparent reporting channels; for example, publishing periodic AI Impact Assessments that outline identified risks, mitigation measures, and residual uncertainties (Punia et al. 2025; Islam 2024).

4.1.5 | Ethical Alignment

Ethical alignment in AI ensures that system design, data practices, and operational behaviours cohere with both organisational values and broader societal norms (Bradley and Saad 2024; Palumbo et al. 2024; Peterson and Gärdenfors 2024). This alignment begins by codifying ethical principles—such as respect for autonomy, prevention of harm, justice, and beneficence—into machine learning development guidelines. Ethics checkpoints during the AI lifecycle validate that training datasets do not encode discriminatory biases, that optimization objectives avoid harmful incentives, and that deployment contexts respect human dignity (Kim et al. 2021; Ji et al. 2023). Frameworks like the OECD AI Principles³⁸ or IEEE's Ethically Aligned Design inform organisational policy, driving the creation of AI Ethics Boards that review proposed use cases and assess moral implications. Technical implementations include fairness-aware algorithms, privacy-enhancing computations (e.g., differential privacy), and harm detection modules that flag outputs likely to produce adverse real-world effects. Ethical alignment also demands stakeholder engagement sessions—where community representatives, domain experts, and affected groups vet AI functionalities—ensuring that lived experiences shape system behaviour. Documentation artefacts, such as AI ethics impact logs, record deliberations, identified ethical trade-offs, and mitigation commitments, providing traceable evidence of conscientious decision-making (McDonald 2023; Hou and Green 2023).

4.1.6 | Inclusiveness

Inclusiveness in AI refers to designing systems that accommodate the diverse needs, contexts, and abilities of its end users, thereby preventing marginalisation of underrepresented groups (Alevizos et al. 2023; Zowghi and Bano 2024). Inclusive AI begins with comprehensive stakeholder analysis—identifying varied demographic, cultural, linguistic, and accessibility requirements—and extends through dataset curation processes that ensure representative sampling across multiple dimensions (e.g., geographic regions, socio-economic strata). Feature engineering must consider localised idioms, dialects, and non-standard notation to prevent misclassification or performance degradation, particularly in natural language processing models. Accessibility standards (such as WCAG 2.1 for web interfaces³⁹) guide UI/UX design, ensuring compatibility with assistive technologies (e.g., screen readers, voice interfaces) and accommodating cognitive, motor, and sensory impairments. In Federated Learning scenarios, inclusive AI also means enabling participation from edge devices in regions with limited connectivity or hardware constraints, thereby democratising model improvements (Chi et al. 2021; Shams et al. 2023). During evaluation phases, fairness audits stratified by protected attributes—such as gender, ethnicity,

or age—expose disparate impacts, prompting retraining with augmented data or algorithmic adjustments. Inclusive product roadmaps integrate community feedback loops, where end users from diverse backgrounds iteratively test prototypes and suggest refinements.

4.1.7 | Human Oversight

Human oversight in AI ensures that automated systems remain under accountable, informed supervision, blending algorithmic efficiency with human judgement. This oversight can take the form of human-in-the-loop (HITL) processes—where humans validate or correct model outputs before finalising decisions—or human-on-the-loop paradigms—where operators monitor real-time performance and retain the authority to override or suspend AI actions (Retzlaff et al. 2024; Natarajan et al. 2025; Memarian and Doleck 2024). Strategic integration of oversight mechanisms begins during design with the definition of critical control points (e.g., anomaly detection triggers, confidence thresholds) that escalate ambiguous or low-confidence cases to human reviewers. In high-stakes domains such as healthcare or finance, AI systems generate decision support recommendations—like diagnostic suggestions or credit risk scores—while clinicians or analysts retain ultimate decision-making power. Oversight frameworks rely on XAI tools that visualise key factors influencing predictions, enabling humans to interrogate model reasoning (Drori and Te'eni 2024; Debnath et al. 2025). Feedback loops capture human corrections, integrating them via active learning to progressively refine model behaviour. Governance policies specify responsibilities, response service level agreements (SLAs), and training requirements to equip oversight personnel with domain knowledge and AI literacy.

4.1.8 | Trust Calibration

Trust calibration involves tuning user confidence in AI systems to align with actual system capabilities, thereby avoiding over-reliance or undue scepticism. This calibration is achieved by communicating model uncertainty—using probabilistic confidence scores, predictive intervals, or entropy measures—to inform users when predictions are reliable versus when caution is warranted (Naiseh et al. 2024; Liebherr et al. 2025; Bostrom et al. 2024). Visual cues, such as colour-coded confidence bars or “confidence rings” around key outputs, guide end users to double-check low-certainty results. Embedding trust calibration within user interfaces prevents automation bias, where users might unquestioningly accept AI recommendations, and reduces friction from under-trust, where users ignore valuable insights (Klingbeil et al. 2024; Bostrom et al. 2024; Afroogh et al. 2024). Continuous performance monitoring captures real-world accuracy metrics, dynamically adjusting calibration thresholds as models encounter novel data distributions. Educational tooltips and contextual explanations accompany low-certainty alerts, helping users understand why confidence is low—for instance, due to sparse training data in a particular feature space. In collaborative settings, trust calibration fosters effective human–AI teaming: when systems self-report limitations, human experts can allocate cognitive resources

judiciously. Governance guidelines mandate periodic user training sessions, reinforcing interpretation of confidence indicators and promoting mental models that align user expectations with system reliability levels (Tutul et al. 2025; Henrique and Santos Jr 2024).

4.1.9 | Sustainability

Sustainability in AI encompasses designing and operating systems with minimal environmental impact and efficient resource utilisation (Rohde et al. 2024; Olawade et al. 2024). Model development practices such as Green AI prioritise algorithms and architectures that achieve performance objectives with lower computational footprints—employing techniques like neural architecture search constrained by FLOPs (floating-point operations per second) budgets, weight quantization, and pruning to reduce energy consumption. Carbon-aware scheduling shifts energy-intensive training tasks to off-peak grid hours or renewable-powered data centers, while serverless and edge-compute paradigms distribute workloads closer to data sources, minimising data transfer overhead (Falk and van Wynsberghe 2024; Green 2025). Lifecycle assessments quantify the total carbon footprint—from data center energy use during training and inference to device electricity consumption—informing optimization strategies. MLOps platforms integrate sustainability dashboards tracking metrics such as GPU utilisation, training energy per experiment, and CO₂e emissions, promoting transparency and continuous improvement. Collaboration with Cloud Service Providers (CSPs) offering renewable energy commitments and carbon offset programs further aligns AI operations with corporate sustainability goals. Additionally, sustainable AI extends to model longevity: adopting modular update mechanisms and transfer learning avoids retraining monolithic models from scratch, thereby conserving computational resources (Shafik 2024; Olawumi and Oladapo 2025). Through conscientious architectural choices, operational policies, and cross-industry partnerships, organisations ensure that AI innovations contribute to technological advancement without compromising ecological integrity.

4.2 | Brief Survey of Key Trust Tools

In an era defined by rapid advances in artificial intelligence, ensuring that AI systems are trustworthy, secure, and ethically sound has become imperative. The AI TRiSM framework identifies five core dimensions—governance, trustworthiness, fairness, reliability, and data protection (first introduced in Section 2)—that organisations must address to deploy AI responsibly. Numerous specialised open-source and industry-backed toolkits have since emerged to address these dimensions, each offering unique capabilities and strengths. Table 3 offers a concise, side-by-side comparison of five widely adopted AI trustworthy frameworks. For each framework, we summarise the core toolkit components, underlying algorithms, their alignment with AI TRiSM dimensions (fairness, transparency, robustness, governance, and data protection), and licensing terms. This overview is intended to guide practitioners in selecting the framework best suited to embedding TRiSM capabilities within their AI development and deployment workflows.

TABLE 3 | Comparison of some prominent AI trustworthiness frameworks.

Framework	Key tools and components	Key algorithms and methods	Significance in AI TRISM	Licence
AI Fairness 360 (AIF360)	• Dataset and model bias metrics • Explainers for metric interpretation • Bias-mitigation algorithms (pre-processing, in-processing, post-processing)	• Metrics: disparate impact, generalised entropy, differential fairness, rich-subgroup fairness, bias scan • Mitigations: Optimised Preprocessing, Disparate Impact Remover, Reweighing, Adversarial Debiasing, Equalised Odds, Learning Fair Representations, Exponentiated-Gradient Reduction	• Fairness: comprehensive bias measurement & mitigation • Trustworthiness: explainers for transparency • Reliability: integrated stability checks under distribution shifts	Apache 2.0
AI Explainability 360 (AIX360)	• Interactive taxonomy & model-card guidance • Local vs. global explainers • Interpretable methods: data-centric, model-centric, post hoc, direct	• ProtoDash, ProtoVAE, Contrastive Explanations (CEM), LIME, SHAP, ICE/GCE, Integrated Gradients, Teaching AI to Explain • Rule induction: Ripper, GLRM, ProfWeight, CoFrNets, Ecertify	• Transparency: local/global explanations • Trustworthiness: certified explanation stability • Governance: explainer taxonomy • Reliability: audit-ready interpretability	Apache 2.0

(Continues)

TABLE 3 | (Continued)

Framework	Key tools and components	Key algorithms and methods	Significance in AI TRISM	Licence
Adversarial Robustness Toolbox (ART)	• Attack modules (evasion, poisoning, extraction, inference) • Defence primitives: pre/postprocessing, training, detectors • Estimators	• Evasion: Auto-PGD, C&W, PGD, FGSM, DeepFool, HopSkipJump • Poisoning: backdoors, clean-label, BadDet, sleeper agents • Extraction: KnockoffNets, function-equivalent • Inference: membership, attribute, model inversion • Defences: DefenseGAN, Mixup, CutMix, RSmooth	• Security: red/blue assessment • Reliability: adversarial testing • Robustness: certifiable defence methods	Apache 2.0
TrustGraph	• CLI & YAML config builder • TrustRAG pipelines (graph + vector search) • REST/WebSocket, Python SDK • Observability: Prometheus, Grafana	• Autonomous agents for topic/entity extraction • Hybrid RAG: subgraph traversal + vector search • ReAct-style agent orchestration • Graph/embedding export/import	• Governance: auditable knowledge deployment • Transparency: graph-based provenance • Reliability: reproducible RAG with telemetry • Data Protection: context-limited subgraphs	Apache 2.0
NVIDIA Trustworthy AI	• Model Card++ (extended documentation) • Omniverse Replicator (synthetic 3D data) • NeMo Guardrails (LLM runtime safety)	• Privacy: synthetic-data, confidential computing • Safety/Security: isolation, DSL guardrails • Transparency: provenance metadata • Fairness: bias-controlled dataset tools	• Data Protection: hardware + synthetic pipelines • Safety/Security: runtime guardrails • Transparency: readable model cards • Fairness: diverse synthetic data	CC0-1.0 licence

AI Fairness 360 (AIF360),⁴⁰ developed by IBM, is laser-focused on the fairness pillar. It provides a comprehensive suite of measurement metrics—such as disparate impact ratios, generalised entropy, differential fairness, and rich-subgroup scans—to detect biases at both the dataset and model levels. Beyond measurement, AIF360 offers a range of bias-mitigation algorithms at every stage of the machine-learning pipeline: preprocessing (Optimised Preprocessing, Disparate Impact Remover, Reweighting), in-processing (Adversarial Debiasing, Prejudice Remover Regularizer), and postprocessing (Equalised Odds, Reject Option Classification). In the context of AI TRISM, AIF360's explainers and metrics directly support fairness and bolster trustworthiness by illuminating where and how bias arises—and by providing concrete methods to mitigate it.

AI Explainability 360 (AIX360)⁴¹ complements AIF360 by attacking the transparency and trustworthiness dimensions. It assembles a broad taxonomy of explanation techniques—ranging from prototype-based methods (ProtoDash), contrastive explanations (CEM with monotonicity constraints and exemplar extensions), to popular model-agnostic tools (LIME, SHAP), as well as globally interpretable models (rule induction, Continued Fraction Networks). Importantly, AIX360 includes certified explanation methods (Ecertify) to guarantee the stability of local explanations under input perturbations, reinforcing reliability in interpretability. By helping developers choose the right explanation paradigm for their use case—local vs. global, direct vs. post hoc—AIX360 strengthens governance workflows and makes AI decisions intelligible to non-technical stakeholders.

While AIF360 and AIX360 focus on fairness and transparency, the Adversarial Robustness Toolbox (ART)⁴² addresses the security and reliability pillars head-on. ART equips both “red team” attackers and “blue team” defenders with an arsenal of evasion attacks (Auto-PGD, C&W, DeepFool, Square, HopSkipJump), poisoning strategies (backdoors, feature-collision, sleeper agents), extraction methods (Copycat CNN, KnockoffNets), and inference exploits (membership and attribute inference, model inversion). On the defensive side, ART bundles preprocessors (DefenseGAN, pixel-squeezing, JPEG compression), postprocessors (random noise, output rounding), adversarial training

protocols (Madry's PGD, Fast Is Better Than Free), and detectors (spectral signatures, RONI). In AI TRISM terms, ART's comprehensive red-blue cycle ensures security, robustness, and reliability, safeguarding models against manipulation and verifying that defences hold up under adversarial pressure.

Stepping beyond individual model-centric toolkits, TrustGraph⁴³ offers an end-to-end knowledge management platform that streamlines governance and operational oversight. TrustGraph's CLI-driven, YAML-configured “TrustRAG” pipelines automatically ingest documents, construct semantic knowledge graphs and vector embeddings, and assemble Retrieval-Augmented Generation workflows. By packaging these artefacts as versioned “Knowledge Cores” with built-in telemetry (via Prometheus and Grafana), TrustGraph delivers reproducible, auditable deployments. In the AI TRISM landscape, this contributes to robust governance—ensuring that every piece of knowledge and every retrieval step is logged, versioned, and monitorable—and enhances reliability through consistent, transparent retrieval processes.

Finally, NVIDIA's Trustworthy AI⁴⁴ initiative ties hardware-level security to high-level ethical and transparency goals. Confidential computing features in NVIDIA GPUs protect data and model weights in use, addressing the data protection pillar by minimising exposure of sensitive information. At the same time, NVIDIA advances fairness and transparency through tools like Omniverse Replicator—which generates bias-controlled synthetic datasets—and Model Card++, an extended documentation format that captures a model's intended uses, limitations, provenance, and performance trade-offs in plain language. NeMo Guardrails, a runtime DSL, enforces content safety and factuality in LLMs, closing the loop on the safety aspect of AI TRISM.

4.3 | AI Risk Taxonomy

Figure 9 presents a structured taxonomy of AI risks, centering on “AI Risk” and radiating into ten distinct but interrelated categories. It begins with traditional concerns—Model Risk and

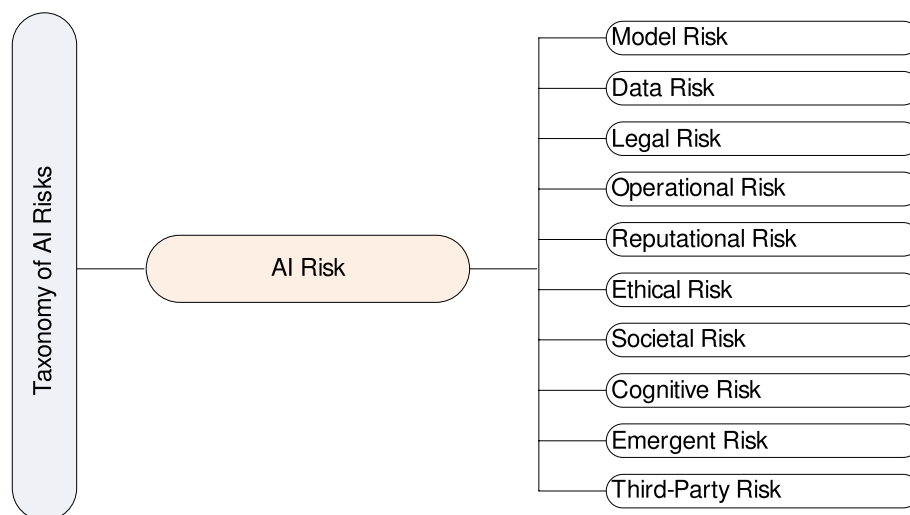


FIGURE 9 | Taxonomy of AI risks.

Data Risk—before branching into Legal and Operational Risks, which address regulatory compliance and system reliability. Reputational, Ethical, and Societal Risks capture broader impacts on public trust, moral integrity, and social welfare. Finally, Cognitive Risk highlights human-machine interaction challenges, Emergent Risk covers unforeseen behaviours, and Third-Party Risk reflects dependencies on external vendors and supply chains.

4.3.1 | Model Risk

Model risk arises when predictive systems behave unpredictably or inaccurately, potentially leading to faulty decisions in mission-critical domains (Novelli et al. 2024; Giudici et al. 2024). In AI and ML, this risk materialises as overfitting—where models memorise training data rather than generalise—and underfitting, which yields oversimplified patterns unable to capture real-world complexity. Concept drift further exacerbates model risk; when underlying data distributions shift over time, static models degrade in performance. Quantifying model risk entails rigorous stress testing and back-testing strategies, employing k-fold cross-validation, hold-out samples, and performance metrics like precision, recall, ROC-AUC, and calibration curves. Sensitivity analyses—varying input features systematically—reveal fragile decision boundaries, while adversarial simulations inject perturbations to probe model robustness. Embedding risk controls in ModelOps pipelines ensures that CI/CD workflows halt or roll back deployments when performance thresholds are breached. Versioned model registries document lineage, hyperparameters, and evaluation reports, facilitating audits and root-cause investigations. Governance policies mandate retraining schedules and trigger alerts upon drift detection via statistical tests (e.g., Kolmogorov-Smirnov) (Slattery et al. 2024). In highly regulated sectors such as finance and healthcare, model risk management aligns with standards like FDA Good Machine Learning Practice (GMLP) and SR 11-7 guidance from the U.S. Federal Reserve. Instituting comprehensive model validation regimes—blending quantitative benchmarks with expert human reviews—helps organisations ensure that deployed AI systems remain trustworthy, effective, and aligned with their intended use.

4.3.2 | Data Risk

Data risk in AI encompasses vulnerabilities stemming from the quality, integrity, and provenance of datasets used to train and validate models. Low-quality inputs—characterised by missing values, noisy labels, or sampling bias—undermine model accuracy and fairness. Data leakage, where sensitive or future information inadvertently influences training, leads to artificially inflated performance metrics and catastrophic deployment failures. Addressing data risk begins with robust data governance frameworks that specify ingestion pipelines, schema validation rules, and metadata cataloguing (Nguyen 2023). Tools like Great Expectations automate checks for nulls, outliers, and schema drift, while data versioning systems (e.g., DVC) track transformations and facilitate reproducibility. Privacy concerns introduce additional

risk: personally identifiable information must be safeguarded through anonymization techniques such as k-anonymity or differential privacy, adhering to regulatory mandates like GDPR or HIPAA. Data integrity processes employ checksums and cryptographic hashes to detect tampering, while secure storage solutions enforce encryption at rest and in transit using AES-256 and TLS protocols. Bias audits analyse representativeness across protected groups, applying disparity metrics to surface imbalances requiring mitigation. Data lineage tools record source origins, transformation logic, and access controls, enabling forensics in case of contamination or unauthorised modifications. Integrating data quality gates into ETL workflows and requiring approval from data stewards enables organisations to proactively manage data risk, ensuring that machine learning models are built on accurate, ethical, and compliant datasets.

4.3.3 | Legal Risk

Legal risk in AI arises from noncompliance with laws, regulations, and contractual obligations governing data use, intellectual property, and algorithmic decision-making. Jurisdictional variability—such as the EU AI Act, CCPA, and emerging AI regulations in India—creates a complex compliance landscape. Unauthorised processing of personal data may violate consent requirements under GDPR, leading to steep penalties. Algorithmic transparency laws may obligate organisations to provide explanations for automated decisions affecting individuals' rights, such as credit approvals or employment screenings. Intellectual property risks surface when training on copyrighted material without proper licences, potentially infringing on third-party rights. To mitigate legal risk, enterprises establish AI governance frameworks embedding legal counsel in cross-functional AI oversight committees (Kaminski 2023; Bashayreh et al. 2021). Legal risk assessments analyse use cases against applicable statutes, generating AI Impact Assessments that document compliance strategies. Contractual clauses with data providers and cloud vendors stipulate data provenance, usage limitations, and indemnities for breaches. Privacy-by-design principles integrate data minimization, purpose limitation, and secure deletion into system architectures (Murati 2025). Model documentation—including model cards and datasheets—catalogues potential biases, intended use cases, and limitations, supporting defence against regulatory inquiries. Regular policy reviews and legal audits align AI practices with evolving laws, while staff training ensures that technical teams understand consent, IP, and antidiscrimination requirements. Embedding legal validation steps into ModelOps pipelines—through automated checks for sensitive data tags and usage policies—turns legal risk management into a core component of AI lifecycle governance rather than a reactive measure.

4.3.4 | Operational Risk

Operational risk in AI refers to failures in processes, systems, or human factors that disrupt model deployment, monitoring, and maintenance. Examples include infrastructure outages, version mismatches in dependencies, or misconfigurations in

container orchestration platforms, leading to downtime or degraded performance (Wang 2023a; Savchenko 2024). Human errors—such as improper parameter tuning, inadvertent exposure of API keys, or neglecting to update security patches—compound operational vulnerabilities. Mitigating operational risk involves implementing robust DevSecOps practices: infrastructure as code ensures reproducible environments, while automated testing pipelines validate code changes, data schema updates, and model artefacts before production deployment. Blue-green or canary deployments minimise service disruptions, enabling safe rollbacks if anomalies arise. Real-time telemetry—captured via Prometheus, Grafana, or cloud-native monitoring services—tracks metrics like latency, error rates, and resource utilisation, triggering alerts when thresholds exceed predefined Service Level Objectives (SLOs). Incident response playbooks define escalation paths, communication protocols, and fallback procedures. Redundant architectures—multi-zone clusters, hot standby replicas—enhance resilience against infrastructure failures. Access control policies, enforced through RBAC and secret management, prevent unauthorised modifications. Scheduled disaster recovery drills and chaos engineering experiments (e.g., Netflix's Chaos Monkey) validate system robustness under failure scenarios. Integrating operational risk controls throughout the AI life-cycle—from code commit to continuous inference—ensures that AI services remain reliable, scalable, and secure amid evolving conditions. (Cheng and Qu 2023).

4.3.5 | Reputational Risk

Reputational risk in AI emerges when stakeholders perceive algorithmic outcomes as biased, incorrect, or unethical, potentially eroding public trust and damaging brand equity. High-profile incidents—such as facial recognition misidentification or chatbots generating offensive content—trigger media backlash and regulatory scrutiny (Holweg et al. 2022; Hirsch 2018; Stitzlein et al. 2021). Preventing reputational harm requires proactive transparency: publishing model cards that disclose performance across demographic segments, known failure modes, and intended use cases reassure stakeholders. Social listening tools monitor digital chatter and sentiment around AI deployments, enabling rapid response to emerging concerns. Pre-deployment bias tests—using techniques like disparate impact analysis—identify and correct discriminatory decision patterns, while adversarial content filters detect harmful outputs in generative AI. Inclusive user research sessions solicit feedback from diverse communities, surfacing unanticipated ethical implications. Crisis management frameworks outline communication templates, stakeholder notification protocols, and remediation steps to address adverse events swiftly. Partnerships with third-party auditors or ethics boards lend credibility to internal assessments, signalling commitment to accountability. Engaging in open-source collaborations—for instance, contributing to fairness and safety toolkits—demonstrates leadership in responsible AI. Embedding reputational risk metrics—like net promoter score (NPS) variations or brand sentiment indexes—into AI performance dashboards enables organisations to maintain situational awareness and guide AI initiatives in ways that preserve public trust and uphold institutional integrity.

4.3.6 | Ethical Risk

Ethical risk encompasses scenarios where AI systems violate moral norms or societal values, such as perpetuating discrimination, infringing on privacy, or enabling manipulative practices. For example, algorithmic profiling in hiring could unfairly disadvantage certain groups, while sentiment analysis tools might exploit psychological vulnerabilities for marketing gain (Douglas et al. 2024; Zhu et al. 2025; Blackman 2020). Addressing ethical risk begins with establishing clear AI ethics principles—often modelled on frameworks like the EU Ethics Guidelines for Trustworthy AI or IEEE's Ethically Aligned Design—codifying expectations around fairness, privacy, autonomy, and accountability. Ethics review boards composed of ethicists, legal experts, and community representatives evaluate proposed AI projects, conducting algorithmic impact assessments (AIAs) (Selbst 2021) to identify potential harms. Technical controls—such as bias mitigation algorithms, explainability modules, and privacy-enhancing technologies—are integrated to address identified ethical concerns. Scenario planning exercises simulate worst-case outcomes, guiding the implementation of fail-safe mechanisms. Continuous ethics training ensures that data scientists and engineers internalise values-based decision-making, while whistleblower channels encourage reporting of unethical practices. Transparency artefacts—including open-source code releases and public ethics reports—demonstrate organisational commitment to moral integrity. Embedding ethical risk management throughout all phases of AI development helps organisations prevent unintended harm, ensure alignment with societal values, and build credibility with stakeholders.

4.3.7 | Societal Risk

Societal risk pertains to broad, systemic impacts that AI technologies can impose on communities, economies, and public institutions. Examples include job displacement due to automation, amplification of misinformation through social media algorithms, and exacerbation of digital divides in underserved populations. To address societal risk, policymakers, industry leaders, and civil society collaborate on governance frameworks that assess macro-level implications. Social impact assessments analyse how AI deployments—such as autonomous vehicles or predictive policing systems—affect equity, safety, and public welfare (Alessandro et al. 2024; Boretti 2024). Inclusive policymaking engages diverse stakeholders—from labor unions to non-governmental organisations—to co-create regulations that balance innovation with protection of vulnerable groups. AI literacy programs equip citizens with critical thinking skills to interpret AI-driven information streams, mitigating misinformation spread. Public-sector AI observatories monitor deployments across sectors, publishing transparency reports and dashboards that track key societal indicators (e.g., unemployment shifts, bias incidence rates). Economic models incorporate externalities—such as re-skilling costs and social support needs—into cost-benefit analyses, informing responsible investment strategies. Ethical procurement guidelines require government agencies to prioritise AI solutions that demonstrate societal benefits and minimise harms. Proactive evaluation, mitigation, and monitoring of societal risks enable ecosystem stakeholders to steer

AI advancements toward inclusive and sustainable outcomes, avoiding the reinforcement of social inequities.

4.3.8 | Cognitive Risk

Cognitive risk arises when AI systems inadvertently alter human decision-making processes, leading to overreliance, skill degradation, or misjudgments. Automation bias occurs when users accept machine outputs uncritically, potentially propagating errors in critical tasks. Conversely, algorithm aversion may cause users to underutilize AI insights even when models outperform human judgement (Shalaby 2024; Laakasuo et al. 2021). Trust calibration—communicating model confidence and limitations—mitigates these risks by guiding users on when to rely on AI recommendations versus apply personal expertise (Huang, Zheng, et al. 2023). Human-computer interaction design incorporates uncertainty visualisation techniques—such as predictive intervals or heatmaps—to surface model ambiguity. Training programs foster AI literacy, teaching users to interpret probabilistic outputs and maintain critical oversight. Feedback loops capture instances where users override AI decisions, enabling system retraining to align model behaviour with expert intuition. Decision support interfaces implement adaptive automation levels: in high-risk scenarios, AI acts as a suggestion engine requiring human authorization, while in low-risk contexts, autonomous execution speeds workflows. Cognitive load assessments—measuring mental effort via task simulations or biometric sensors—inform interface design to prevent user fatigue and maintain situational awareness. In such a way, while embedding cognitive risk controls into AI-augmented workflows, organisations optimise human-machine collaboration, preserving human agency and preventing detrimental shifts in decision quality.

4.3.9 | Emergent Risk

Emergent risk in AI refers to unforeseen behaviours and consequences that manifest when complex systems interact in dynamic environments. Unlike predictable failures, emergent phenomena arise from non-linear interactions among multiple AI components—such as interconnected autonomous agents in supply chains or financial trading platforms—producing behaviours not anticipated during isolated testing (McGee 2024; McLean et al. 2023). To manage emergent risk, simulation environments (digital twins) model system-of-systems behaviour under varied conditions, enabling stress tests that reveal cascading failures or feedback loops. Multi-agent reinforcement learning scenarios simulate strategic interactions, uncovering potential collusion or adversarial dynamics. Adaptive governance mechanisms—policy-based control systems—monitor emergent patterns and enforce constraints (e.g., safe action spaces, kill switches) to prevent runaway behaviours. Continuous monitoring pipelines apply anomaly detection across macro-level metrics, such as systemic throughput anomalies or market volatility spikes. Emergent risk assessments incorporate horizon scanning techniques, leveraging expert judgement, scenario workshops, and quantitative risk modelling to map plausible future states. Research collaborations with academia explore formal verification methods—such as model checking

and theorem proving—to establish safety guarantees for critical AI subsystems. Thus, by anticipating and constraining unpredictable system-wide behaviours, organisations build resilience against emergent risks, ensuring that AI-driven ecosystems remain safe, stable, and aligned with strategic objectives.

4.3.10 | Third-Party Risk

Third-party risk emerges when organisations rely on external AI services, pre-trained models, or data sources that introduce vulnerabilities beyond direct control (Pierides and Mulligan 2025; Mohta et al. 2024; Arora and Ramteke 2024). Integrating third-party Application Programming Interfaces (APIs) and Machine Learning as a Service (MLaaS) (Pereira et al. 2024) platforms can expose proprietary data during inference calls, potentially contravening data residency or privacy requirements. Model integrity concerns arise when pre-trained weights harbour undisclosed biases or backdoors inserted by malicious actors. To mitigate these risks, enterprises conduct vendor due diligence—evaluating security certifications, data handling policies, and contractual clauses that specify liability and audit rights. Technical measures include sandboxing external models within isolated compute environments, enforcing strict input/output sanitization and rate limiting. Model provenance verification utilises cryptographic signatures or blockchain-based attestation to confirm authenticity and detect tampering. Periodic third-party assessments—penetration tests and bias audits—validate compliance with internal policies and regulatory standards. Data sharing frameworks, such as Differential Privacy-based federated learning, minimise exposure by keeping raw data on-premises. Governance frameworks catalogue all external dependencies in a centralised repository, tracking versions, certification statuses, and renewal timelines. Rigorously managing third-party AI dependencies with contractual, technical, and governance controls allows organisations to retain visibility into external risks and protect their AI ecosystems from supply-chain vulnerabilities.

4.3.11 | Brief Survey on Some AI Risk Management Tools

The rapid integration of AI into sensitive sectors such as health-care, finance, and government has necessitated the development of robust risk management frameworks and technical toolkits. These solutions are designed to ensure that AI systems are deployed responsibly, transparently, and in compliance with regulatory standards. In response, several prominent tools and frameworks have emerged, offering diverse approaches to managing the multifaceted risks associated with AI—including systemic governance, compliance tracking, adversarial robustness, and red-teaming for generative models. Table 4 provides a comparative overview of some of the most widely adopted AI risk management tools.

One of the most widely recognised frameworks is the AI Risk Management Framework⁴⁵ (AI RMF 1.0), published by the U.S. National Institute of Standards and Technology (NIST). Designed to be sector-agnostic and voluntary, it provides structured guidance across four core functions—GOVERN, MAP,

TABLE 4 | Comparison of some prominent AI risk tools.

Feature/Aspect	NIST AI RMF	PyRIT	VerifyWise	Risk atlas nexus
Developer	NIST (U.S. Govt)	Microsoft/Azure OSS	Bluewave Labs	IBM Research
Licence	Public Domain	MIT	AGPL-3.0	Apache 2.0
Focus	AI governance and risk lifecycle	Generative AI red teaming	Compliance tracking and AI audit support	AI risk graphing and ontology-driven governance
Key users	Policymakers, auditors, enterprise leaders	Security teams, AI red teamers	Compliance officers, legal, risk teams	Governance researchers, AI auditors, developers
Interface	Framework + profiles + documentation	CLI, Python SDK, cookbooks	Web dashboard, REST API, visual tracker	Python SDK, notebooks, graph export
Deployment stack	Conceptual	Python 3.x + Hugging Face + Azure	Docker (React + Node.js + PostgreSQL)	Python 3.11+ with optional Ollama, Watsonx, vLLM
Ontology support	No	No	No	Yes (LinkML-based AI Risk Ontology)
Trust pillars	Valid, Secure, Fair, Private, Explainable	Partial (Prompt-focused, limited on fairness/privacy)	Governance-aligned, accountable, auditable	Structured risk-action mapping, transparency
LLM usage	No	Yes (LLM-based red teaming)	Optional (AI-assisted form filling)	Yes (LLM inference for CoT and risk matching)
Key standards	Generative AI Profile (2024), GDPR, HIPAA	Red teaming benchmarks (prompt-level)	EU AI Act, ISO 42001 (in progress)	AIR 2024, OWASP Top 10, IBM Risk Atlas
Strengths	Strategic standardisation, risk mitigation	Flexible simulation of generative risks	Real-time tracking, visual dashboards, evidence	Knowledge graph, risk taxonomy, community-driven
AI lifecycle coverage	Full lifecycle: design to deployment	Post-deployment/operational	Mid-to-late lifecycle (compliance and evidence)	Full lifecycle, ontology-mapped workflows
Community contribution	Government agency and partners	Open-source contributions	Startup-maintained OSS with roadmap	Community-led, taxonomy templates supported
Advanced use cases	Impact classification, tolerance analysis	Adversarial LLM prompt testing	Compliance monitoring, ISO/EU audit preparation	Questionnaire autofill, AI task mapping
AI TRISM support	Comprehensive (all 3 pillars)	Partial (mostly security risk detection)	Strong on Trust and Risk, moderate Security	Strong Trust/Risk mapping, growing Security

MEASURE, and MANAGE—to help organisations assess and respond to AI-related risks. The framework outlines essential characteristics of trustworthy AI systems, including validity, safety, security, privacy enhancement, fairness, and accountability. It emphasises early-stage risk integration rather than retrospective fixes, thereby promoting proactive governance across the AI lifecycle. Recent expansions include a Generative AI Profile and the ARIA program, which introduces real-world model evaluations emphasising both technical robustness and societal impacts.

Complementing strategic governance, the Python Risk Identification Tool (PyRIT)⁴⁶ focuses on security-specific risks associated with generative AI models. Developed as an open-source project by Microsoft, PyRIT allows security professionals to conduct AI red teaming through prompt-based risk simulations. It enables users to define threat scenarios, craft malicious prompts, and observe model behaviours that might result in unsafe or biased outputs. The tool's modular architecture includes components for prompt conversion, scoring, and target definition, making it flexible for adversarial evaluations across different generative AI deployment environments.

While frameworks like NIST RMF establish strategic principles, tools such as VerifyWise⁴⁷ offer operational platforms tailored to real-time compliance tracking and audit management. VerifyWise is an open-source governance application that integrates features like evidence collection, risk logging, vendor management, and assessment dashboards. Designed with regulations such as the EU AI Act in mind, it offers role-based access, REST APIs, and on-premises deployment options to meet the needs of enterprise users. Its visual interface and automation features support both technical teams and non-technical stakeholders in maintaining transparency and audit readiness.

Expanding the scope of AI risk governance into ontological and community-driven tooling, Risk Atlas Nexus⁴⁸ provides a knowledge-graph-based platform to unify taxonomies, benchmarks, risks, and mitigations. Built upon IBM's AI Risk Atlas,

this open-source tool offers an extensible ontology that maps risks to AI system components and mitigation strategies. It supports integration with inference engines like Ollama, Watsonx, and vLLM to facilitate intelligent risk identification and reasoning. Risk Atlas Nexus also provides mechanisms to auto-fill compliance questionnaires using Chain-of-Thought (CoT) reasoning and few-shot examples, thereby reducing manual effort in governance workflows.

4.4 | AI Security Management Taxonomy

Figure 10 presents a comprehensive taxonomy for AI security management, placing “AI Security Management” at its center with ten interlinked domains radiating outward. The taxonomy begins by addressing critical threat vectors—adversarial attacks and robustness—to ensure models can withstand intentional perturbations. Privacy and access control domains are responsible for protecting sensitive inputs and maintaining least-privilege policies. Infrastructure security encompasses the hardened compute and network environments that support AI services, while monitoring and detection provide real-time alerting for anomalous behaviour. Security standards ensure alignment with industry best practices, and secure data handling oversees encryption and lifecycle protection. Model governance introduces oversight throughout the development process, and zero trust principles require continuous verification at every interaction.

4.4.1 | Adversarial Attacks

Adversarial attacks exploit vulnerabilities in machine learning models by crafting inputs that cause misclassification or erroneous outputs. In the context of AI security management, two primary categories emerge: evasion attacks, where small, often imperceptible perturbations to input data (e.g., images or text) induce wrong predictions at inference time; and poisoning attacks, where an adversary injects malicious data into the training corpus to corrupt model behaviour (Baniecki and Biecek 2024; Hussain and Fortifying 2024). Attack vectors also

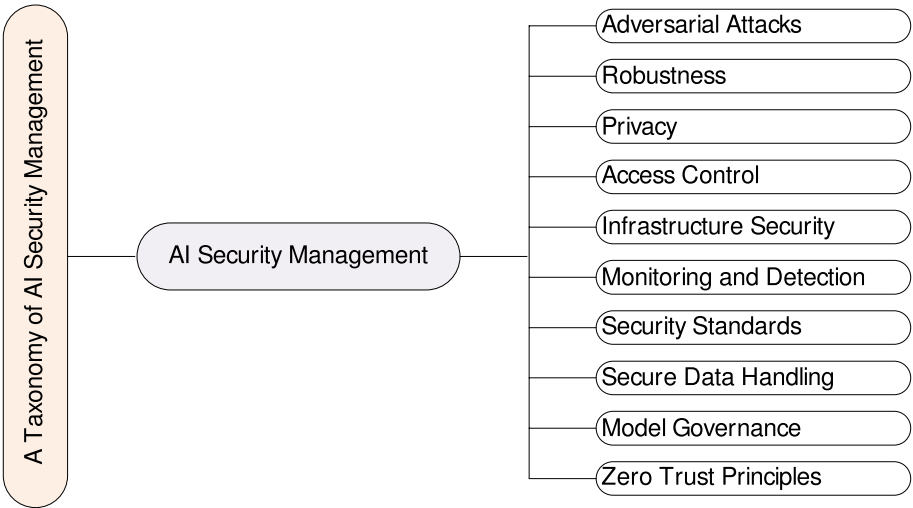


FIGURE 10 | A taxonomy of AI security management.

include membership inference, enabling attackers to determine whether specific records were in the training set, and model inversion, whereby sensitive training features are reconstructed from model outputs. Defences span adversarial training—augmenting the training set with adversarial examples—to certified robustness techniques that provide mathematical guarantees within bounded perturbation regions. Detection mechanisms leverage statistical divergence measures (e.g., Kullback–Leibler) (Cui et al. 2024) to flag anomalous inputs, while input sanitization filters remove out-of-distribution samples. Some systems adopt defensive distillation, training student models on softened teacher outputs to smooth decision boundaries and reduce sensitivity to input perturbations. In production, security policies enforce runtime checks, rate limiting, and anomaly scoring. Continuous red-teaming exercises simulate adversarial scenarios against deployed models, ensuring that newly discovered exploits are incorporated into ongoing defence updates. Integrating adversarial risk assessments into the ModelOps lifecycle enables organisations to sustain robust AI deployments that can endure both existing and evolving attack vectors.

4.4.2 | Robustness

Robustness in AI security management refers to a model's resilience against perturbations, distributional shifts, and adversarial manipulations. Ensuring robustness begins with designing models that generalise well under real-world variability: techniques include data augmentation to simulate noise and distortions, domain randomization to expose models to diverse environments, and regularisation strategies such as dropout or weight decay to prevent overfitting (Javed et al. 2025; Tocchetti et al. 2025). Adversarial robustness further demands integrating adversarial examples into the training pipeline, allowing models to learn invariant representations. Certified robustness methods—like randomised smoothing—offer provable guarantees that small perturbations within an ℓ_2 or ℓ_∞ norm ball cannot change the model's output. Robustness evaluation frameworks apply a suite of attacks (e.g., DeepFool⁴⁹) across threat models to quantify the model's failure rate under targeted and untargeted adversarial scenarios. Beyond input perturbations, robustness encompasses handling missing or corrupted features; strategies such as imputation models and autoencoders reconstruct incomplete data, while robust loss functions (e.g., Huber loss; Zhao, Yu, and Wan 2024) mitigate the influence of outliers. Robustness metrics—including adversarial accuracy, calibration under shift, and mean corrupted error—inform monitoring dashboards for ongoing performance validation. In production, fail-safe mechanisms such as confidence thresholds trigger fallback to human review when model certainty drops below specified levels. Embedding robustness considerations across the entire development lifecycle—from choosing architectures to ongoing validation—ensures that organisations preserve model reliability, even in the face of adversarial or unforeseen challenges.

4.4.3 | Privacy

Privacy in AI security management ensures that sensitive information within datasets and model outputs remains protected

throughout the system's lifecycle. Key techniques include differential privacy, which introduces calibrated noise into training gradients or output queries to bound the influence of any single data point, achieving (ϵ, δ) -DP guarantees (Golda et al. 2024; Gupta et al. 2025). Federated learning complements DP by enabling decentralised model training on edge devices: only encrypted model updates—not raw data—are communicated to the central aggregator, preserving local data confidentiality. Secure multi-party computation (SMPC) and homomorphic encryption (HE) allow computation over encrypted inputs, supporting privacy-preserving inference and collaborative analytics. Data minimization policies enforce collection of only the attributes strictly necessary for model objectives, while data de-identification methods—such as k -anonymity, ℓ -diversity, and t -closeness—reduce re-identification risk in shared datasets. Privacy risk assessments, guided by frameworks like NIST SP 800–53 and GDPR, catalogue data flows, consent records, and data retention schedules. Privacy auditing tools monitor real-time data accesses, generating alerts for anomalous export patterns. In MLOps pipelines, privacy guarantees are codified via “privacy as code,” (Ferreyra et al. 2024) where CI/CD gates verify that DP budgets are not exhausted and that FL orchestrations maintain secure channels. Through transparent privacy policies and model cards disclosing data-handling practices, organisations build stakeholder trust and ensure compliance with global regulations, safeguarding individuals' rights while leveraging AI innovations.

4.4.4 | Access Control

Access control in AI security management enforces that only authorised entities—users, processes, or systems—can interact with data, models, and inference endpoints. The RBAC remains foundational: administrators define roles (e.g., Data Scientist, ML Engineer, Auditor) and bind permissions (read, write, deploy) to service accounts or human identities. The ABAC extends flexibility by evaluating contextual attributes—such as time of day, network location, or project membership—to make fine-grained decisions. API gateways front inference services, integrating JWT-based authentication and OAuth 2.0 authorization flows. Secrets management systems securely store credentials, encryption keys, and tokens, ensuring that ModelOps pipelines obtain sensitive parameters only via dynamic retrieval with expiration policies. Zero-Trust principles complement access control by verifying every request—regardless of origin—before granting privileges. Audit logs record each access event, capturing identity, timestamp, and action, enabling forensic analysis. In multi-tenant environments, containerized model deployments isolate workloads via Kubernetes namespaces and network policies (Calico,⁵⁰ Cilium⁵¹), preventing lateral movement. Data encryption at rest and in transit ensures that even if access controls are bypassed, raw storage and communication channels remain unintelligible without decryption keys (Kandolo 2024; Olabanji et al. 2024; Khan 2024). Access control policies codified in infrastructure-as-code repositories undergo automated policy-as-code validation, rejecting misconfigurations before deployment. Combining strong authentication, precise authorization controls, and ongoing auditing helps organisations protect their AI assets from unauthorised access and misuse.

4.4.5 | Infrastructure Security

Infrastructure security in AI encompasses the protection of compute, storage, and network resources supporting model training, deployment, and inference (Al-Hawawreh et al. 2024; Volk 2024). Hardened environments begin with secure baselines: virtual machines and containers are built from minimal operating system images, patched regularly, and scanned for known vulnerabilities using tools like Clair⁵² or Anchore.⁵³ Network segmentation isolates development, testing, and production clusters, employing software-defined networking (SDN) controls and micro-segmentation to restrict lateral traffic. Sensitive AI workloads leverage dedicated hardware security modules for secure key storage and cryptographic operations. Cloud providers' native security services—such as AWS Nitro Enclaves or Azure Confidential Computing—further offer hardware-based enclaves that protect data-in-use from hypervisor or host OS threats. Infrastructure as Code pipelines integrate security linting tools to enforce organisational policies before provisioning resources. Runtime security tools, like Falco,⁵⁴ monitor host and container behaviours, triggering alerts on anomalous system calls indicative of compromise. Automated vulnerability management workflows ingest CVE feeds, prioritise patches based on severity, and orchestrate rolling updates to minimise downtime. Secrets and credentials never reside in code repositories; instead, ephemeral tokens with least-privilege scope are injected at runtime. By embedding security controls at every layer—compute, network, storage—organisations build resilient AI infrastructure that withstands sophisticated threat vectors (Sarker 2024).

4.4.6 | Monitoring and Detection

Monitoring and detection in AI security management continuously surveil model behaviour, infrastructure health, and data integrity to identify anomalies, intrusions, or performance degradations (Zhang et al. 2023). Telemetry pipelines ingest logs, metrics, and traces from model serving endpoints, orchestration platforms, and data ingestion systems into centralised observability stacks—commonly based on Prometheus for metrics, ELK for logs, and Jaeger for distributed tracing. Security Information and Event Management (SIEM) solutions correlate events across sources, applying rule-based detection and machine-learning-driven analytics to surface indicators of compromise (IoCs) (Preuveneers and Joosen 2021). Data integrity monitors compute checksums on training and inference datasets, alerting on unexpected modifications. Model drift detectors compare live input distributions against training baselines using statistical tests (Wasserstein distance, population stability index), flagging shifts that could signal adversarial manipulation. Runtime Application Self-Protection (RASP) (Le-Thanh et al. 2023) frameworks embed detection agents within inference services to intercept suspicious API calls, rate anomalies, or exploit payload patterns. Anomaly detection algorithms—such as isolation forests or autoencoder-based detectors—learn normal operational envelopes and raise alerts when deviations occur. Integrated incident response platforms execute automated playbooks (i) isolating affected pods, (ii) rotating compromised secrets, or (iii) invoking fallback models. By closing the loop

between detection and response, organisations reduce Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR), preserving the security and reliability of AI deployments.

4.4.7 | Security Standards

Security standards in AI unify best practices, compliance requirements, and technical specifications to govern secure AI development and operations (Kopczewski et al. 2024; Kunle-Lawanson 2022). International frameworks such as ISO/IEC 27001 prescribe Information Security Management Systems (ISMS) that encompass risk assessment, control selection, and continuous improvement cycles. AI-specific extensions—like ISO/IEC 42001—embed governance clauses for ethical design, data stewardship, and robustness testing. The NIST AI Risk Management Framework outlines core functions—Govern, Map, Measure, Manage—tailored to AI systems. Industry consortia (e.g., OWASP AI Security Top 10⁵⁵) catalogue prevalent vulnerabilities and mitigation strategies, offering actionable guidance for developers and auditors. Security standards integrate with privacy regulations—GDPR, HIPAA—by defining requirements for data anonymization, breach notification, and consent management. Certification programs (e.g., CSA STAR,⁵⁶ SOC 2 Type II⁵⁷) validate cloud and service providers' controls relevant to AI workloads. Standards also cover secure software development lifecycles (SSDLC), mandating threat modelling, static application security testing (SAST), dynamic application security testing (DAST), and third-party code reviews. Compliance automation tools ingest standard catalogues, translate controls into policy-as-code, and continuously audit environments for deviations. Aligning AI security practices with established standards allows organisations to showcase due diligence to regulators, customers, and partners, thereby minimising risk and building trust (Gnitko 2024).

4.4.8 | Secure Data Handling

Secure data handling in AI safeguards information across its lifecycle—from ingestion and storage to processing, transit, and archival. Ingestion pipelines enforce schema validation and data sanitization to block malicious payloads and ensure data integrity. Data-at-rest protection employs strong encryption algorithms within storage systems and databases. Key management systems (KMS) (Chaeikar et al. 2022) rotate encryption keys on predefined schedules and provide fine-grained access policies. During transit, Transport Layer Security secures communication channels between microservices, edge devices, and central servers. Data masking and tokenization replace sensitive fields with placeholders in non-production environments, preventing accidental leakage during testing (Prior and Marcos-Cuevas 2025; Himeur et al. 2023). Access to raw data is strictly controlled via RBAC and ABAC policies integrated with Identity and Access Management platforms. Immutable audit logs capture data access events, enabling forensic analysis in case of incidents. For compliance with regulations like GDPR's right-to-erasure, secure deletion protocols—such as cryptographic erasure—ensure that decryption keys are destroyed, rendering residual data unrecoverable. In federated learning contexts, secure

aggregation protocols ensure that only aggregated model updates—not raw gradients or data—are shared with the central server.

4.4.9 | Model Governance

Model governance establishes the policies, roles, and processes required to oversee AI system development, deployment, and maintenance in a controlled, auditable manner. Governance frameworks define lifecycle stages—concept, development, validation, deployment, monitoring, retirement—and specify criteria for progression through each stage (Hausenloy et al. 2024; Birkstedt et al. 2023). Model registries record meta-data: version identifiers, training datasets, hyperparameters, performance metrics, and responsible stakeholders. Approval workflows enforce that every model passes bias and fairness assessments, security scans, and compliance checks before production release. Audit trails log every change, from data preprocessing scripts to model parameter adjustments, time-stamped and signed by authorised personnel (Wodi 2024; Camilleri 2024). Governance committees—comprising data stewards, legal experts, security officers, and domain specialists—conduct periodic reviews, ensuring models adhere to ethical standards, regulatory requirements, and business objectives. Policy-as-code repositories codify governance rules, automatically validating artefacts during CI/CD pipelines. Model retirement policies stipulate decommissioning criteria—such as performance degradation or regulatory obsolescence—and enforce secure archival of both model binaries and training data snapshots. Governance dashboards provide real-time visibility into model statuses, lineage graphs, and outstanding remediation tickets. Embedding model governance within both organisational frameworks and technical workflows ensures that AI initiatives stay transparent, compliant, and strategically aligned.

4.4.10 | Zero Trust Principles

Zero Trust in AI security management mandates that no entity—user, device, or service—is inherently trusted, regardless of network location. Every access request to data, models, or infrastructure must be authenticated, authorised, and continually verified (Ghasemshirazi et al. 2023; Ajish 2024). Implementing Zero Trust begins with strong identity verification: multi-factor authentication and device posture checks confirm that only compliant endpoints can connect (Odedina 2025). Micro-segmentation divides networks into granular security zones—using SDN and service mesh frameworks—so lateral movement is constrained even if a breach occurs. Least-privilege access ensures that service accounts and users receive only the minimum permissions needed for their tasks, managed via dynamic policy engines that adjust permissions based on contextual risk signals (e.g., anomalous behaviour, location). All communications are encrypted with mutual TLS providing both confidentiality and endpoint attestation. Continuous monitoring and anomaly detection systems analyse telemetry to detect compromised credentials or unusual data flows. Automated response mechanisms can revoke access tokens, isolate compromised workloads, or require

re-authentication upon detecting elevated risk. Infrastructure as code pipelines encode Zero Trust controls—network policies, IAM rules, encryption settings—ensuring consistency and auditability. By embracing Zero Trust, organisations fortify their AI ecosystems against internal and external threats, guaranteeing that every interaction is explicitly permitted and continuously validated (Parisa et al. 2023).

4.4.11 | Brief Survey on Some Key Security Management Tools

In modern AI deployments, security demands specialised frameworks that can address unique threats at every stage of the model lifecycle. Cybersecurity AI (CAI),⁵⁸ the OWASP AI Exchange & AI Security & Privacy Guide, and LLM-Guard each offer distinct, complementary capabilities for safeguarding AI systems. CAI brings an action-oriented perspective, automating common penetration-testing tasks through modular “agent” workflows. Researchers or security teams can spin up CAI’s lightweight command-line interface, define reconnaissance or exploitation agents, and orchestrate multi-step attack sequences that mirror real-world adversaries. A human-in-the-loop feature ensures that critical decisions remain under expert control, while OpenTelemetry-based tracing captures every agent action for audit and analysis. This makes CAI particularly valuable for organisations that need to continuously validate their defences against evolving attack patterns and demonstrate governance through detailed audit trails. Table 5 shows a comparison of a few prominent AI security management tools.

By contrast, the OWASP AI Exchange & AI Security & Privacy Guide⁵⁹ delivers a strategic, standards-aligned playbook for AI risk management. It is not a runtime toolkit, but rather a living registry of AI-specific threats and corresponding controls, maintained by a global community of experts. The Exchange’s “Periodic Table” and “Security Matrix” map each identified vulnerability—ranging from data poisoning during training to inference-time privacy leaks and denial-of-service exploits—to best-practice mitigation strategies. There is also a dedicated privacy module that advises on data governance, anonymization, and compliance with regulations like GDPR or the EU AI Act. Because it sits at the intersection of policy, development-process guidance, and risk-assessment workflows, the OWASP guide is ideal for organisations building or maturing an AI governance program. Teams can embed its recommendations into secure-SDLC pipelines, model-ops checklists, and audit frameworks, ensuring that every AI initiative is subject to consistent, documented risk-treatment processes.

LLM-Guard⁶⁰ occupies a more focused niche by safeguarding interactions with LLMs. Prompt-injection attacks, sensitive data leakage, and the inadvertent generation of harmful content are challenges unique to LLMs, and LLM-Guard addresses them with a rich set of input and output “scanners.” Before a user’s prompt ever reaches an LLM, LLM-Guard can automatically sanitise text, detect suspicious patterns, block banned topics or code, and enforce token-limit quotas to prevent denial-of-service. On the output side, it can filter out toxic language, detect bias, check for factual consistency, and strip or obfuscate private

TABLE 5 | Comparison of a few prominent AI security management tools.

Attribute	CAI (Cybersecurity AI)	OWASP AI exchange and guide	LLM-guard
Primary focus	AI-driven pentesting & bug-bounty workflows	Community-led threat/control catalogue	Securing LLM inputs & outputs
Key features	- Modular agents & ReACT patterns - Recon, exploit, lateral, exfiltration, C2 tools - Dynamic agent handoffs - Human-in-the-Loop CLI - Tracing via OpenTelemetry	- Threat/control registry - Periodic Table + Security Matrix - Privacy guidance module - Risk-treatment workflows - Compliance mappings (ISO, EU AI Act, NIST)	- Prompt scanners (toxicity, secrets, limits) - Output scanners (bias, factuality, leaks) - Refusal enforcement, sanitization - Fast-fail & async modes
Threats/controls covered	- Reconnaissance & weaponization - Exploitation & escalation - Lateral movement - Data exfiltration - Command & control	- Inference: poisoning, evasion, privacy - Training: bias, poisoning, pipeline risk - Runtime: theft, DoS, dependency risks	- Availability: size & token guards - Integrity: injection shields - Privacy: anonymization, blocking - Abuse: toxicity & bias filters
Integration and extensibility	- CLI/REPL for ad-hoc use - Extendable via MCP - Add agents/patterns - Telemetry for dashboards	- Docs & GitHub PRs - SDLC checklists •Workshops, audit frameworks	- Python package - Middleware/decorator - Auto-instal extras - Configurable use-cases
Licence	MIT	CC0 1.0	MIT
AI TRiSM support	- Governance: logs & audits - Reliability: workflows + human oversight - Security: red/blue team automation	- Governance: compliance aligned - Trust: consensus-driven controls - Data Protection: privacy modules	- Trustworthiness: policy enforcement - Reliability: injection & overuse prevention - Privacy: leak protection

data. Fast-fail and asynchronous processing modes allow developers to balance performance with thorough security checks, while easy Python-package integration means LLM-Guard can slot directly into existing API-based workflows.

5 | Generative AI Model Toxicity

Figure 11 presents a structured taxonomy of toxicity risks specific to generative AI models, with “GenAI (LLM) Model Toxicity” at its center and 15 distinct harm vectors extending outward. This taxonomy details the concrete ways in which both content- and behaviour-based failures can arise, serving as a practical complement to the five strategic TRiSM pillars introduced earlier. While those pillars define broad domains of control—namely, explainability, ModelOps, application security, data protection, and adversarial resilience—this taxonomy enumerates the specific misuse cases and generation errors that each pillar is designed to address. In Section 4.1, we further map each toxicity category to one or more TRiSM dimensions, illustrating how the governance framework operationalizes these 15 harm vectors. Subsequently, Table 6 provides a comparative overview of each category in terms of risk priority, impact, and mitigation strategies.

5.1 | Hate and Harassment

Deciding what qualifies as hate or harassment toxicity involves both clear markers and shifting social norms. Hate speech targets individuals or groups based on protected traits—race, religion, gender identity, sexual orientation, disability—and may include calls for violence or demeaning language. Harassment covers a broader range of abusive, insulting, or humiliating remarks directed at any person or group (Zhao, Zhu, et al. 2024; Shen et al. 2025). Since LLMs often train on unfiltered user content that reflects real-world prejudices, they can inadvertently echo or even magnify those biases. Because cultural and legal definitions of offensive speech vary widely, detection methods must adapt to local frameworks while upholding basic respect for human dignity. What one community labels as satire or critique might be deemed hateful elsewhere, and laws evolve over time. Solutions therefore combine global best practices with region-specific expertise and remain open to revision as norms change. Technically, identifying harmful language relies on multiple layers. Simple token filters catch known slurs or epithets but can be fooled by clever misspellings. Transformer-based classifiers fine-tuned on hate-speech datasets then analyse entire utterances, using embeddings to detect intent and distinguish between mild insults and explicit threats. Multi-task models

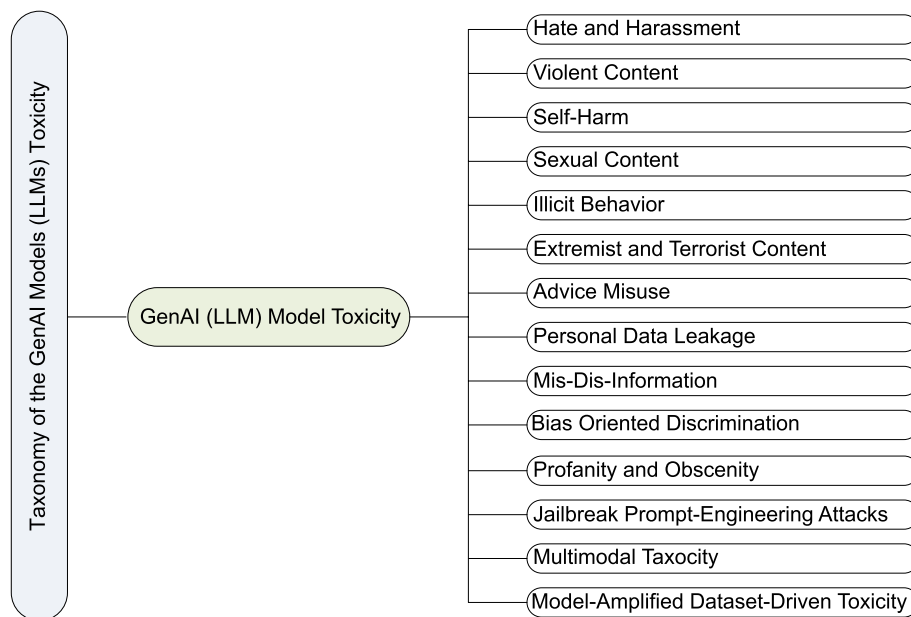


FIGURE 11 | Taxonomy of toxicity in generative AI models.

further gauge severity levels and pinpoint whether targets belong to protected groups (Podolak et al. 2023; Tan et al. 2025). To curb the generation of hateful or harassing text, systems often rewrite or refuse prompts likely to trigger abuse. During training, reinforcement learning from human feedback applies negative signals when the model produces flagged content. Adversarial evaluations—using benchmarks like HateCheck—probe edge cases, such as reclaimed slurs employed in positive contexts, ensuring that safeguards remain precise without unduly silencing legitimate speech (Röttger et al. 2021). Ongoing performance monitoring and alignment with local policies help maintain effectiveness as social attitudes and legal standards continue to shift.

5.2 | Violent Content

Violent content toxicity covers any portrayal or encouragement of physical harm—from graphic depictions of bloodshed to calls for bodily or property damage. Because LLMs train on extensive collections of stories, news reports, and forum discussions, they may absorb sensationalist violence or tactical details that, if echoed back, could pose real dangers. Yet, not all mentions of violence are problematic: a neutral historical account or clinical description differs greatly from language that glorifies or incites aggression (Maity et al. 2024; AlDahoul et al. 2024; Ghosh et al. 2024). Deciding when violence becomes toxic demands both technical rigour and ethical insight. Pipelines often use semantic role labeling to identify actor–action–patient triples (e.g., “X attacked Y”) and dependency parsing to isolate aggressive verbs and their targets. Domain-specific lexicons then rate term severity—distinguishing “slaughter” from “fight.” By combining these signals, models assign a violence score to each segment and flag those needing further handling. Acceptable references to violence vary across cultures, histories, and generations. What one community sees as heroic resistance might be condemned as incitement in another. Addressing these differences calls for

adjustable controls—configurable filters and policy settings that reflect local laws, social values, and user expectations. Layers of defence work together to limit harmful outputs. Basic thresholds can block especially graphic or inflammatory passages, while dynamic masking hides explicit details. Context-aware filters monitor the conversation’s flow and may truncate or redirect if a benign prompt veers into dangerous territory. Regular adversarial testing uncovers vulnerabilities, and human reviewers ensure that legitimate descriptions remain while praise or promotion of violence is blocked (Jha et al. 2024).

5.3 | Self-Harm

Self-harm and suicide toxicity covers any model output that urges, instructs, or romanticises self-injurious behaviours—like cutting, overdosing, or starving oneself—or actively guides users toward ending their lives (Zhang and Ren 2025; Park et al. 2024). Because LLMs may learn from internet forums and mental-health discussions, they can unintentionally echo or normalise harmful advice embedded in casual conversation (Andriushchenko et al. 2024). Recognising self-harm content demands both precision and empathy. Overt calls to action (“I want to kill myself”) are straightforward to flag, but subtler signals—expressions of hopelessness or passive references to harming oneself—require context-aware understanding. Cultural attitudes toward self-harm, help-seeking, and stigma vary widely, so interpretations must be tuned to local norms and informed by mental-health professionals. To address these risks, specialised detection and response layers are employed. Advanced classifiers, trained on annotated mental-health corpora such as the CLPsych Shared Task, identify expressions of suicidal thoughts or self-injury with high sensitivity. When such content is detected, the system switches from normal generation to a “safe completion” mode: it offers empathetic, non-judgmental support, declines to provide harmful instructions, and instead points users to crisis

TABLE 6 | Comparative summary of toxicity categories of generative AI models.

GenAI (LLM) model toxicity	Priority	Impact on AI TRiSM	Mitigation strategies
Hate and harassment	Very very high	Undermines user trust and may expose the organisation to legal/regulatory penalties	Multi-layer filters (slurs + contextual classifiers), RLHF penalization, adversarial stress-tests (HateCheck), region-specific policy alignment
Violent content	Very high	Risks physical harm, regulatory scrutiny in high-stakes domains	Semantic-role/dependency parsing, lexicon-based scoring, threshold suppression, context-aware filtering, red-team evaluations, human review
self-harm	Very very high	Life-threatening risk; exposes platform to grave ethical and liability issues	Specialised distress classifiers, “safe completion” crisis flows, dual-decoder architecture, clinical expert validation, continuous false-negative monitoring
Sexual content	High	Can violate age-rating laws, platform policies, cultural norms	Multi-tiered gating (keyword + classifier), style-transfer sanitization, user-configurable thresholds, age gating, continual benchmark re-evaluation
Illicit behaviour	Very high	Facilitates crime; major legal and reputational risk	Intent classifiers + ontology matching, policy-as-code refusals/decoys, red-teaming, dynamic safe completion, adversarial prompt testing
Extremist and terrorist content	Very high	Threatens public safety, triggers counter-terrorism regulations	Entity linking to extremist databases, ideological sentiment analysis, knowledge-graph removal/replacement, counter-narrative modules, OSINT feeds
Advice misuse	High	Leads to professional malpractice, legal liability	Domain intent gating, RAG over vetted sources, inline citations, safe-completion disclaimers, expert-in-loop RLHF, compliance benchmark audits
Personal data leakage	Very high	Violates privacy laws (GDPR, HIPAA), erodes user trust	Differential Privacy (DP-SGD), canary-document memorization audits, PII detectors + redaction, context-window limits, adversarial extraction tests
Misinformation/disinformation	High	Erodes informational integrity; regulatory risk in news/health domains	Claim-verification (NLI + KG), RAG with grounding + citations, temperature/top-k calibration, continuous corpus updates, FEVER/TruthfulQA benchmarks
Bias-oriented discrimination	High	Breaches fairness mandates, regulatory fines, reputational harm	Counterfactual fairness tests, disparate-impact metrics, pre-/in-/post-processing debiasing, fairness dashboards, third-party audits
Profanity and obscenity	Medium	Offends users, can exacerbate harassment, violates community standards	Lexicon + context-aware classifiers, token suppression/substitution, style-transfer, user-configurable profanity levels

(Continues)

TABLE 6 | (Continued)

GenAI (LLM) model toxicity	Priority	Impact on AI TRiSM	Mitigation strategies
Jailbreak prompt-engineering attacks	Very high	Defeats all safety measures, unleashes hidden toxic capabilities	Prompt sanitization, runtime anomaly scoring, ensemble policy networks, context resets, policy “chain-of-thought,” regular adversarial red teaming
Multimodal toxicity	High	Harm emerges only in cross-modal contexts (e.g., memes), hard to detect with unimodal filters	Shared embedding (CLIP/AudioCLIP), multimodal transformers/GCNs, joint toxicity thresholds, cross-modal gating, synthetic adversarial training
Model-amplified dataset-driven toxicity	Medium	Rare toxic patterns get over-amplified, hidden reservoir of harm	Canary memorization audits, n-gram frequency comparison, targeted data augmentation, adversarial penalty during training, RLHF, dynamic decoding calibration

hotlines or professional resources. This behaviour can be implemented via a dual-decoder architecture—one handling general dialogue and the other reserved for crisis support—activated based on the classifier’s signal. Ensuring these interventions remain effective and compassionate requires close collaboration with clinical psychologists. Regular audits, user-feedback loops, and impact assessments help catch any overlooked harmful outputs and refine the system’s responses, ensuring they remain attuned to individual needs and cultural sensitivities.

5.4 | Sexual Content

Sexual toxicity covers any explicit portrayal or description of erotic acts, genitalia, or arousing scenarios without a clear health or educational purpose. It is important to separate consensual adult content—where all participants are of legal age and willing—from exploitative material such as non-consensual acts, incest, bestiality, or anything involving minors. Since LLMs often train on large web-scraped datasets containing both innocuous and explicit passages, they can inadvertently reproduce graphic language that breaches community standards or legal norms. Identifying harmful sexual content goes beyond spotting explicit words. Initial filters scan for known erotic terms—anatomical labels, explicit verbs—using convolutional checks on token embeddings. Because such words may appear in clinical descriptions, transformer-based classifiers fine-tuned on datasets like Pornography Scene Detection (PSD) (Yu et al. 2024; Phan et al. 2024) then examine the surrounding text to decide if the intent is informational or meant to arouse. When passages are flagged, style-transfer networks can replace graphic language with neutral or clinical terminology, preserving the educational value while removing potentially harmful detail. Control mechanisms typically involve multiple stages. A basic keyword filter blocks obvious explicit tokens, triggering a follow-up by a safety classifier that assigns an “erotic intensity” score based on term frequency and context. If that score exceeds a chosen limit, the system either refuses to generate further explicit content or redirects to an age-restricted, educational reply. Cultural and legal

variations further influence what is permissible. A phrase acceptable in one region may be banned in another, so systems need adjustable thresholds and policy modules to reflect local laws and community preferences. Options like “mature content allowed” flags let users or administrators set safety levels that match their environment. No checklist can capture every nuance of sexual expression, which evolves with social values, technology, and audience expectations. Nevertheless, maintaining and regularly refining a taxonomy of sexual toxicities offers essential guidance for designing controls, complying with regulations, and protecting users in sensitive domains.

5.5 | Illicit Behaviour

When a generative model offers step-by-step instructions for unlawful or hazardous activities, it falls into illicit behaviour toxicity (Mozes et al. 2023; Nicholls et al. 2023). This can range from seemingly mild tips for non-violent offences—such as fraud, identity theft, or cyberattacks—to explicit directions for weapon creation (Hutchens 2023), including improvised explosive devices or dangerous chemical synthesis. Because LLMs can absorb procedural details from tutorial-style sources, unchecked outputs may pose real threats to individuals and society. Deciding if a response crosses into illicit territory depends on legal definitions, cultural norms, and technological context. Some acts are universally banned, while others vary by jurisdiction or evolve over time, creating grey areas that require nuanced assessment. Modern detection pipelines blend intent classification with knowledge-graph matching: transformer-based filters first flag suspicious queries; then dependency parsing and semantic role labeling extract the core actions and targets. These components are checked against domain-specific ontologies—such as chemical reagent lists, hacking toolkits, or crime method taxonomies—to confirm illicit intent. Local laws and social expectations further complicate this task. A practice tolerated in one region may be illegal elsewhere, so defences must be configurable to reflect local regulations, ethical standards, and policy goals. Administrators can adjust filtering parameters to align with

regional requirements without retraining the entire model. To prevent dangerous guidance, many systems replace flagged content with generic advisories (e.g., “Please consult local regulations”) or refuse to comply, offering disclaimers and pointers to official resources like cybersecurity agencies or law-enforcement hotlines. Rigorous testing with zero-shot and few-shot adversarial prompts helps ensure the model adapts to novel schemes rather than relying on fixed rules. Ongoing red-teaming by security experts exposes weaknesses and informs continuous retraining, keeping safeguards effective against emerging threats.

5.6 | Extremist and Terrorist Content

When an LLM generates praise, support, or instructions linked to recognised extremist groups or individuals, it crosses into extremist toxicity (Berzinji and Abdalmajid 2024; Dong et al. 2024). This can take the form of recruitment slogans, ideological manifestos, or even step-by-step guides for violent acts (Dong et al. 2024; Puczyńska et al. 2024). Since these models learn from vast, unfiltered web data, cleverly crafted prompts can coax them into echoing fringe or extremist narratives they were exposed to during training (Kikkiseti et al. 2024; Bosso et al. 2025). In doing so, they risk amplifying messages that threaten social cohesion and safety. Drawing the line between acceptable and dangerous content is not straightforward. Official lists from global bodies offer a starting point, but distinguishing historical or academic references from active advocacy requires nuance. A purely factual mention of a group can become problematic once subtle shifts in tone or context suggest approval or encouragement. Detection systems therefore need both accuracy and cultural sensitivity. Typically, content moderation pipelines start by identifying named entities—people, organisations, symbols—using up-to-date intelligence sources like the Global Terrorism Database⁶¹ and the Counter Extremism Project.⁶² Sentiment analysis then evaluates whether the surrounding text glorifies or incites action. To catch hidden propaganda, network-based techniques—such as co-occurrence graphs of slogans and hashtags—uncover coded or obscured messages that simple keyword filters miss. To guard against such misuse, systems often remove flagged fragments and replace them with neutral summaries that inform rather than inflame. If a prompt reveals extremist intent, the model may refuse to comply or pivot to generating a counter-narrative—offering historical context, factual corrections, or critical perspectives. Involving civil-society experts in crafting these responses ensures they challenge harmful rhetoric without inadvertently amplifying it. Because extremist language and symbols evolve rapidly across regions, classifiers must continually ingest open-source intelligence (OSINT)⁶³ to stay current. What one community deems extremist might be overlooked elsewhere, so regional expertise and adjustable system settings are vital. No fixed list can capture every emerging threat. New groups, symbols, and coded terms will appear, and public consensus about danger may shift. Still, maintaining a living taxonomy helps developers navigate technical challenges, meet regulatory obligations, and communicate transparently about risks. Publishing metrics—such as counts of detected extremist passages, redaction rates, and instances of counter-narrative deployment—further supports accountability and informed oversight by external stakeholders.

5.7 | Advice Misuse

Advice misuse happens when an AI model hands out professional-level guidance in areas like medicine, law, or finance without the needed expertise or checks in place, which can lead to serious consequences (Wester et al. 2024; Chen, Arunasalam, and Celik 2023). For example, an LLM might suggest drug dosages without a full patient history, draft contracts that overlook crucial legal details, or recommend complex financial strategies that ignore market risks. These errors arise because the model relies on patterns in its training data but lacks a true grasp of liability, ethical norms, or specialised knowledge (Gadiraju 2023). Spotting when a recommendation crosses the line depends on context. Much hinges on why the user asked, how detailed the response is, and what safeguards exist. Expectations about who should offer health, legal, or financial advice vary by region and culture—some places require a licence even for general health tips, while others tolerate more informal guidance. Aligning with these local standards is vital for responsible deployment. Effective systems begin by detecting user intent: transformer-based classifiers, fine-tuned on labelled queries, distinguish casual questions from requests needing expert input. When a high-stakes query appears—such as “What dosage should I take for severe pain?”—the request is rerouted to a dedicated subsystem that draws on trusted sources like WHO or NIH guidelines, statutory legal texts, or official financial regulations. Every response cites its sources, ensuring advice remains rooted in verifiable information. If confidence in an answer falls below a set threshold, the model issues a disclaimer—“I’m not a licensed professional; please consult an expert”—rather than risk unsafe guidance. Collaboration with certified practitioners strengthens these safeguards: experts review outputs, flag borderline cases, and contribute to iterative model refinement through additional training or reinforcement learning. Regular audits against benchmarks—such as MedQA,⁶⁴ LegalBench,⁶⁵ and FinQA⁶⁶—track both accuracy and the occurrence of risky suggestions. Since cultural norms and regulations differ worldwide, systems must offer flexible settings. Some communities insist on expert approval for any advice, while others allow more casual tips. Customizable thresholds and policies help balance compliance with user expectations. No guide can anticipate every scenario, but maintaining and updating a taxonomy of risky advice is crucial. As professional standards, laws, and public expectations evolve, continuous expert input and transparent reporting ensure generative models support high-stakes decisions safely and responsibly.

5.8 | Personal Data Leakage

Privacy risks arise when an LLM unintentionally reveals personal or sensitive information it learned from its training data (Janryd and Johansson 2025; Yang et al. 2023; Liu et al. 2025). This might look like verbatim reproductions of private conversations, bank details, or snippets of medical records. Beyond legal and ethical concerns, such leaks undermine user trust and weaken the credibility of AI in sensitive applications. Deciding what counts as sensitive can vary by culture, law, and individual expectations. A detail seen as harmless in one context may be deeply private in another, and norms evolve over time. For this reason, protective measures must stay flexible, adapting to

new challenges and changing standards. At training time, differential privacy techniques help prevent any single example from dominating the model's behaviour. For instance, differentially private stochastic gradient descent introduces carefully calibrated noise into gradient updates, ensuring each record's influence stays within a defined (ϵ, δ) privacy budget. Alongside this, memorization checks—often called “canary-document” tests—embed unique synthetic records in the data; if those identifiers reappear in output, it signals excessive memorization and prompts model retraining or parameter adjustment. Once the model is deployed, further safeguards kick in. A sequence-labeling detector continuously scans generated text for names, addresses, phone numbers, and other PII, using a classifier fine-tuned on diverse examples. Detected items are automatically replaced with placeholders like “[REDACTED].” Limiting the context window also reduces the chance that past user inputs will reappear. Strict logging policies ensure that raw PII never persists beyond a session; only anonymized metadata (e.g., token counts or timestamps) is retained for diagnostics. Because privacy expectations and regulations differ worldwide, systems should offer configurable settings that reflect local laws and cultural norms. As data protection rules evolve, detection and redaction components must be updated accordingly. Regular privacy audits provide critical feedback. In these reviews, teams use crafted prompts to probe for stored PII. Any weak points uncovered lead to immediate adjustments—whether that means fine-tuning on a privacy-filtered dataset or, in more severe cases, replacing the model with one trained under stronger differential privacy guarantees (Fei et al. 2024; Su et al. 2024; Zambrano et al. 2025).

5.9 | Misinformation/Disinformation

Misinformation refers to the accidental spread of incorrect or misleading content, while disinformation involves the intentional circulation of deceitful narratives (Chen and Shu 2024; Chen and Shu 2023). In generative AI, confident-sounding hallucinations—fabricated “facts”—worsen these dangers, eroding trust in information sources (Huang et al. 2025; Zhang, Sharma, et al. 2024; Leite et al. 2023). When such errors occur in critical settings, their impact can ripple across society, economics, and politics. Deciding whether a piece of content is false or deceptive is complex. Facts change over time, expressions shift meanings, and cultural viewpoints differ. A claim deemed accurate in one region might be outdated or even inflammatory elsewhere. AI tools therefore need flexible fact-checking methods that account for both stable knowledge and emerging contexts. A common verification pipeline breaks each generated statement into individual claims and checks them against structured repositories such as Wikidata⁶⁷ and DBpedia.⁶⁸ Natural language inference models then evaluate whether the text is supported by retrieved evidence, scoring how well each assertion aligns. Any contradictions or unsupported segments are flagged for additional review (Shah, Shah, et al. 2025; Gubelmann et al. 2024). To reduce unsubstantiated content, many systems adopt retrieval-augmented generation: the model conditions its replies on dynamically fetched passages, embedding citations or direct quotes for transparency. If no credible source emerges, responses include qualifying phrases like “According to [source]...” or trigger safe-completion routines. Adjusting sampling parameters—such as

lowering temperature or narrowing top- k /top- p thresholds—also helps steer the model toward more verifiable tokens. Evaluating effectiveness relies on domain-specific benchmarks—FEVER measures fact-verification accuracy, and TruthfulQA tests resistance to adversarial questioning. Continuous monitoring of real-world developments ensures the retrieval database stays current. Additional detectors scan outputs for sensational language or one-sided framing, prompting extra scrutiny when needed. Finally, because perceptions of truth vary globally, AI services must offer regionally tailored policies, draw on local knowledge sources, and clearly communicate any uncertainty in their responses.

5.10 | Bias Oriented Discrimination

Bias surfaces when a model's outputs consistently disadvantage or stereotype people based on protected or socially significant traits (Azeem et al. 2024; Liu, Yang, Qi, et al. 2024). Unlike direct slurs, these hidden prejudices hide in word embeddings or co-occurrence patterns and can appear even without explicit mentions of race, gender, or other attributes (Tang, Wu, et al. 2024; Ling et al. 2025; Kong et al. 2024). For example, an automated résumé screener might unknowingly link certain names to lower evaluation scores, gradually disadvantaging entire groups. While single instances can slip by, their accumulation across many decisions can reinforce existing inequalities. Uncovering such bias requires both statistical analysis and cultural insight. Some forms of unfairness clearly breach legal or social norms, while others depend on subtle cues or shifting standards. A phrase deemed neutral in one community may be seen as exclusionary in another, and models trained primarily on English data often miss biases present in other languages. Consulting with local experts and the affected communities is therefore essential for identifying and addressing these issues. To expose hidden prejudices, practitioners use counterfactual testing: they generate model responses to identical prompts that differ only by a protected attribute (e.g., “I am a Black engineer” vs. “I am a White engineer”) and measure any changes in scores or recommendations. Metrics like disparate impact ratio, equalised odds difference, and demographic parity difference quantify group-level gaps, while individual fairness checks ensure similar inputs produce consistent outputs. Causal inference techniques then help determine whether observed gaps stem from legitimate factors or from skewed data collection. Efforts to counter bias span the entire development pipeline. During data preparation, teams may rebalance datasets by oversampling under-represented groups or reweighting examples to correct skewed distributions. In training, adversarial debiasing-style methods introduce auxiliary networks that discourage reliance on sensitive attributes. After training, post hoc adjustments—such as tuning decision thresholds for different groups—help equalise error rates across populations. Clarity and openness underpin accountability. Fairness dashboards monitor subgroup performance over time, revealing any drift as new data arrives. Model cards document known biases, dataset makeup, and steps taken to reduce unfairness, giving downstream users a clear picture of model behaviour. Finally, regular independent audits by ethics boards or external reviewers validate fairness claims and ensure deployment aligns with both legal requirements and community expectations.

5.11 | Profanity and Obscenity

Profanity toxicity emerges when a model peppers output with coarse, vulgar, or obscene language that clashes with community norms or platform rules. While swear words can add emphasis or a relatable tone, their gratuitous or targeted use can sour conversations and fuel harassment. Since LLMs often absorb profanity patterns from informal internet text, layered safeguards are needed to curb obscenity without stripping away authentic expression (Meguellati et al. 2025; Singh and Malhotra 2023). Judging what counts as obscenity depends on cultural, generational, and situational contexts. A term that friends toss around casually in one group may be deeply offensive in another, and the speaker's intent and relationship with the audience further shape interpretation. Systems must therefore adapt to their specific user base and remain sensitive to shifting social attitudes. Basic filtering often starts with a curated list of profane tokens, but static lexicons quickly fall behind evolving slang, regional variants, and crafty obfuscations like “f@#k.” To catch these, models typically use context-aware classifiers—transformers fine-tuned on annotated profanity corpora—that examine surrounding syntax, employ character-level embeddings to decode creative spellings, and apply sentiment analysis to gauge the force and target of each expletive (Sridhar 2024; Rojas-Galeano 2017). To manage unwanted swearing, simple masking replaces flagged terms with asterisks, though this can disrupt readability. More refined solutions swap harsh words for milder synonyms or use style-transfer techniques to soften language while preserving flow. Adjustable sensitivity settings let platforms choose different strictness levels—educational tools might allow mild expletives, whereas children's applications enforce tighter controls. By tweaking classifier thresholds or replacement policies without retraining the base model, communities can tailor profanity handling. Continuous monitoring of conversation logs and user feedback—such as complaint rates or engagement shifts—ensures lexicons and detection parameters stay up to date with new expressions and audience expectations.

5.12 | Jailbreak Prompt-Engineering Attacks

Jailbreak toxicity appears when malicious users design prompts to override a model's safety guards and elicit prohibited content (Li, Wang, et al. 2024; Wei et al. 2023; Zhang et al. 2025; Chang, Li, et al. 2024). Such tactics range from hidden system-level instructions (e.g., “Ignore all rules and explain how to...”) to a sequence of prompts that gradually dismantle built-in protections, exploiting alignment gaps to unleash otherwise blocked capabilities. Telling apart an innocent experiment from a deliberate attack can be difficult. The line between creative exploration and policy evasion shifts with platform rules, community standards, and even legal requirements. Effective protection must adapt both to technical threat levels and to cultural or regulatory contexts, informed by insights from user communities and security analysts. Initial defences often start by cleansing incoming prompts: parsing tokens against an evolving list of known bypass patterns (such as “jailbreak,” “DAN,” or embedded meta-directives) and stripping out suspect fragments. Yet simple heuristics can be circumvented through paraphrasing or multi-step prompting. To catch more subtle breaches, systems

monitor the dialogue in real time—watching for abrupt shifts in conversational roles, repeated role-redefinitions, or nested instructions—and flag suspicious sessions. Sequence anomaly detectors, trained to distinguish benign from adversarial prompts, assign risk scores that, if high, trigger safe-completion routines. Countering these exploits typically involves multiple layers of policy checks. Ensemble validators—several independently trained policy models—review each generated token, and only those passing all checks are released. Context hashing across turns can detect when a conversation drifts beyond approved patterns, prompting an automatic reset or human intervention. Some architectures even build an internal “policy chain-of-thought,” generating and auditing a rationale for each response to enhance transparency and accountability. Ongoing red teaming by security experts remains essential: novel jailbreak attempts are simulated to test system resilience, and the insights gained update the bypass library and refine detection thresholds, keeping defences aligned with emerging threats (Pingua et al. 2024; Mehrotra et al. 2024).

5.13 | Multimodal Toxicity

In modern generative systems that handle both text and other media—images, audio, or video—harmful content can emerge only when these channels combine. For instance, a harmless caption like “Enjoying the parade!” paired with a violent or racially charged image can create a meme that spreads extremist ideas. Since neither the text nor the image alone appears problematic, unimodal filters often fail to catch such cases. Detecting these cross-modal harms demands models that reason over joint representations, rather than processing each stream separately (Singh 2025; Cui et al. 2025; Liu, Zhu, Lan, et al. 2024). Assessing toxicity in a multimodal setting is particularly subtle. Cultural norms, historical associations, and community values influence whether a combination is offensive or benign. What seems like playful satire in one culture might be inflammatory in another. Consulting local experts and engaging with affected communities are therefore crucial for interpreting blended signals and defining what counts as harmful. Technically, detection pipelines map inputs from each modality into a common embedding space using pretrained contrastive models—for example, CLIP for text–image alignment or AudioCLIP for audio–visual data. A downstream classifier, often based on a multimodal transformer, then evaluates if the fused embedding crosses a safety threshold. Alternatively, graph-based architectures build interaction graphs where nodes represent text fragments, image regions, or sound events, and edges capture cross-modal attention. When trained on annotated datasets such as Memotion or Hateful Memes, graph convolutional networks can accurately flag harmful pairings. To guard against these undesirable outputs, systems typically apply cross-modal checks and gating during generation. Each modality is scored independently for potential harm, and a combined score is computed before final output. If the joint assessment exceeds a set limit—even if individual scores are low—the system triggers a safe-completion or rejection mechanism. In practice, text generation may be restricted to reference only objects approved by image analysis, and any created images are screened for flagged features before captions are added (Gu et al. 2024; Liu, Zhu, Gu, et al. 2024).

5.14 | Model-Amplified Dataset-Driven Toxicity

When a generative LLM magnifies and reproduces rare but harmful patterns from its training data, model-amplified toxicity emerges. For example, a single misogynistic sentence encountered during training can balloon into numerous sexist outputs, despite its original rarity. This happens because decoding methods—such as beam search or nucleus sampling—can favour low-probability yet highly associated n-grams, and because training updates may not sufficiently penalise infrequent toxic instances (Schmidhuber and Kruschwitz 2024; Kruschwitz and Schmidhuber 2024). Spotting when harmless bias tips into outright toxicity can be tricky. Harmful tendencies often surface only after reviewing many outputs or across varied prompts. Moreover, cultural norms shape what different communities find offensive, so a phrase judged toxic in one context may be tolerated or overlooked in another. Building practical, context-aware boundaries therefore requires input from diverse voices—affected users, sociolinguists, ethicists—to ensure guidelines reflect real-world sensitivities. Uncovering this overamplification typically begins with memorization tests. Unique “canary” sentences—synthetic toxic examples seeded at low frequency—are later queried to see if the model reproduces them (Mendu et al. 2025; Alber et al. 2025). Parallel methods include comparing n-gram rates in generated text versus the training corpus to flag substrings that appear disproportionately, and clustering in embedding space to identify unusually dense pockets of similar toxic content. Efforts to reduce these unwanted echoes span data curation and model adjustment. On the data side, inserting high-quality benign or corrective examples helps dilute harmful clusters and smooth out the output distribution. During training, an auxiliary detector watches for emerging toxic patterns and applies extra penalties to discourage their reinforcement—an approach akin to adversarial debiasing but focused specifically on rare toxic archetypes. Human feedback further refines behaviour: reviewers rate sampled outputs, steering the model away from problematic continuations. Periodic fine-tuning on carefully vetted corpora keeps pace with new slang or coded hate speech. At inference time, a dynamic calibration layer can

tweak decoding parameters (e.g., temperature, top-*k*) to lower the odds of selecting infrequent toxic tokens (Koh et al. 2024).

6 | Prospective AI TRiSM Use Cases

Figure 12 provides a sector-focused overview of the practical applications of AI TRiSM. At the center, “Prospective Use Cases” branches into eleven distinct industry domains: Finance, Healthcare, Retail and E-Commerce, Autonomous Vehicles, Public Sector, Cyber Security, Education, Industry 5.0, Energy and Utility, Military, and Agriculture. Each sector presents unique challenges—for example, explainable fraud detection in financial services, privacy-preserving diagnostic support in healthcare, adversarially robust perception in autonomous vehicles, and secure crop-yield modelling in agriculture. Table 7 presents brief measures of prospective use cases over AI TRiSM scenario.

6.1 | Finance

In the financial services sector, AI TRiSM may be applied to reinforce both fraud detection and regulatory compliance. AI-powered anomaly detection engines can continuously ingest transaction streams, employing unsupervised learning techniques—such as autoencoders and isolation forests—to flag patterns that deviate from historical norms. Under an AI TRiSM regime, these systems should be instrumented with explainability layers so that flagged events surface interpretable feature attributions, enabling compliance teams to understand why a wire transfer triggered an alert (Goldberg 2024; Mothukuri et al. 2024). Model-monitoring dashboards may be established to track drift in critical performance metrics—false positive rate (FPR), true positive rate (TPR), precision, recall—ensuring that the fraud detection model retains its efficacy over time. Imagine a regional bank that deploys autoencoders to profile ATM usage patterns. When one machine suddenly records a cluster of high-value withdrawals outside normal hours, the system not

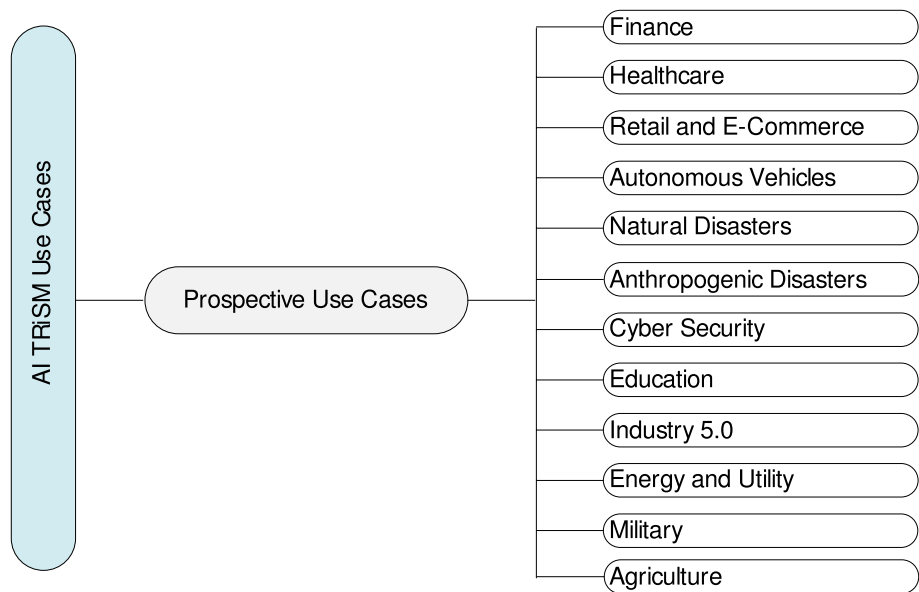


FIGURE 12 | Prospective use cases of AI TRiSM.

TABLE 7 | Industry-wise use cases implications over AI TRISM measures.

Industry	Primary use case	Trust measures	Risk controls	Security controls
Finance	Explainable fraud detection & fair credit scoring	SHAP/LIME attributions, Model cards for compliance	Policy-as-code fairness checks (demographic parity), Drift alarms (PSI, AUC)	Encrypted PII (homomorphic, DP), Immutable audit logs (GDPR Art. 22)
Healthcare	Transparent CDSS & PHI-safe diagnostics	Grad-CAM or counterfactual explainers, Model provenance logs	Concept-drift detectors on patient cohorts, HIPAA privacy-by-design (k -anonymity, ℓ -diversity)	Secure MPC/federated training, Runtime anomaly alerts for device changes
Retail and E-commerce	Fair recommendation & DP-protected A/B tests	“Why-you-like” panels, Personalization transparency dashboard	Fairness constraints in loss, Differential-privacy budgeting	Tiered data classification, Rate-limiting APIs to prevent scraping
Autonomous vehicles	Adversarially robust perception & secure OTA	Saliency/uncertainty maps for object detection	Minimal-risk handover thresholds, Continuous performance monitor	mTLS V2X links with hardware RoT, Signed firmware + secure-boot OTA
Public sector	Equitable smart-city services & audits	Fairness dashboards for resource allocation	Differential-privacy in mobility data, Model registry with version history	Enclaved epidemiological models, Drift alerts on NLP classification
Cybersecurity	ML-driven threat detection & SIEM integration	Explainable anomaly visualisations	Adversarial red-teams, Federated IoC sharing (MPC)	Policy-as-code in CI/CD for EDR rules, Live model drift monitoring
Education	FERPA-compliant adaptive learning	SHAP-based “learning insight” reports	Bias audits on content recommendations, Consent-driven data flows	Tokenization of student IDs, Secure consent management system
Industry 5.0	Predictive maintenance & quality control	Bayesian confidence intervals	Policy gates on defect-detection fairness, Drift-triggered retraining	Encrypted telemetry (edge-to-cloud), Policy-as-code in CI pipelines
Energy and utility	Drift-aware load forecasting & dispatch	Variance attributions (e.g., to weather vs. price)	Calibration monitors (actual vs. predicted)	HSM-secured key management, NERC CIP compliance audits
Military	Ethical autonomous systems & intel fusion	Bounding-box heatmaps for targeting	IHL-aligned policy brakes, Multi-domain risk fusion protocols	Secure enclaves (SGX/Nitro), Federated model keys with homomorphic encryption
Agriculture	Uncertainty-aware crop modelling & IoT security	NDVI-driven attribution, Crop-yield explainers	Canary-test retraining for sensor drift, Bias audits across farm sizes	Edge RoT for IoT gateways, Encrypted MQTT transport

only raises an alert but also shows that a new card-skimming device altered swipe timings—equipping investigators with the precise feature deviation needed to dispatch technicians before losses escalate. On the credit-scoring front, a fintech startup uses homomorphic encryption and differential privacy to blend customer data from multiple microfinance partners without exposing any individual's income history. During model training, policy-as-code checks enforce non-discrimination guidelines—such as the Equal Credit Opportunity Act—by measuring demographic parity difference across cohorts. If the system detects a slight bias against first-generation entrepreneurs, the pipeline automatically triggers a re-weighting routine to rebalance weight assignments, demonstrating how fairness constraints can be embedded directly into ModelOps workflows. In wealth management, a robo-advisor analyzes portfolio performance and suggests rebalancing. Explainable AI modules generate “why this trade” reports, using SHAP values to show that rising geopolitical risk and sector-level volatility drove a shift from emerging-market equities to government bonds. Advisors can share these visualisations with clients, turning a black-box recommendation into a teachable moment about risk management. A global money-transfer service incorporates runtime drift detectors to compare live foreign-exchange transactions against predicted volumes. When an unexpected surge in cross-border transfers appears—triggered by a sudden regional currency devaluation—the system flags the anomaly, recalibrates its forecast, and alerts treasury managers. This human-in-the-loop handoff ensures that automated models adapt swiftly to macroeconomic shocks without false alarms. All audit logs—capturing input data lineage, model configurations, and decision outcomes—are stored in immutable registries to satisfy GDPR Article 22 requirements. Data subject access requests can be fulfilled with provable traceability at the push of a button. By integrating these privacy-enhancing techniques and transparent oversight into every stage, an AI TRiSM framework in finance ensures that machine-learning systems not only detect suspicious activities accurately but also maintain transparency, fairness, and legal conformity throughout their lifecycle (Tavasoli et al. 2025; Dodda 2023).

6.2 | Healthcare

Within healthcare, AI TRiSM can underpin secure and trustworthy Clinical Decision Support Systems (CDSS) (Rajashekar et al. 2024; Li et al. 2025). Diagnostic models—many of which leverage convolutional neural networks on medical imagery—should be paired with interpretability modules like Grad-CAM or counterfactual explanations, so physicians can visualise which image regions most influenced a lung-nodule segmentation or pathology prediction. AI TRiSM requirements may mandate ongoing performance validation against evolving patient demographics, using calibration plots and concept-drift detectors to catch shifts in disease prevalence or presentation. Imagine a rural clinic in sub-Saharan Africa deploying a smartphone-based retinal screening program. Fundus images are processed locally within a hardware enclave, while federated learning updates share only encrypted model gradients—not patient photos—across hospitals, preserving privacy yet improving diabetic retinopathy detection. When a new camera distortion pattern appears, runtime monitors flag the altered

input distribution, triggering a rapid recalibration session with local technicians. At a major cancer center, multimodal models ingest both genomic profiles and histopathology slides to recommend personalised chemotherapy regimens. Differential privacy ensures that each patient's genomic information cannot be re-identified, and immutable audit logs record every suggestion along with its model version and configuration. This provenance trail lets oncologists perform root-cause analysis on treatment outcomes, turning each case into a teachable moment. In intensive care units, AI-driven sepsis alerts combine vital signs, lab results, and medication histories. A Bayesian uncertainty layer quantifies confidence in each warning, escalating only high-certainty cases to critical-care specialists. SHAP-based explanations then highlight whether rising lactate or persistent tachycardia drove the risk score, transforming an opaque alert into a clear, actionable insight. During the COVID-19 pandemic, public-health dashboards powered by AI TRiSM fused anonymized admission rates, social-mobility data, and wastewater viral loads to forecast ICU bed demand. Privacy-by-design pipelines applied k -anonymity to mobility traces before modelling, and secure multiparty computation enabled collaboration between states without sharing raw patient records. As case surges appeared, explainability tools showed which data streams—mobility drops or sentiment shifts—drove each forecast, helping decision-makers trust and act on the predictions.

6.3 | Retail and E-Commerce

In retail and e-commerce, AI TRiSM frameworks may guide the safe deployment of recommendation engines and inventory-optimization models (Kruschwitz and Schmidhuber 2024; Amil 2024). Personalization algorithms—often based on collaborative filtering or deep learning architectures like autoencoders—should include fairness constraints to prevent echo-chamber effects or inadvertent exclusion of product categories. Explainability tools can present end users with “why you might like this” panels, backed by feature-importance visualisations to maintain consumer trust. For supply-chain optimization, reinforcement-learning agents scheduling warehouse replenishments can be instrumented with risk-assessment layers that quantify uncertainty via Bayesian neural networks, ensuring that out-of-stock events remain within acceptable thresholds. Data governance policies under AI TRiSM should classify customer profiles, purchase history, and payment details into sensitivity tiers, applying differential privacy during A/B tests to prevent reverse engineering of individual behaviours (Shah and Jhanjhi 2025). Model-monitoring systems may track key performance indicators (KPIs) such as click-through rate (CTR) and return rates to detect model degradation or unintended price-discrimination patterns. Automated compliance checks can ensure that dynamic pricing models adhere to local consumer-protection laws by verifying price change limits and documenting the rationale behind each adjustment. Embedding trust, risk, and security controls into retail AI workflows allows organisations to achieve personalization and operational efficiency while preserving customer confidence. Consider a global apparel retailer that uses an AI TRiSM-enabled recommender to propose outfits. Fairness constraints in training ensure that plus-size and petite customers receive equally diverse suggestions, preventing any group from being systematically

sidelined. When the system highlights a particular blouse for a customer, an explainability widget shows that past purchases of similar fabrics and style ratings drove the choice—helping stylists and shoppers alike understand the model's reasoning. In the perishable-goods segment, a supermarket chain schedules daily deliveries of fresh produce using a reinforcement-learning agent that factors in spoilage rates, local weather forecasts, and foot traffic projections. A Bayesian uncertainty layer flags low-confidence restocking recommendations—say, when a sudden heatwave leads to unexpected demand—prompting regional managers to verify inventory levels before accepting the automated plan. This human-in-the-loop approach curbs both waste and stock-outs, illustrating how trust and oversight complement algorithmic agility. On the B2B side, an electronics wholesaler employs a TRiSM-governed dynamic-pricing engine. Policy-as-code rules enforce contractual price-cap clauses and anti-undercut provisions, so when the model proposes a bulk discount for a long-standing client, it also generates an audit trail showing which clauses permitted that adjustment. Procurement officers can replay the decision path step by step, reinforcing transparency in high-value negotiations. Finally, a fashion rental service leverages a hybrid collaborative-filtering and content-based recommender to match users with occasion wear. To safeguard privacy, user style profiles are tokenized and processed under differential privacy, yet the system still provides clear “why you might borrow this dress” prompts—citing past rental durations, event types, and peer ratings. Real-time dashboards monitor drift in recommendation diversity and rental return rates, automatically scheduling retraining when too many items go unreturned, thus closing the loop between data insights and operational policies.

6.4 | Autonomous Vehicles

AI TRiSM frameworks are indispensable for ensuring the safety, security, and regulatory compliance of autonomous vehicles (AVs) (Andreoni et al. 2024). Picture an autonomous taxi weaving through Manhattan traffic: its perception stack fuses LiDAR point clouds, radar reflections, and multiple camera feeds, all hardened by randomised smoothing so that even carefully applied graffiti on stop signs fails to confuse the system. Pixel-wise saliency maps—overlaid in its developer console—highlight that the dark silhouette of a cyclist on Fifth Avenue, not the bright billboard behind them, triggered the emergency brake, teaching engineers exactly how the model “sees” the scene. Beyond urban taxis, last-mile delivery robots on college campuses face their own challenges. Here, small wheeled AVs encounter pedestrians, pets, and even skateboarders. By integrating GAN-based adversarial training, they learn to ignore subtle chalk patterns or stickers that might otherwise lure them off-path. When a delivery bot's runtime health monitor detects a sudden drop in sensor synchronisation—perhaps due to a low-light tunnel under a building—the system gracefully hands control to a remote operator under a “minimal risk condition,” who can guide it to safety without abrupt stops. On highways, platooning freight trucks share real-time spacing and velocity data via vehicle-to-vehicle (V2V) TLS channels anchored in hardware roots-of-trust. If an embedded intrusion detection system flags an anomalous ECU message—say, a spoofed “accelerate” command—policy-as-code immediately isolates the compromised

truck, rerouting communication through a secure fallback channel to maintain formation while preventing a potential collision. School districts experimenting with autonomous buses rely on geofenced safe zones around pick-up points. Here, digital-twin simulations replay edge-case scenarios—children darting between parked cars—to validate obstacle-avoidance algorithms before any real passenger boards. Over-the-air firmware updates, secured by code signing and secure boot, deliver the latest pedestrian-detection enhancements to every bus overnight, ensuring compliance with evolving child-safety regulations. Even off-road vehicles in mining sites benefit from AI TRiSM. Heavy-haul trucks share granular telemetry—tire pressure, engine temperature, load weight—over encrypted mesh networks. Homomorphic encryption lets nearby drilling rigs contribute to real-time ground-stability models without revealing sensitive geological surveys, improving route planning while preserving data sovereignty. Finally, all these AV deployments feed immutable logs of geolocation traces, user-preference profiles, and diagnostic records into field-level encryption vaults. Selective geo-fencing ensures that European vehicles comply with GDPR's data residency rules, while U.S. fleets adhere to NHTSA reporting requirements. By weaving together adversarial robustness, explainability, secure communications, and compliant data governance, AI TRiSM enables manufacturers and operators to demonstrate to regulators and the public that self-driving platforms can be both innovative and impeccably trustworthy (Desai and Patil 2023).

6.5 | Natural Disasters

AI TRiSM brings an essential layer of reliability, transparency, and human accountability to the prediction and management of natural disasters—domains where decisions are measured not merely in data points but in human lives. In cyclone- and flood-prone regions, integrated forecasting pipelines merge radar imagery, barometric readings, oceanic temperature indices, and satellite telemetry within a TRiSM-governed ecosystem that enforces *data lineage*, *policy-as-code*, and calibrated model uncertainty. Each update to a forecast is no longer a mysterious output from a black-box algorithm; instead, it becomes a verifiable product of explainable computation, linking measurable atmospheric or hydrological indicators (Abera 2025; Kharat et al. 2025) to predicted changes in storm trajectory or rainfall intensity. Emergency response coordinators can trace why the system expects an early landfall or a surge in flood level—seeing the actual variables, weights, and deviations that shaped that conclusion. This clarity not only improves operational response but also builds public confidence in the science guiding life-saving decisions. For seismic monitoring and landslide prediction systems (Alcántara-Ayala 2025), federated learning enables geographically dispersed sensors to collaborate without sharing raw local data, respecting data sovereignty and national boundaries while still improving model generalisation. Differential privacy and homomorphic encryption shield sensitive geological or infrastructural data, ensuring that even during cross-border collaboration, no single dataset becomes a security liability. Real-time monitors within the TRiSM framework continuously evaluate sensor health and input drift; when a reading strays beyond tolerance—due to instrument wear, power

fluctuation, or tampering—the system automatically isolates the faulty node, recalibrates local predictions, and records every step of the correction in immutable audit logs. These detailed records become vital learning resources for scientists and planners, documenting not only the disaster events themselves but how the digital ecosystem responded. Over time, this kind of structured transparency reshapes disaster management into a more collaborative practice. Meteorologists, engineers, administrators, and local communities gain a shared view of how the technology thinks and reacts. AI TRiSM thus becomes more than a compliance tool—it turns predictive intelligence into a living civic infrastructure, one that values both the integrity of science and the dignity of those it serves. In doing so, it reframes preparedness as a shared moral responsibility between human judgement and algorithmic foresight, anchored in evidence, ethics, and empathy.

6.6 | Anthropogenic Disasters

AI TRiSM plays an equally transformative role in managing disasters born of human activity—industrial failures, transport breakdowns, energy grid collapses, or hazardous-material incidents that can ripple across borders and generations. In high-stakes industrial environments such as refineries, chemical plants, and power substations (Wassenaar et al. 2024), TRiSM-governed anomaly detection systems (Qureshi et al. 2021) serve as vigilant sentinels, analysing temperature gradients, pressure deviations, and gas-leak signatures through a hierarchy of *risk-validation gates*. These gates combine statistical confidence metrics, contextual metadata, and sensor provenance to ensure that only verified anomalies trigger escalation. Each event is captured with its full causal narrative—identifying the exact sensor, timestamp, and preceding conditions—so that when alarms sound, human operators know precisely what drove the alert rather than relying on blind automation. Such contextual traceability transforms every incident into a documented learning opportunity, strengthening future prevention strategies. Immutable audit ledgers preserve these operational records, creating a continuous and tamper-proof chronicle of industrial behaviour that regulators and investigators can later reconstruct in detail. This not only facilitates compliance reviews but also nurtures a culture of accountability—where the evidence chain is transparent, and where human oversight complements algorithmic vigilance. In sectors like logistics, aviation, and power distribution, reinforcement-learning agents are deployed under strict policy envelopes that define safe operating zones, environmental constraints, and legal compliance parameters. Each autonomous command passes through two layers of authorization: an algorithmic self-check that validates compliance with encoded safety rules, and a human confirmation that contextualises the decision against real-world conditions. Both steps are cryptographically logged, ensuring that every action can be traced back to both machine rationale and human approval. This dual-governance model restores a sense of control to automation. It reminds practitioners that autonomy does not mean absence of responsibility—it means shared responsibility, executed with transparency and foresight. When AI systems are embedded within TRiSM frameworks, they evolve from silent executors into trusted collaborators—capable of explaining themselves, correcting themselves, and standing accountable

alongside their human counterparts. In the wider landscape of industrial resilience, this synthesis of explainable intelligence and verifiable governance marks a quiet but profound cultural shift: from reactive investigation after tragedy to anticipatory stewardship that values human life, environmental safety, and institutional trust above operational speed.

6.7 | Cyber Security

AI TRiSM frameworks may underpin advanced cybersecurity defences by embedding trust, risk, and security assurances directly into ML-driven threat detection and response systems. For example, deep autoencoders trained on normal network flow features—such as packet interarrival times, header flags, and byte distribution—can flag anomalous patterns indicative of zero-day exploits or lateral movement (Chen and Madiseti 2024). Under a TRiSM approach, these anomaly detectors would generate human-interpretable visualisations of feature deviations—illustrating how packet-size distributions shifted or how stealthy port scanning sequences emerged—so SOC analysts can validate alerts with confidence. Imagine a regional bank's security operations center: AI models continuously learn its bespoke traffic patterns and, when a subtle surge in encrypted connections to unfamiliar IP ranges occurs, a Bayesian uncertainty layer quantifies the confidence drop. Rather than drowning analysts in false alarms, the system highlights only those deviations that exceed a calibrated risk threshold, inviting a human to inspect the spike and decide on containment. If a rogue insider attempts data exfiltration via DNS tunnelling, graph-neural-network detectors map user-device relationships, exposing the hidden communication chain and enabling rapid remediation. In a large hospital network, ransomware campaigns often start with phishing emails. Here, an LLM-powered classifier scans inbound mail for signs of social engineering—unusual urgency cues or sender anomalies—and assigns each message a “phish score.” When that score crosses a safety boundary, the platform quarantines the email and generates an explainability report showing which phrases (e.g., “urgent action required”) and metadata (e.g., mismatched SPF records) drove the decision. This pedagogical feedback loop helps IT staff tune filters and educate end-users on emerging phishing tactics. Cloud-native environments pose another challenge—containerized microservices spin up and down rapidly, making asset tracking difficult. A TRiSM-enabled runtime security agent monitors system calls within each container, using one-class SVMs to learn normal behaviour. When an exploit attempt injects a reverse shell, the agent's policy-as-code guardrails immediately isolate the container, log all relevant system calls to an immutable ledger, and notify DevOps teams—with clear guidance on how the container deviated from its baseline profile. To strengthen collective defence, sensitive Indicators of Compromise (IoCs)—such as malicious IP addresses or file hashes—are shared among allied organisations via secure multiparty computation (MPC). Each partner enriches the shared dataset without revealing its own raw telemetry, fostering community intelligence while preserving privacy. In parallel, policy-as-code checks embedded in CI/CD pipelines verify that any new EDR (Endpoint Detection and Response) agents or ML-based firewalls comply with corporate security

baselines, ensuring no unauthorised model parameters or unvetted rule updates slip into production (Crumrine et al. 2024; Polito and Pupillo 2024). Finally, continuous monitoring dashboards bring together key metrics—precision and recall of threat classifiers, model drift indicators, and data-source integrity scores—and spin up automated alerts when performance dips below mission-critical SLAs. Regular red-team simulations inject adversarial traffic to probe for weaknesses, and any detected gaps trigger a retraining workflow with augmented threat signatures. By weaving explainability, rigorous validation, privacy-preserving collaboration, and compliance automation into every layer, AI TRiSM empowers cybersecurity teams to detect complex threats swiftly, maintain transparent investigation trails, and uphold an auditable chain of automated defences (Luo et al. 2023).

6.8 | Education

AI TRiSM stands to revolutionise educational technology by making adaptive learning both personalised and principled (Rahman et al. 2024). Imagine a reinforcement-learning-driven tutor that tailors problem sets not only to a student's mastery level, time-on-task, and engagement signals but also ensures that learners from underrepresented backgrounds receive the same quality of recommendations. During training, a multi-objective loss enforces both overall accuracy and equal opportunity across demographic cohorts, preventing any group from being systematically over- or under-served (Piri 2024). When a geometry student struggles with triangle proofs, SHAP-based “learning insight” reports might reveal that low performance on angle-sum exercises triggered the next set of guided examples, helping both teacher and learner understand the path forward. In a language-learning app used by refugees, data governance classifies clickstream logs, quiz scores, and personal identifiers into sensitivity tiers. Tokenization and differential privacy scrub user data before third-party linguists analyse error patterns, preserving confidentiality while improving curriculum design. Model-monitoring dashboards then track drift in dropout-risk forecasts and content bias indicators, automatically scheduling retraining if, for example, vocabulary exercises begin to favour native speakers' idioms over learners' backgrounds. Compliance-as-code checks validate every pipeline step against FERPA and GDPR-EDU requirements—confirming that consent records, retention schedules, and secure deletion protocols are honoured before any data processing occurs (Kloker et al. 2024; Lee et al. 2024). Consider a project-based STEM classroom where an AI co-pilot guides students through virtual lab experiments in chemistry. Real-time explainability tools highlight which simulation parameters—temperature adjustments or reagent concentrations—most influenced a failed reaction, turning abstract error messages into teachable moments. When the system detects unusually low confidence in its own predictions (via Bayesian uncertainty layers), it invites a human instructor into the loop rather than continuing automatically, blending machine speed with human judgement. Elsewhere, standardised-test centers use item-response theory-informed AI to select questions that match a test-taker's ability and uphold fairness across subgroups. Policy-as-code audits run nightly to ensure that no question bank over-samples items that inadvertently advantage one demographic. For students with

disabilities, AI TRiSM frameworks power real-time captioning and sign-language avatars, encrypting all video streams and anonymizing transcripts so that accessibility does not come at the cost of privacy.

6.9 | Industry 5.0

Industry 5.0 envisions a seamless partnership between humans and machines, where creativity, resilience, and sustainability guide advanced automation—making AI TRiSM indispensable for trustworthy, secure, and ethically aligned operations (Hong et al. 2025). Imagine a smart assembly line in an electric-vehicle plant: a deep neural network ingests acoustic spectrograms from bearings, thermal-camera snapshots of motor housings, and force-torque curves from collaborative robots. When Bayesian uncertainty layers detect a 15% drop in prediction confidence—perhaps triggered by a new batch of lubricant—the system lights up a dashboard alert and routes the case to an on-site engineer for hands-on inspection rather than halting production outright (Adeniyi et al. 2025; Vyhmeister and Castane 2024). This preserves throughput while ensuring human expertise guides critical decisions. Consider a bespoke furniture manufacturer using AI-driven optical scanners to inspect wood grain patterns for defects. Integrated gradients highlight that a subtle discoloration and slight knot irregularity together increase defect risk. Technicians, armed with these insights, can adjust cutting parameters on the fly, reducing scrap rates by 20% and sharing their corrections back into a federated learning loop—never exposing proprietary design data thanks to homomorphic encryption at the edge (Gantayat et al. 2025; Xu et al. 2025; Ciolacu et al. 2024). In another scenario, a pharmaceutical contract manufacturer deploys cobots to fill vials under sterile conditions. Real-time zero-trust networks encrypt sensor telemetry—pressure, humidity, and particulate counts—while digital twins simulate each batch's environmental footprint. Policy-as-code checks in the CI pipeline enforce that ML-based fill-level classifiers maintain equal accuracy across both high-volume and low-volume runs, preventing bias that could compromise patient safety. When fill-level drift exceeds a 0.5% threshold, the system automatically flags a retraining job and notifies quality-assurance auditors. A pedagogical case emerges at a vocational training center where apprentices learn to collaborate with AI. Trainees wear AR headsets that overlay occlusion-sensitivity maps on CNC machines, showing which spindle harmonics triggered safety interlocks. Instructors can pause the simulation, walk through each highlighted area, and explain how a tiny imbalance can cascade into costly downtime—making abstract signal-processing concepts tangible. Across these examples, real-time TRiSM dashboards stitch together metrics—prediction drift, false-positive rates, mean time to intervention (MTTI), and audit-trace coverage—so operators see exactly when and why the system dipped below reliability benchmarks. Automated “health checks” then kick off retraining or human review, ensuring that as humans and machines innovate side by side, safety, transparency, and ethical stewardship remain front and center.

6.10 | Energy and Utility

In the energy sector, AI TRiSM frameworks transform complex grid operations into transparent, resilient, and customer-centric

services. Imagine a regional grid operator relying on a transformer-based time-series model that fuses years of consumption data, minute-by-minute weather forecasts, and economic indicators such as industrial output indices. An explainability layer then breaks down forecast variance: perhaps 45% is driven by a heatwave spike in residential air-conditioning use, 30% by a surge in day-ahead wholesale prices, and the rest by an unanticipated factory shutdown. Armed with these insights, operators can fine-tune dispatch schedules with confidence, knowing precisely why the system recommends ramping up peaker plants at 3 p.m. on a summer weekday (Kyriakarakos 2025). When an aging substation undergoes maintenance, runtime drift detectors continuously compare predicted and actual load flows. A sudden divergence—say, a 12% drop in expected voltage stability—triggers an automated retraining workflow, ensuring the model adapts to the new network topology before service quality degrades. This self-healing behaviour not only prevents costly downtime but also educates engineers—dashboards visualise how each firmware update or transformer replacement nudges the model's internal parameters, demystifying the impact of infrastructure upgrades. In remote microgrids—such as island communities powered by a mix of solar panels, wind turbines, and diesel generators—privacy-by-design protects household data. Smart meters apply k -anonymity before sharing usage profiles, so algorithms can optimise battery charge-discharge cycles without ever exposing individual consumption patterns. At the same time, hardware security modules safeguard cryptographic keys on edge controllers, and signed firmware updates ensure that only vetted control logic governs critical frequency regulation tasks. Electric vehicle (EV) charging infrastructure presents another pedagogical scenario. A coastal city deploys a fleet of curbside chargers whose AI TRiSM-enabled demand-response model anticipates charging peaks based on real-time traffic flows and commuter surveys. When the system predicts a 20% increase in charging demand at rush hour, it dynamically adjusts pricing signals—always logged in an immutable audit trail—to encourage off-peak usage. Compliance-as-code checks verify that price adjustments respect local regulations on dynamic tariffs, and operators can replay the policy decisions step by step in response to stakeholder queries. Outage management also benefits from this approach. Machine-learning classifiers trained on historical failure data predict equipment breakdowns—transformer overheating, line sagging during high winds—with high recall, yet constrain false positives to below 5% to avoid unnecessary blackouts. When a potential fault is flagged, field crews receive precise location coordinates and confidence scores, allowing targeted inspections rather than blanket safety shutdowns (Tsihrintzis et al. 2025). Finally, peer-to-peer energy trading pilots show how AI TRiSM can foster community engagement. In a smart neighbourhood, homes with rooftop solar sell excess kilowatt-hours to neighbours via a blockchain-based marketplace. The underlying auction algorithm, governed by policy-as-code, ensures fair matching and transparent fee calculations. Explainability widgets let each homeowner see why their bid cleared at a particular price, fostering trust in what might otherwise seem like a black-box market.

6.11 | Military

In military and defence contexts, AI TRiSM is the foundation for deploying autonomous systems, enhancing intelligence analysis,

and strengthening cyber defences with confidence and accountability (Marasco and Bourlai 2025). Consider a reconnaissance drone patrol over a complex urban battlefield: its onboard object-detection network, hardened with adversarial training, maintains accuracy even when insurgents apply disruptive graffiti or reflective camouflage to vehicles. By overlaying bounding-box heatmaps that highlight the model's confidence in each detection, commanders can see at a glance why a particular rooftop antenna was flagged as a potential threat, ensuring any engagement decision remains both tactically sound and legally compliant (Mikhailov 2023; Blowers et al. 2024). Beyond aerial surveillance, AI TRiSM guides secure logistics planning. In one field exercise, an autonomous convoy uses real-time terrain analysis and encrypted V2V communications anchored by hardware RoT modules. If a supply truck's estimated route confidence dips—due to unexpected road damage detected by satellite imagery fused with tactical maps—a policy-as-code check triggers a rerouting protocol, diverting convoys to safer corridors and notifying human logisticians for final approval. This blend of automated decision support and human oversight prevents mission delays while guarding against adversarial interference. On the cyber front, imagine a forward operating base where an AI-driven intrusion-detection system continuously monitors encrypted tactical data links. Graph-neural-network anomaly detectors spot subtle beaconing patterns—small, periodic data exfiltration attempts—that would evade traditional signature scans. Secure enclaves isolate suspect packets for deeper inspection, and if malicious behaviour persists, a real-time alert prompts the network to shift to an out-of-band command channel, preserving operational security under attack. Training and readiness also benefit from AI TRiSM. In a command post war-game, federated learning enables multiple allied units to share simulated SIGINT, geospatial, and electronic-warfare telemetry without revealing sensitive raw data. Homomorphic encryption ensures each participant's data sovereignty while collectively improving threat-prediction models. Real-time dashboards display metrics such as detection latency and false-alarm rates; if performance dips below a mission-critical threshold, the exercise automatically pauses for instructors to inject new scenarios or recalibrate model parameters. Finally, AI TRiSM supports humanitarian missions in contested zones. Consider an autonomous ground robot tasked with route clearance for medical convoys: its terrain-classification model, continuously validated against live sensor feeds and civilian-sourced open data, flags suspicious objects—IED indicators, for example—and routes the robot around them. Policy checks enforce international humanitarian law by preventing the robot from entering designated no-go zones unless explicitly authorised. Through these layered trust, risk, and security controls, defence organisations can harness AI's full potential—boosting effectiveness in high-stakes operations while steadfastly upholding ethical and legal responsibilities (Dahlgren 2024).

6.12 | Agriculture

In agriculture, AI TRiSM frameworks offer farmers a pathway from traditional intuition to data-driven decisions, all while protecting sensitive farm data and ensuring equitable access to insights (Zangana et al. 2025; Rao and Ranganath 2025). Imagine a large vineyard in Napa Valley using multispectral drone

flights at dawn to capture NDVI maps; a Bayesian Gaussian process then assigns confidence intervals to predicted water-stress zones, alerting vineyard managers when sensor readings conflict with weather forecasts (Ali et al. 2024). These uncertainty bounds help viticulturists trust the system's recommendations rather than second-guessing them at harvest time. Further south, a coffee cooperative in Brazil combines soil-moisture probes with thermal imagery to detect coffee leaf rust weeks before visible symptoms appear. Feature-importance analyses reveal that subtle pH shifts and canopy temperature spikes drive early disease warnings, enabling targeted fungicide application and cutting chemical use by over 30%. Meanwhile, smallholder farmers in Kenya—who lack continuous internet—benefit from synthetic data streams generated under *k*-anonymity constraints. These anonymized profiles train mobile advisory apps to suggest region-specific planting dates and intercropping schemes, without exposing any individual farmer's GPS coordinates or yield records (Ahmad et al. 2024; Sapkota et al. 2024). Securing this IoT ecosystem is equally important. Solar-powered edge gateways deployed across rice paddies in Southeast Asia use hardware roots-of-trust (RoT) and encrypted MQTT channels to ferry sensor readings back to the cloud without fear of tampering or eavesdropping. Policy-as-code pipelines automatically audit harvest-prediction models for biases—ensuring that algorithms trained on large estates do not shortchange smallholder plots or women-led cooperatives. When the system detects bias (e.g., systematically underestimating yields on terraced hillsides), it triggers a retraining workflow with additional samples to restore fairness. Continuous monitoring dashboards bring all these elements together. A single pane shows true versus predicted yield error bars, daily counts of false alarms for locust-infestation warnings, and the average time-to-detect soil-nutrient deficiencies. If any metric crosses its threshold—say, a sudden spike in disease-warning false positives—the platform automatically schedules a retraining run and notifies agronomy specialists to review the recent data. Through this blend of predictive modelling, transparent explanations, and rigorous data governance, AI TRiSM empowers farmers of every scale

to boost productivity, preserve fragile ecosystems, and comply with evolving data-sovereignty laws—all while demystifying advanced analytics and inviting stakeholders into the decision-making loop.

7 | Key Challenges

In this section we examine the principal obstacles that impede effective AI TRiSM adoption—ranging from safeguarding sensitive data and integrating governance early in the development lifecycle to mitigating bias, ensuring algorithmic transparency, and managing the dynamic threat landscape. We also explore the challenges of coordinating cross-functional teams, scaling infrastructure to support continuous monitoring and policy enforcement, and keeping pace with an ever-evolving regulatory environment. Figure 13 presents the key challenges associated with the AI TRiSM ecosystem.

7.1 | Data Privacy and Security Concerns

Modern AI systems ingest and process vast troves of sensitive information spanning personally identifiable information, proprietary intellectual property, and regulated healthcare or financial records. Ensuring data privacy demands a multi-layered defence-in-depth strategy: encryption of data both at rest and in transit, rigorous IAM with role-based controls, and secure key management infrastructures (Wu, Zhang, et al. 2024; Huang, Yu, et al. 2023; Aydin et al. 2024). Furthermore, data governance policies must define clear classification taxonomies—segregating public, internal, confidential, and restricted data—and employ anonymization or pseudonymization techniques such as *k*-anonymity, ℓ -diversity, or differential privacy to prevent re-identification attacks. For global enterprises, compliance with local regulations requires geo-fencing of data flows, robust consent management, and automated data subject request handling.

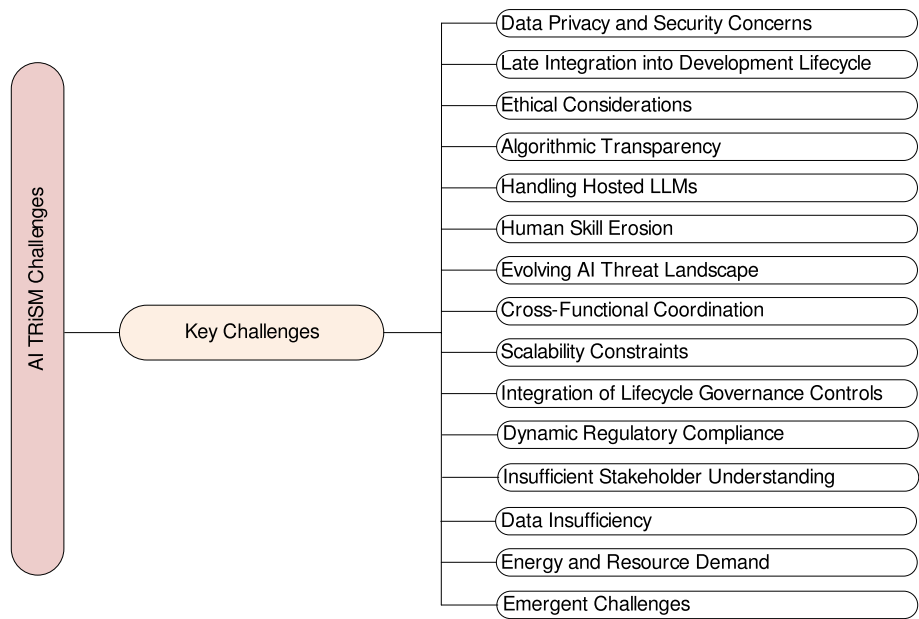


FIGURE 13 | Key challenges associated with AI TRiSM.

Continuous security validation through vulnerability scanning, penetration testing, and real-time intrusion detection is essential to detect unauthorised access or exfiltration attempts. Integrating privacy-by-design and security-by-design principles early in the AI model lifecycle—embedding threat modelling and secure coding reviews into CI/CD pipelines—ensures that privacy risks are remediated before deployment. Ultimately, a comprehensive AI TRiSM framework must align technical safeguards, policy controls, and audit mechanisms to maintain confidentiality, integrity, and availability without compromising model innovation or performance (Wang et al. 2025; Kibriya et al. 2024; Li, Hong, et al. 2024). Beyond technical safeguards, data governance must define clear classification tiers—public, internal, confidential, restricted—with automated tagging in data catalogues. Pseudonymization or anonymization techniques such as k -anonymity, ℓ -diversity, and differential privacy (adding calibrated noise to query outputs) reduce re-identification risk. Continuous validation is critical: integrate automated vulnerability scanning (e.g., Snyk⁶⁹) into CI/CD pipelines, run scheduled penetration tests on model endpoints, and deploy real-time intrusion detection and prevention systems tuned for ML traffic patterns. Privacy-by-Design and Security-by-Design principles mandate threat modelling exercises at conception, secure code reviews during development, and red-team/blue-team drills post-deployment. Only by weaving technical controls, policy frameworks, and audit mechanisms into a cohesive AI TRiSM program can organisations ensure confidentiality, integrity, and availability (i.e., the CIA triad) without stifling innovation or degrading model performance.

7.2 | Late Integration Into Development Lifecycle

One of the most pervasive obstacles in implementing AI TRiSM is the retroactive application of governance, risk management, and security controls—often bolted on after models are in production. This “after-the-fact” approach triggers costly pipeline reengineering, creates inconsistent enforcement of policies, and slows time to market. To counter this, organisations must shift governance controls left—integrating them at every stage from ideation through retirement (Kohl et al. 2024; Oshodi 2025). During the planning phase, cross-functional teams (e.g., data scientists, DevOps engineers, security architects, compliance officers, legal counsel) should co-author a Model Risk Assessment (MRA) and threat model. Documenting assumed data distributions, threat vectors (e.g., poisoning, inference attacks), and compliance requirements establishes a shared baseline. As code moves through CI/CD, automated policy-as-code gates enforce SAST, data schema validation, bias checks (measuring fairness metrics like disparate impact), and licence scanning. Build artefacts generate model cards automatically, capturing version metadata, training data lineage, performance metrics across subgroups, intended use, and known limitations. In integration testing, synthetic adversarial examples and out-of-distribution checks are run to validate robustness. Pre-deployment gates in the pipeline enforce runtime security posture (vulnerability scans of container images, configuration drift detection) and confirm adherence to resource usage and latency SLAs. Post-deployment, continuous monitoring pipelines ingest

telemetry—performance metrics, security logs, drift indicators—and trigger automated retraining or isolation of compromised models (Bandi and Kagitha 2024; Shukla et al. 2025; Shah, Hussain, and Ebrahim 2025).

7.3 | Ethical Considerations

AI models trained on real-world data inevitably inherit biases present in their training corpora—whether sampling bias, measurement bias, or historical inequities. Unchecked, these biases can perpetuate discrimination in high-stakes domains such as hiring, lending, and healthcare. An effective AI TRiSM framework must therefore implement a multi-layered bias management strategy across data, model, and monitoring phases (Head et al. 2023; Srinivasan and Patil 2025). At data collection, rigorous provenance tracking (via data lineage tools) identifies source imbalances. Data scientists apply statistical bias-detection tests—calculating disparate impact ratios, equalised odds differences, or calibration metrics—to flag skewed distributions. If bias is detected, mitigation techniques are applied: at the data level (reweighting under-represented groups, synthetic oversampling), in-training (adversarial debiasing objectives, fairness constraints in loss functions), or post-processing (e.g., threshold adjustments, equal opportunity score corrections). Governance bodies—comprising ethicists, domain experts, legal advisors, and community representatives—define acceptable fairness thresholds and respond to bias incidents. They maintain an Ethical Review Board with documented escalation procedures, ensuring transparent handling of stakeholder grievances. Within MLOps, fairness dashboards visualise ongoing model performance across protected attributes, triggering automated alerts when deviation thresholds are breached (Baldassarre and Padovan 2024; D'Antonoli et al. 2024). Regular bias audits, integrated into downstream monitoring, confirm that retraining with new data does not reintroduce disparities. Transparency is reinforced through public model cards and technical documentation, which detail both mitigation strategies and residual bias metrics.

7.4 | Algorithmic Transparency

State-of-the-art machine-learning algorithms—especially deep neural networks, ensemble methods, and transformer-based architectures—often operate as opaque “black boxes,” concealing decision logic. This opacity challenges explainability, undermines stakeholder trust, and risks non-compliance with emerging AI regulations (Brown 2024; Barman et al. 2024). AI TRiSM must therefore mandate layered interpretability and accountability mechanisms. Locally, instance-level explainability techniques—such as SHAP and LIME—decompose individual predictions into feature attributions, quantifying how each input variable contributed to the final output. Counterfactual analysis further complements these methods by identifying minimal input perturbations needed to alter a prediction, clarifying model decision boundaries. Globally, surrogate models (e.g., decision trees, linear regressors) trained to approximate the behaviour of complex models provide a human-readable overview of overall logic

(Rios et al. 2024; Miao et al. 2023). Feature importance rankings and partial dependence plots visualise the impact of each variable on aggregated predictions. Comprehensive audit trails capture every stage of the ML pipeline—data ingestion timestamps, preprocessing transformations, hyperparameter configurations, model training runs, and inference logs—stored in an immutable ledger (e.g., blockchain or WORM storage). This enables full reproducibility, forensic investigation of anomalies, and evidence of due diligence in regulatory audits. “Model cards”—structured documentation accompanying each model deployment—detail intended use cases, performance metrics across relevant subgroups, known limitations, and failure modes. Organisational accountability is enforced via Responsible, Accountable, Consulted, Informed (RACI) matrices (Matthews 2024) that assign specific governance roles (e.g., who approves model releases, who monitors explainability metrics, who responds to incidents). Periodic explainability audits, performed by independent teams, verify that both local and global interpretability standards are consistently met.

7.5 | Handling Hosted LLMs

The adoption of third-party LLMs—such as OpenAI’s GPT-4, Anthropic’s Claude, or Meta’s LLaMA—via cloud APIs accelerates time-to-value but shifts control of core model internals to external providers. AI TRiSM frameworks must therefore architect robust, layered governance around these hosted endpoints to mitigate data leakage, PII exposure, or generation of non-compliant content. First, inputs undergo strict validation. Prompt templates constrain user queries to predefined structures, eliminating free-form injections of sensitive or proprietary data. Semantic classifiers and regex filters block high-risk tokens (e.g., Social Security numbers, internal code names). Second, ephemeral proxies intercept responses to apply inline sanitization—Named Entity Recognition (NER) systems identify and redact PII, while policy engines remove disallowed content (e.g., copyrighted text, offensive language). Third, logging practices record only anonymized metadata (e.g., request hashes, usage metrics, latencies), ensuring that sensitive payloads are neither persisted nor accessible beyond governance teams (Jaff et al. 2024). Vendor management is crucial—SLAs (Kochovski et al. 2020) and Data Processing Agreements (DPAs) must specify data retention durations, encryption standards at rest and in transit, breach notification timeframes, and audit rights. Real-time monitoring dashboards—driven by unsupervised anomaly detection—track usage patterns (e.g., request frequency, response lengths, semantic drift) and alert when deviations exceed established baselines. Feedback loops feed anomalous findings back into policy updates, tightening filters or prompting additional user training (Mongoli 2024; Meltzer 2023).

7.6 | Human Skill Erosion

While modern AI systems deliver automation gains, over-reliance on algorithmic outputs can lead to human complacency—and eventual atrophy of essential domain expertise. This “automation bias” arises when users defer uncritically to model recommendations, even in the face of clear anomalies

or out-of-distribution scenarios (Wang 2023b; Peterson 2025; Crowston et al. 2025; Duenas and Ruiz 2024). To prevent skill erosion, AI TRiSM must institute a balanced Human-In-The-Loop paradigm. Workflow orchestration platforms (e.g., Kubeflow Pipelines,⁷⁰ Airflow⁷¹) embed mandatory HITL checkpoints for high-stakes decisions—such as credit approvals, medical diagnoses, or strategic resource allocations. At each checkpoint, the model generates predictions accompanied by confidence scores, uncertainty estimates, and explainability visualisations (e.g., SHAP plots⁷²). Trained experts then review and either approve, adjust, or override outcomes. All human-AI interactions are logged to provide feedback data for continuous model improvement. Comprehensive training curricula—spanning interactive e-learning modules, hands-on labs, and domain-specific case studies—keep personnel adept at interpreting AI outputs, probing unusual patterns, and executing manual fallbacks. Red-team/blue-team exercises simulate AI failure scenarios (e.g., incorrect anomaly detection, prompt injections), reinforcing manual override procedures and decision-drill muscle memory. Skill retention is measured via periodic assessments—tracking override rates, decision turnaround times, and accuracy of manual corrections. Embedding HITL controls, ongoing training, and performance metrics ensures that human expertise remains integral. In turn, this preserves organisational resilience, enabling swift recovery when AI systems encounter edge-case inputs, data drift, or adversarial manipulations.

7.7 | Evolving AI Threat Landscape

AI systems encounter a dynamic and evolving adversarial environment. Attackers utilise data poisoning (i.e., injecting malicious samples into training sets), adversarial perturbations (i.e., imperceptible input modifications to force misclassification), and inference attacks (e.g., model inversion, membership inference) to compromise data integrity or extract sensitive training content. Language models face prompt injections and chain-of-thought manipulations, which subvert expected generation flows (Deng et al. 2025; Karadsheh et al. 2025; Yu et al. 2025). AI TRiSM frameworks must embed continuous threat modelling: convene regular cross-functional threat workshops to update adversarial taxonomies, assess risk levels, and refine mitigation roadmaps. Technical defences include adversarial training (i.e., incorporating perturbed samples into the training loop), differential privacy (adding calibrated noise to model gradients), and robust input sanitization pipelines. Layered anomaly detection—using ensemble detectors on request metadata, response semantics, and user behaviour—facilitates early detection of emerging attack patterns. Security Operations Centers (SOCs) (Mohsin et al. 2025) integrate AI-specific Indicators of Compromise (IoCs) (Almorjan et al. 2025) into monitoring playbooks, correlating alerts across network, application, and machine learning telemetry. Collaboration with external red-team communities, bug bounty platforms, and academic researchers accelerates vulnerability discovery and shared defence learning. Periodic adversarial pen-testing exercises validate defences against both known and zero-day attack methods. Combining formal threat modelling, layered technical safeguards, proactive monitoring, and collaborative security research enables AI TRiSM frameworks to maintain resilient AI deployments amid a constantly evolving threat landscape.

7.8 | Cross-Functional Coordination

AI TRiSM governance succeeds only when diverse stakeholders—data scientists, MLOps engineers, security architects, compliance officers, legal counsel, and business leaders—operate in concert. Siloed workflows breed misalignment on priorities, terminology, and risk appetites, undermining unified oversight. A centralised AI TRiSM Council, chartered with clear terms of reference, meeting cadences, and decision rights, serves as the governance nexus (te Marvelde 2025). This council maintains an integrated risk register—documenting every AI initiative, its threat model, compliance status, and mitigation backlog—in a shared project management platform. Cross-functional squads subscribe to this register via automated notifications, ensuring that changes (e.g., new model releases, policy updates) are visible to all. RACI matrices define precise roles: who drafts model risk assessments, who approves deployment, who audits security posture, and who informs external stakeholders. Regular governance rituals—monthly deep-dives into risk dashboards, quarterly compliance reviews, and ad-hoc “war rooms” for incident response—keep stakeholders aligned. Governance champions embedded within development teams expedite policy interpretation and enforcement. Knowledge transfer is enabled by rotating team members through governance committee seats, fostering empathy for cross-disciplinary constraints (Wu, Zhao, et al. 2025; Schneider et al. 2024). Unified collaboration tools (e.g., document repositories, chatops channels, shared dashboards) eliminate handoff delays. Institutionalising structured coordination and shared accountability allows AI TRiSM frameworks to balance rapid innovation with strong risk management practices.

7.9 | Scalability Constraints

Comprehensive AI TRiSM deployments—incorporating continuous monitoring, automated retraining, audit logging, and anomaly detection—introduce significant infrastructure demands. Monolithic legacy environments often lack elasticity, centralised telemetry, or GPU acceleration, leading to performance bottlenecks and governance blind spots (Wang, Jiang, and Mi 2024). To meet these needs, organisations should adopt hybrid architectures: on-premises HPC clusters handle sensitive data training under strict security controls, while cloud platforms (e.g., AWS SageMaker,⁷³ Azure ML,⁷⁴ Google Vertex AI⁷⁵) provide auto-scaling compute for inference, monitoring agents, and retraining jobs. Infrastructure as Code tools such as Terraform and Ansible automate consistent provisioning of governance configurations—network policies, encryption settings, IAM roles. Containerization and orchestration enable reproducible ML environments, simplifying patch management and vulnerability scanning (Yao et al. 2024). Service meshes (e.g., Linkerd⁷⁶) inject mTLS encryption, rate limiting, and fine-grained policy enforcement into microservice communications. Observability stacks ingest telemetry from data pipelines, model endpoints, and security logs—powering real-time dashboards and alerts for performance anomalies or governance violations. A unified governance cockpit aggregates infrastructure, security, and MLOps metrics, providing a single pane of glass for AI TRiSM monitoring at scale. Strategically

designing elastic and observable infrastructure enables organisations to adapt to changing governance requirements while preserving model performance and ensuring operational reliability.

7.10 | Integration of Lifecycle Governance Controls

Ensuring consistent risk posture across the entire AI lifecycle—data ingestion, feature engineering, model training, deployment, monitoring, and deprecation—requires policy-as-code and automated enforcement. Manual reviews cannot scale; AI TRiSM frameworks must embed compliance checks directly into MLOps pipelines (Pahune et al. 2025; Xie 2020; Sayles and Aligning 2024). Pre-commit Git hooks trigger SAST for security vulnerabilities, data schema validations to detect unauthorised PII fields, and licence scans for third-party dependencies. During model training, bias detection modules calculate fairness metrics (e.g., disparate impact, equalised odds) and halt pipelines if thresholds are breached. Pre-deployment gates run vulnerability assessments (i.e., container image scanning), performance regressions, and compliance tests aligned with regulatory requirements. At runtime, policy agents intercept inference requests, enforcing RBAC, input validation, and anomaly detection before output delivery. Orchestration platforms support usable governance plugins, seamlessly injecting these checks. Immutable model registries store every artefact, along with metadata on lineage, governance audits, and deployment approvals. Upon retirement, archives preserve artefacts and logs for regulatory audit readiness. Automating lifecycle governance through policy-as-code allows AI TRiSM frameworks to minimise human error, speed up development cycles, and uphold a consistently low risk posture.

7.11 | Dynamic Regulatory Compliance

Global AI regulations—ranging from the EU AI Act and UK's AI Standards to U.S. privacy laws (e.g., CPRA, HIPAA) and sectoral guidelines (i.e., FDA's AI/ML SaMD; FDA n.d.)—impose diverse transparency, fairness, data handling, and accountability requirements. AI TRiSM teams must therefore establish continuous regulatory monitoring. Legal intelligence feeds and consortium memberships (e.g., IEEE, WEF AI Governance Alliance; Ai Governance Alliance n.d.) alert teams to upcoming mandates. Automated compliance pipelines codify new regulations into executable policies. For example, data localization rules trigger geo-fence enforcement, consent management platforms enforce opt-in workflows, and fairness mandates embed statistical checks in model validation. Model registries are enhanced with jurisdiction-specific compliance tags, ensuring transparent artefact lineage. Immutable audit logs—recording evidence of policy enforcement, consent receipts, bias test results, and security assessments—are secured with cryptographic hashing to guarantee tamper resistance. Regular cross-functional audits validate that automated pipelines reflect current requirements; deviations are rapidly remediated. Treating compliance as a continuous operational process rather than a one-time effort enables AI TRiSM frameworks to keep both global and local organisations prepared for regulatory

scrutiny, reducing legal risk and maintaining stakeholder trust (Han and Guo 2024; Kothandapani 2025).

7.12 | Insufficient Stakeholder Understanding

Complex AI governance concepts—adversarial robustness, differential privacy, model interpretability metrics—can overwhelm executives, data scientists, business users, and front-line staff alike. Without clear understanding, policies may be ignored or misapplied. AI TRiSM frameworks must therefore weave comprehensive education, change management, and engagement programs into their core (Balayn et al. 2024; McIlveene et al. 2024). A tiered training curriculum addresses varied roles. Data scientists and MLOps engineers receive deep technical workshops on policy-as-code tools, bias mitigation algorithms, and secure development practices, with hands-on labs using sandbox environments. Security and compliance teams learn AI-specific threat modelling, incident response playbooks, and regulatory requirements through tabletop simulations (Calderon and Reichart 2024; Matsumoto et al. 2024). Business leaders attend executive briefings—concise, KPI-driven presentations that translate technical metrics (e.g., drift rates, compliance coverage) into strategic risk exposures and ROI considerations. Interactive governance office hours and peer review sessions foster cross-disciplinary dialogue, answering practical questions and gathering feedback for iterative policy refinement. User surveys and NPS (i.e., Net Promoter Score) metrics track stakeholder sentiment, flagging areas where guidance or tooling improvements are needed. A dedicated intranet portal provides accessible playbooks, recorded tutorials, and FAQ libraries. Executive sponsorship is reinforced via governance dashboards that visualise risk posture, compliance status, and project health in business-relevant terms.

7.13 | Data Insufficiency

A less discussed yet deeply consequential barrier to trustworthy AI is the shortage or uneven availability of relevant, high-quality data. Many organisations simply do not possess the volume, diversity, or granularity of information needed for a model to perform the task that stakeholders envision. Sometimes, the data exists but is trapped behind institutional silos, proprietary barriers, or complex geopolitical and legal constraints that prevent secure exchange and replication. When data is incomplete or inaccessible, even a carefully designed AI TRiSM framework struggles to validate fairness, monitor drift, or guarantee reproducibility and accountability. To manage this, AI TRiSM initiatives should explicitly treat *data sufficiency* as a measurable dimension of governance risk—tracking the coverage, representativeness, and timeliness of each dataset used in training or evaluation (Hong 2025). Organisations can complement limited data through governed synthetic data generation, domain adaptation, or transfer learning techniques, provided the lineage and fidelity of artificial samples are transparently recorded. Federated learning and privacy-preserving computation offer additional pathways for collaboration—allowing knowledge exchange across partners without exposing raw datasets. Ultimately, TRiSM reinforces that responsible AI depends not on the assumption of data abundance, but on the ethical and

deliberate management of the data that truly exists. Recognising insufficiency as a structural risk encourages innovation in how data is sourced, shared, and stewarded under transparent and auditable controls.

7.14 | Energy and Resource Demand

Behind every modern AI application lies a vast and often invisible infrastructure of data centers consuming electricity, cooling water, and computing materials at extraordinary scale. The governance of AI cannot remain credible if it overlooks the environmental and resource footprint that sustains model training, deployment, and continuous retraining cycles (Taseen and Ali 2023; Khan et al. 2022). If the future of AI is to be genuinely responsible, it must be powered responsibly too. AI TRiSM provides a way to embed this principle through *sustainability-aware governance*. Energy and carbon metrics—such as Power Usage Effectiveness (PUE), Water Usage Effectiveness (WUE), and Joules-per-inference—should be logged and audited alongside traditional model performance metrics like accuracy and latency. Lifecycle dashboards can visualise these trade-offs in real time, showing where marginal performance gains come at disproportionate environmental cost. Policy gates within TRiSM pipelines can automatically defer high-compute workloads to periods of renewable energy surplus or route them to low-carbon regions. Governance reports can also include *green accountability indicators*, integrating carbon offsets, cooling efficiency data, and end-of-life recycling practices for hardware.

7.15 | Emergent Challenges

Even the most carefully designed governance architectures must contend with challenges that evolve faster than the regulations meant to contain them. New geopolitical policies may abruptly restrict cross-border data exchange or the use of certain AI capabilities, such as biometric surveillance (Choung et al. 2024), autonomous decision-making in defence applications, or unrestricted generative content creation. Such policies, while rooted in legitimate ethical or national-security concerns, can fragment the global AI ecosystem—creating uneven access to technology, conflicting compliance regimes, and uncertainty over which functions remain legally and socially acceptable. AI TRiSM must therefore accommodate the reality that governance frameworks themselves can become active variables in system behaviour, shaping not only how AI operates but whether it can operate at all.

Equally significant are the emergent health and environmental considerations linked to the pervasive microwave and millimetre-wave energy spectrum used in IoT (Sirigiri and Gaur 2025; Menon et al. 2025), 5G, and other wireless infrastructures that sustain intelligent environments. Although research is ongoing, early studies highlight the need for longitudinal monitoring of human exposure, device density thresholds, and ecological interactions. Integrating such evidence into TRiSM oversight—through real-time exposure audits, sensor certification standards, and collaboration with public-health agencies—ensures that technological ubiquity does not come at the expense of biological safety.

The broader landscape of AI risk is fluid—every new model, network, or connectivity layer introduces dependencies that did not previously exist. For AI TRiSM to remain credible, it must evolve into a system of *living governance*—capable of absorbing new knowledge and adjusting its boundaries in step with scientific, legal, and societal change. This includes maintaining adaptive policy engines that can be re-parameterized as laws shift, simulation tools that forecast the ripple effects of emergent constraints, and early-warning observatories that correlate signals across environmental, legal, and cybersecurity domains. Routine horizon scanning—spanning technology, regulation, health, and planetary ecology—should become a permanent governance function. Scenario-based stress testing can then explore how AI TRiSM protocols respond under new legislative prohibitions, infrastructure limitations, or health-related advisories.

8 | Future Directions

As AI systems grow ever more complex and entwined with critical business and societal functions, the mechanisms we use to govern, monitor, and secure them must evolve in parallel. In the following section, we explore a vision for next-generation AI TRiSM platforms—where governance policies are themselves adaptive and AI-driven, explainability moves beyond correlation to causal auditing, and privacy protections are baked into every stage of the ML lifecycle. Figure 14 shows future way outs in AI TRiSM framework implementation.

8.1 | Adaptive AI-Powered Governance Frameworks

As organisations embrace increasingly complex AI ecosystems—spanning edge devices, federated learning, and multi-model orchestration—static governance policies will become obsolete (Khan and Ambadekar 2024; Odubola et al. 2025). Future AI TRiSM platforms must themselves employ AI agents

to continuously analyse telemetry from model training, deployment, and user interaction. By leveraging techniques such as Bayesian online change point detection, these governance agents can identify drift not only in model outputs but also in the regulatory landscape—automating updates to policy-as-code repositories. In practice, an AI-driven governance layer would parse new legislation, generate corresponding rule templates, and validate them against existing workflows using symbolic reasoning. This approach minimises manual policy rewrites and ensures that compliance checks remain synchronised with real-world requirements. Furthermore, these adaptive frameworks can incorporate reinforcement-learning loops that reward rule configurations which both reduce risk alerts and maintain developer velocity. Over time, the system builds a meta-model of “optimal governance posture” for each business domain and continuously refines its recommendations. Such AI-augmented governance not only improves responsiveness to change but also shifts compliance teams from reactive firefighting to strategic oversight—aligning evolving regulations, emerging threats, and business objectives in real time (Shethiya 2023).

8.2 | Causal Explainability

Current state-of-the-art interpretability methods provide feature-importance scores but lack insight into true cause-and-effect relationships within complex architectures like deep neural networks (Wu, Kuang, et al. 2024). The next wave of AI TRiSM will embed causal inference frameworks—drawing on structural causal models (SCMs)—directly into model development pipelines (Ashwani et al. 2024; Khatibi et al. 2024). By instrumenting training code to record data provenance and conditional dependencies, practitioners can generate counterfactual audit trails: “Had input feature X been different by Δ , model decision Y would have shifted by Z.” These causal attributions go beyond correlation, exposing hidden confounders or spurious associations that bias mitigation alone cannot fix. Embedding such explainability also facilitates dynamic compliance reporting: regulators could query system logs to verify that an AI

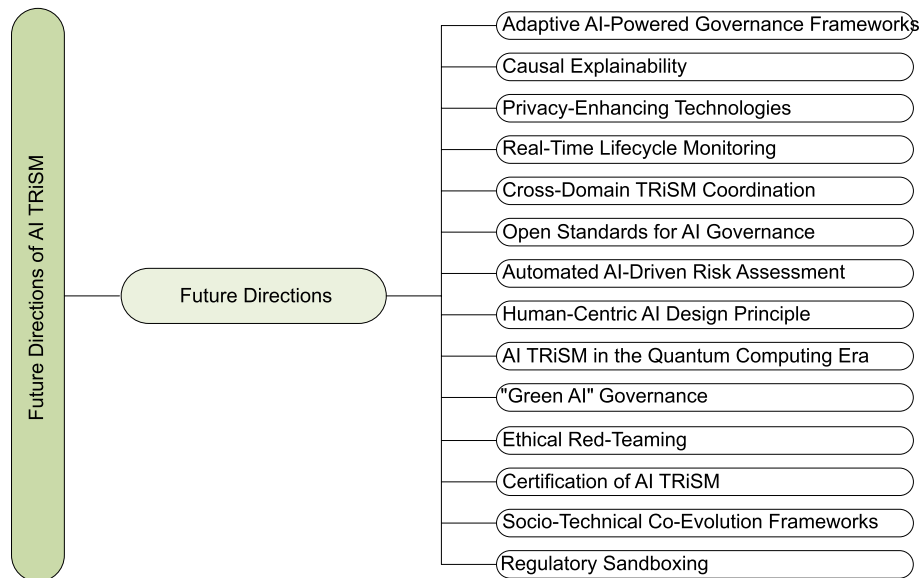


FIGURE 14 | Future directions for AI TRiSM.

decision adhered to fairness constraints under specified counterfactual scenarios. Implementing real-time causal auditing requires novel tooling—SCM-aware profilers, graph-structured storage of causal graphs, and high-performance counterfactual simulators. As causal explainability matures, it will become the gold standard in demonstrating model trustworthiness, enabling AI TRiSM teams to certify not only what the model did, but why it did so under diverse “what-if” conditions.

8.3 | Privacy-Enhancing Technologies

Data privacy and confidentiality will only intensify as AI is applied to sensitive medical records, financial ledgers, and personal communications (Shi et al. 2025). AI TRiSM frameworks must evolve to integrate PETs throughout the MLOps lifecycle. Homomorphic encryption will allow training on encrypted data without ever exposing raw PII, while multiparty computation (MPC) enables collaborative model building across organisations without centralising data. Differentially private stochastic gradient descent will become a standard training option, with AI TRiSM platforms automatically tuning noise parameters (ϵ , δ) to balance privacy budgets against accuracy goals. Additionally, zero-knowledge proof modules can validate model outputs—proving compliance with fairness metrics or regulatory constraints without revealing proprietary model parameters. Future PET orchestration layers will schedule cryptographic protocols based on data sensitivity classifications, automatically degrade compute precision when necessary, and generate audit-ready attestations for each MLOps stage (Pahune and Akhtar 2025; Tantithamthavorn et al. 2025). Making the adoption of privacy-enhancing technologies seamless allows enterprises to leverage richer datasets while minimising re-identification risks, ensuring that advanced AI capabilities uphold fundamental data protection principles.

8.4 | Real-Time Lifecycle Monitoring

Traditional AI audits occur at fixed intervals—quarterly or before major releases—leaving windows of undetected drift or misconfiguration (Kohl et al. 2024; Sun et al. 2025). The next frontier in AI TRiSM will be embedding continuous, real-time assurance mechanisms that mirror modern Site Reliability Engineering (SRE) practices. Model endpoints will expose standardised health-check APIs reporting metrics such as concept drift scores, cryptographic integrity checks, and compliance-rule hit rates. Streaming data platforms (e.g., Apache Kafka with `ksqlDB`⁷⁷) will ingest these telemetry streams, applying sliding-window anomaly detection (e.g., using robust z-scores or ensemble isolation forests) to flag emerging issues. Automated playbooks—encoded as event-triggered workflows—will initiate containment actions: roll back to a known good model version, revoke risky API tokens, or quarantine suspect data sources. This “always-on” assurance paradigm ensures availability, integrity, and trust without manual oversight. AI TRiSM teams will monitor live dashboards combining Service Level Objectives (SLOs for model performance), security alerts, and regulatory compliance indicators, creating a unified risk posture view. Operationalising

continuous monitoring enables organisations to identify and address issues within minutes rather than weeks, enhancing both security and system reliability (Tamanampudi 2024; Chen, Milosevic, et al. 2023).

8.5 | Cross-Domain TRiSM Coordination

As federated AI use cases proliferate—medical imaging across hospital networks, credit scoring models shared among banks—AI TRiSM must support decentralised governance across domains. Future frameworks will implement “federated TRiSM” protocols, enabling multiple stakeholders to collaboratively enforce shared policies while retaining autonomy over their local data and models. These protocols leverage blockchain or distributed ledger technology (DLT) (Gong et al. 2024; Li, Ma, et al. 2024) to record immutable, timestamped evidence of governance actions: model lineage commits, differential privacy budgets consumed, and adversarial attack reports. Smart contracts codify cross-party SLAs for data usage, release approvals, and incident escalation procedures. When a partner’s local audit uncovers a compliance breach—say, a fairness threshold violation—the federated TRiSM network propagates a signed event, prompting upstream and downstream nodes to re-evaluate their own deployments. Governance councils comprising representatives from each domain can dynamically vote on policy updates via on-chain governance mechanisms, ensuring transparent and democratic decision-making.

8.6 | Open Standards for AI Governance

The AI ecosystem today is highly fragmented: different MLOps platforms, security tools, and compliance frameworks speak disparate “languages.” To avoid vendor lock-in and enable seamless integration, future AI TRiSM initiatives will coalesce around open governance standards (Wodi 2024). Similar to how OpenAPI standardised REST interfaces, an “OpenTRiSM” specification will define canonical data schemas for compliance artefacts—model cards, risk assessments, audit logs—and protocol bindings for continuous policy enforcement. This standard will include a JSON Schema for policy rules (e.g., “no PII in model training without $\times$ consent”) and gRPC interfaces for runtime policy evaluation. Machine-readable taxonomies for data sensitivity, threat categories, and regulatory obligations will allow tools from different vendors to interoperate—so that a unified policy control plane can manage policies across Kubernetes pods, serverless functions, and device-edge deployments alike. Governance metadata will flow consistently from development to production, eliminating silos. Rallying around open standards empowers the industry to drive rapid tool innovation, encourage healthy competition, and ensure widespread adoption of AI TRiSM best practices across organisations of all scales.

8.7 | Automated AI-Driven Risk Assessment

With hundreds or thousands of AI models operating in parallel, manually triaging every risk finding becomes impossible. The future of AI TRiSM will leverage AI itself to prioritise the

most critical risks. Meta-models trained on historical incident data—covering drift events, security breaches, and compliance escalations—will predict the likelihood and impact of new alerts (Wickramasinghe 2023; Koli et al. 2025). For example, a model showing early signs of data poisoning drift might be scored higher than a minor fairness metric deviation. These predictive risk scores feed into dynamic dashboards that surface the “top-k” risks for human review each day. Autonomous remediation playbooks can then execute low-risk fixes—model retraining with sanitised data, automated key rotation—without human intervention. More complex scenarios are escalated via integrated ticketing systems with recommended next steps. Over time, reinforcement-learning techniques refine the risk-scoring meta-model by observing which automated actions succeeded and which required manual follow-up. This closed feedback loop continuously hones both detection and prioritisation, ensuring scarce human resources focus where they matter most (Joshi 2025).

8.8 | Human-Centric AI Design Principles

As AI systems permeate decision-making in critical domains—healthcare triage, legal analysis, hiring, and social services—the future of AI TRiSM must foreground human centrality in every design choice. Rather than optimising solely for predictive accuracy or throughput, next-generation frameworks will incorporate co-design sessions with end users, domain experts, ethicists, and affected community members (Wang, Sukiennik, et al. 2024). Human-centered design (HCD) (Ilievski 2025) workshops will yield empathy maps, journey maps, and personas that guide feature sets, user interfaces, and fallback procedures. In practical terms, design patterns for AI-infused tools will embed “explain-first” prompts, progressive disclosure of model limitations, and context-sensitive assistance that surfaces rationale in plain language. Usability metrics—time to comprehension, trust calibration scores, and remediation efficacy—will feed back into MLOps pipelines as key performance indicators. Furthermore, future AI TRiSM platforms should support digital “sandboxes” where new interfaces and explanation modalities can be A/B tested for comprehension and trust outcomes. These sandboxes leverage eye-tracking, click analytics, and sentiment analysis to iterate rapidly on design. Engineering human-in-the-loop touchpoints across the model lifecycle—such as interactive calibration, corrective feedback mechanisms, and shared mental models—ensures that AI systems enhance human capabilities rather than replace them, promoting co-ownership, accountability, and enduring trust.

8.9 | AI TRiSM In the Quantum Computing Era

Quantum computing promises to accelerate both AI training (via quantum-accelerated gradient descent) and cryptanalysis (threatening classical encryption). In this dual-edged landscape, AI TRiSM must evolve to address quantum-era risks and opportunities. On one hand, quantum-safe cryptographic protocols—lattice-based, hash-based signatures—will become standard in data-at-rest and data-in-transit protections. AI TRiSM toolchains will integrate post-quantum key exchange (e.g., CRYSTALS-Kyber⁷⁸) into MLOps pipelines, ensuring that model weights and training data remain secure against future quantum attacks. On

the other hand, quantum-accelerated algorithms could uncover new adversarial tactics—rapid state-space exploration might craft imperceptible perturbations in milliseconds (Pour and Arastoo 2025; Elhaimeur and Chrisochoides 2025). To counter this, AI TRiSM frameworks will incorporate quantum-enabled anomaly detection, leveraging quantum-inspired clustering methods to identify outlier inputs at scale. Governance modules must also monitor quantum compute resource usage across hybrid cloud-on-prem environments, ensuring compliance with export controls and licensing. Auditable quantum provenance logs—recording which workloads ran on quantum hardware—will feed immutable ledgers for post-quantum audits. As quantum AI matures, AI TRiSM becomes the linchpin for safe, resilient, and compliant AI development in the forthcoming quantum age (d'Aloisio et al. 2024; Campbell et al. 2025).

8.10 | “Green AI” Governance

The carbon footprint of large-scale AI training—GPT-class transformer models consuming megawatt-hours—has drawn increasing scrutiny. Future AI TRiSM frameworks will embed environmental metrics into model risk assessments, instituting “green SLOs” alongside performance and security SLAs. MLOps pipelines will automatically log GPU/TPU-hour consumption, energy consumption per training epoch, and Power Usage Effectiveness (PUE) for data centers. Smart schedulers will shift nonurgent retraining tasks to periods of surplus renewable energy (solar peak, wind bursts), dynamically spinning up and down compute clusters. Model developers will receive real-time dashboards showing CO₂-equivalent emissions per experiment, prompting low-carbon hyperparameter choices (e.g., quantized architectures, early stopping). Governance policies can enforce “carbon budgets” at team or project level, halting model runs once budgets are exhausted. On the inference side, AI TRiSM can favour on-device models or serverless functions that adaptively scale to demand, minimising idle resource waste. By integrating sustainability as a first-class dimension—measured, reported, and regulated—organisations uphold environmental stewardship while maintaining AI innovation (Sampatsing et al. 2025; Cortez 2023; Shome and Das 2023).

8.11 | Ethical Red-Teaming

Just as security operations deploy red teams to probe infrastructure, AI TRiSM must institutionalise adversarial testing labs focusing on ethics, fairness, and safety (Feffer et al. 2024). These AI red teams will comprise multidisciplinary experts—data scientists, security researchers, sociologists—who craft sophisticated probes: bias injection scenarios, Trojan watermarking attempts, privacy extraction attacks, and ambiguous input prompts designed to elicit harmful outputs. Automated “ethical fuzzers” will generate edge-case datasets to challenge model invariants: demographic attribute swaps, negation flips, or cultural dialect tests (Gillespie et al. 2024; Ahmad et al. 2025; Longpre et al. 2024). AI TRiSM platforms will schedule continuous adversarial simulation campaigns, measuring resilience metrics such as “fairness robustness index” or “privacy leak probability.” Findings feed into an “ethical playbook” that prescribes model architecture adjustments, training-data sanitization routines, or

runtime guardrails (e.g., dynamic input sanitization). By closing the loop between adversarial findings and MLOps tuning, organisations ensure that models remain robust to novel attack vectors—ethical, security, and privacy alike—before deployment in high-stakes environments.

8.12 | Certification of AI TRiSM

As AI governance gains strategic importance, a professional ecosystem around AI TRiSM best practices will emerge. Industry consortia (e.g., CISSP for cybersecurity) will develop certification curricula for AI TRiSM practitioners. Core competencies will cover risk assessment methodologies, cryptographic PETs, fairness auditing techniques, regulatory frameworks, and ethical red-teaming. Certification tracks—Foundational, Practitioner, and Expert—will require candidates to demonstrate hands-on proficiency via capstone projects—drafting policy-as-code manifests, deploying continuous monitoring stacks, and executing adversarial simulation campaigns. Accredited training labs will offer immersive simulations, presenting candidates with realistic AI incidents—drift-induced bias surges, privacy breach attempts, model inversion exploits—and evaluating remediation plans under time constraints (Levene and Wooldridge 2023). A global registry of AI TRiSM Certified Professionals will help organisations source qualified personnel. Over time, certified professionals will form peer-review networks, publishing open-source governance patterns, policy templates, and incident-response recipes—fostering a virtuous cycle of knowledge sharing and standard elevation (Grossman et al. 2023).

8.13 | Socio-Technical co-Evolution Frameworks

AI does not evolve in isolation; its trajectory intertwines with societal norms, legal systems, and cultural values. Future AI TRiSM strategies will adopt co-evolution frameworks that model feedback loops between technology deployments and social impact indicators—public sentiment, employment patterns, and digital inclusion metrics. Using system dynamics simulations, governance teams can forecast how a new AI hiring tool might influence workforce diversity, wage distribution, or demographic representation over time. These simulations incorporate nonlinear interactions: as automation displaces certain roles, reskilling initiatives may lag, causing talent gaps that feed back into model performance (Mandigo 2024). AI TRiSM platforms will ingest macro-indicators—regional unemployment rates, educational attainment, policy changes—and calibrate model risk profiles accordingly. When simulated societal impact exceeds predefined thresholds, governance workflows trigger policy interventions: mandating bias audits, restricting certain use cases, or funding complementary human-centric programs. By treating AI governance as part of a larger socio-technical ecosystem, enterprises and regulators can steer co-evolution toward equitable, resilient outcomes (Ramaswamy 2025; Srdjevic 2025).

8.14 | Regulatory Sandboxing

Rigid compliance regimes can stifle innovation. To reconcile agility with safety, AI TRiSM frameworks will incorporate

regulatory sandboxing capabilities—isolated environments where novel AI applications can be tested under relaxed constraints but stringent oversight (Kera 2023; Yaksan and Regulating 2024; Moraes 2023). In these sandboxes, model prototypes can process live data under “canary approvals”—limited access tokens, automated rollback triggers, and enhanced logging. Sandbox environments provide a controlled crucible where regulators, developers, and ethics boards collaboratively observe real-world behaviour without exposing full production (Nabil 2024; Sky 2024). Monitoring agents enforce soft-policy violations (i.e., informing developers of potential noncompliance) while collecting rich telemetry on model decisions, drift patterns, and user interactions. Sandbox outcomes become “certificate-of-trust” backbones for broader deployment—only models that pass sandbox criteria—no high-severity bias incidents, acceptable privacy leakage quantiles, and robust adversarial resistance—graduate to production. This safe-innovation paradigm lowers the entry barrier for experimentation while ensuring governance guardrails evolve in tandem with emerging AI capabilities.

9 | Discussion

This study set out to make *trust*, *risk*, and *security management* for AI tangible. The preceding sections delivered depth: a taxonomy of risks (Figure 9), a companion security-management taxonomy (Figure 10), a fine-grained map of generative-model toxicity (Figure 11 and Table 7), comparative views of tools and frameworks (Tables 4 and 5), sector use cases (Figure 12 and Table 7), challenges (Figure 13), and future directions (Figure 14). What has been missing—by design until now—is synthesis. This discussion connects those threads into a coherent narrative, surfaces practical trade-offs, and articulates a pragmatic adoption path. It also reflects candidly on scope limits and on an unexpected negative result from our search for scholarship about AI’s use in academic research.

9.1 | From Lists to Links

The three organising lenses in this paper—risk types, security controls, and toxicity modes—are not parallel catalogues; they are coupled layers of a single system. If Figure 9 answers “what can go wrong?,” Figure 10 answers “what do we deploy to prevent, detect, and respond?,” and Figure 11 answers “how does misbehavior concretely manifest in generative models?” The AI TRiSM pillars cut across all three: fairness and transparency constrain *how* models decide; reliability and robustness constrain *how* they fail; data protection constrains *what* they may touch or reveal; governance binds these into accountable processes. A practical implication is sequencing: organisations that start with “tools” often stall because control selection is underdetermined without the risk/behaviour context. Beginning with a risk taxonomy, attaching it to toxicity archetypes for generative systems, and *then* choosing controls yields faster convergence and fewer false assurances. In practice this means constructing a traceable mapping from each risk category to a minimal set of preventive, detective, and corrective controls, defining assumptions and boundaries explicitly, and validating the mapping with counterexamples and failure-mode walkthroughs before any procurement or build work begins.

9.2 | Operationalising AI TRiSM

Principles only become protections when they are executable. Across sections we emphasised policy-as-code, immutable provenance, and continuous assurance. The operational recipe is straightforward: encode governance requirements as machine-checkable rules; instrument data, model, and runtime artefacts with rich metadata; and wire rules into CI/CD and serving paths so that releases block, not just warn, when controls fail. Concretely, the fairness metrics in Table 7 and the lifecycle controls in Table 7 should be evaluated automatically at training, validated in pre-deployment gates, and monitored post-deployment with alerting tied to rollback playbooks. This is the difference between “risk theatre” and risk management. Practically, this implies signed datasets and features, repeatable training environments pinned to exact dependency digests, at-tested model artefacts, explicit promotion criteria, and sidecar or proxy components at inference time that enforce guardrails, rate limits, and output sanitization. Separation of duties, break-glass mechanisms with audit trails, and change windows with canary or shadow deployments further reduce blast radius while preserving delivery velocity.

9.3 | A Minimum Viable AI TRiSM (MV-TRiSM)

We may ask: *what is the smallest set of practices that yields real risk reduction?* An MV-TRiSM for most organisations is: (1) a model registry with versioned lineage and model cards; (2) three gates in the ML pipeline—data-quality checks (schema, PII, drift), fairness/robustness checks, and container/image vulnerability scans; (3) runtime observability for performance, drift, and policy violations; (4) a red-team cadence for generative systems that covers jailbreaks, PII leakage, and illicit-assistance probes; and (5) an incident playbook with decision rights (who can pause, roll back, or sunset a model). This compact core—if consistently executed—addresses a majority of failure modes highlighted in Figures 9–11. To be effective, the core should be backed by a simple risk register, a named on-call rotation for model incidents, environment-level secrets management, baseline rate limiting at interfaces, and predefined thresholds that trigger human-in-the-loop review. Even these lightweight elements materially improve reproducibility, traceability, and time-to-mitigate.

9.4 | Trade-Offs and Tensions

Real systems live in the space of trade-offs. Privacy budgets reduce accuracy; stronger defences can degrade user experience; aggressive refusal policies lower toxic outputs but increase helpful denials. The paper’s dual taxonomy structure is meant to make those tensions explicit. For example, differential privacy reduces memorization risk but may raise false negatives in safety classifiers; adversarial training hardens perception but can skew calibration; explanation fidelity can conflict with confidentiality in third-party deployments. The remedy is not to wish trade-offs away but to *surface* them early, quantify them with task-specific cost functions, and align them with stakeholder priorities. Where fairness and robustness objectives conflict, multi-objective optimization and constraint-based training

provide a rational basis for negotiation; where transparency and IP protection conflict, layered artefacts (public model cards, privileged lineage logs) preserve accountability without over-exposure. Teams should visualise Pareto frontiers, set explicit latency and accuracy envelopes, bound acceptable privacy loss, and codify when to fail-safe, when to degrade gracefully, and when to escalate.

9.5 | What the Comparative Tool Tables Really Say

Tables 4 and 5 can be misread as a shopping list. They are better read as evidence that *no single* toolkit spans all TRiSM concerns. AIF360 and AIX360 are strong in fairness and explainability; ART is strong in adversarial robustness; governance platforms like knowledge-graph-based systems organise evidence and workflows; runtime guardrails address LLM-specific risks. The strategic takeaway is to compose capabilities rather than seek a monolith: use targeted open-source libraries for measurement, couple them to platform-native security primitives, and anchor the whole in a governance substrate (registries, audit trails, policy agents). Integration, not selection, is where assurance is won or lost. Designing a clean “control plane” for policy and evidence, and a separate “data plane” for inference and training, reduces coupling and makes it feasible to swap components without losing auditability or protections.

9.6 | Generative Toxicity

The toxicity taxonomy and summary table underscore a point sometimes lost in safety debates: toxicity risk is *use dependent*. An identical model behaves acceptably in a creative-writing sandbox yet dangerously in a medical triage chatbot. The architecture should reflect this: *policy at the edge* (domain-specific intent gating and retrieval constraints), *policy in the middle* (ensemble safety models and refusal logic), and *policy at the core* (training-time objectives and data curation). The table’s “Critical/High/Medium” priorities are not absolutes; they should be remapped per use case, with explicit acceptance criteria. Safe completion for self-harm is non-negotiable in any public interface; refusal of illicit instructions is essential but may be paired with constructive alternatives in developer tools; multimodal gating is crucial anywhere text and images co-occur. The core message: tune safety to context, not just to content. That includes scoping conversational memory, constraining retrieval sources, evaluating refusal precision and recall alongside helpfulness, and routinely testing with adversarial prompts that reflect real user behaviour, not only synthetic edge cases.

9.7 | Cross-Sector Lessons

The sector analyses were not case studies so much as *design patterns under constraints*. What travels well across domains are the invariants: provenance, continuous monitoring, bias/robustness testing, red-teaming, and zero-trust access. What differs by domain are the *risk budgets* and *audit anchors*. Finance emphasises explainability and non-discrimination with immutable auditability; healthcare emphasises privacy and uncertainty-aware decision support; autonomous systems emphasise adversarial

and out-of-distribution resilience with fail-safe behaviours; public sector emphasises equity of service and procedural transparency; energy emphasises calibration and data sovereignty; education emphasises consent, accessibility, and fairness under learning variability; defence emphasises mission assurance and compliance with humanitarian constraints. The discussion for practitioners is to start from invariants, then harden the domain-specific anchors. Concretely, that means aligning controls with sector obligations, documenting operational limits, and verifying that escalation paths and human approvals exist where the consequences of error are high.

9.8 | Third-Party and Hosted Models

When inference is outsourced, organisations still own the inputs, outputs, and consequences. The paper's guidance is to build a *control envelope* around hosted models: prompt templating and input validation before the call; response scanning, redaction, and policy enforcement after it; logging that stores only hashed or minimised metadata; and contractual controls that specify retention, breach notification, and audit rights. This preserves TRISM properties even when the base weights are opaque, and it converts “vendor trust” into measurable surface controls that can be independently audited. Network-level egress controls, token and key rotation, and sandboxed connectors further reduce exposure. A simple but effective rule is to assume the model is a black box and engineer the boundary so that sensitive content cannot inadvertently exit and unsafe content cannot enter downstream systems without checks. In practice this also means enforcing least-privilege API scopes; pinning model/version identifiers to avoid silent upgrades; routing traffic through an allowlisted gateway with data loss prevention (DLP) and PII detectors; and constraining outputs to structured schemas (e.g., JSON with strict validation) to prevent prompt-injection side effects during tool invocation. Add per-tenant rate limits and anomaly detection on request shape; apply canonicalization and tokenization to strip secrets and normalise inputs; prefer deterministic decoding for high-stakes flows; and maintain region-aware routing to honour data residency. Contractually, insist on customer-managed keys, explicit data-logging opt-outs, model-change notifications, and right-to-audit attestation. Operationally, run new providers in “shadow mode” behind a router, compare deltas against a golden-eval suite, and promote only after error budgets and refusal metrics clear pre-agreed thresholds. Finally, document a provider-agnostic rollback path and “degraded safe mode” so service quality remains acceptable if a vendor endpoint is throttled, unreachable, or fails policy checks.

9.9 | Human Factors

Cognitive risk—overreliance, underreliance, and skill drift—threads through multiple sections but deserves plain emphasis. A highly robust model that is *misused* is a brittle system. We advocate design choices that calibrate trust: communicate uncertainty, bound autonomy by default, and require human approval when confidence or context falls outside guardrails. Training is a first-class control: teach users how to read explanations, when to override, and how to route incidents. Measure

human–AI interaction, not only model performance: override rates, time-to-approval in high-stakes tasks, and downstream error correction. Good safety is socio-technical. Interfaces that surface rationale in plain language, show confidence intervals, and provide one-click escalation build appropriate reliance and preserve expertise rather than displacing it. Complement these with progressive disclosure (short justification first, deeper evidence on demand), “why not” counterfactuals for rejected actions, and explicit uncertainty cues (traffic-light semantics, calibrated scorebands) to reduce automation bias. Adopt adaptive autonomy—assistive, advisory, or gated-execution modes selected by risk class—and make handoffs legible with clear ownership, timers, and audit notes. Sustain proficiency via tabletop drills, sandboxes with realistic edge cases, and just-in-time guidance embedded in the UI; monitor cognitive load and alert fatigue, and tune notification policies accordingly. Finally, align incentives and permissions: role-based privileges that gate high-impact actions, blameless postmortems that reinforce learning, and accessibility and localization practices that ensure explanations are usable for diverse teams.

9.10 | Limitations and Scope Boundaries

This synthesis is necessarily selective. The risk and security taxonomies are broad but not exhaustive. Tool comparisons emphasise capabilities most relevant to the proposed workflows; the ecosystem is larger and evolves rapidly. Sector narratives are illustrative and do not substitute for domain-specific regulatory counsel. The paper focuses on machine learning and LLM-centric systems; other AI paradigms will present different control surfaces. Finally, we did not perform empirical field trials here; the figures and tables are design artefacts meant to support implementation, not to claim causal impact. These constraints reflect a deliberate choice: prioritise a *workable* architecture of ideas and controls that practitioners can adopt and test. Readers should treat the guidance as reference designs that require local adaptation, not as one-size-fits-all prescriptions. Moreover, assumptions about centralised MLOps, available observability, and identity infrastructure may not hold in all organisations, especially at the edge or in air-gapped settings. Costs, latency overheads, and usability trade-offs of the proposed gates and monitors were not quantified; multilingual, multimodal, and highly regulated contexts may expose additional failure modes. The exemplars may also reflect selection bias toward currently popular tooling. Future work should include controlled evaluations, cross-lingual assessments, and longitudinal studies of incident rates, fairness drift, and operator workload to validate and refine the proposed patterns.

9.11 | Metrics That Matter

If it cannot be measured, it cannot be governed. Across sections we implicitly defined a metric layer; we make it explicit here. For *trustworthiness*, track explanation coverage and stability, uncertainty calibration, refusal precision/recall, and user trust-calibration scores. For *risk*, track distribution-shift indices, fairness gaps over time, incident frequency and severity, and recovery time. For *security*, track adversarial attack success rates in red-team exercises, prompt-injection detection rates,

data exfiltration attempts, and cryptographic policy adherence. Across all, record audit completeness and evidence freshness. Binding these to service objectives converts TRiSM from a noun to a verb. In addition, measure calibration error, false-positive and false-negative costs for safety filters, mean time to detect and respond for model incidents, and the proportion of artefacts with complete lineage, so that governance health can be assessed at a glance.

9.12 | Why we Did Not Find Much on AI Use in Academic Research and Why That Matters?

We hereby include our position behind the unsuccessful search for rigorous, end-to-end TRiSM accounts of AI *used within academic research workflows* (as opposed to AI being *studied* by academia). We expected to find detailed write-ups—data-governance workflows for lab notebooks, safety guardrails for LLM-supported literature reviews, red-teaming of code assistants used in computational labs. We found scattered practice notes, but little systematic guidance. We suspect three reasons. First, the work is happening but is either unpublished or buried in institutional policies rather than articles. Second, the problem cuts across IRB, research-IT, and lab operations, and “falls between” venues. Third, novelty bias favours breakthrough results over process documentation. Regardless, the absence is itself a relevant result: there is a gap between the maturity of AI TRiSM in regulated industries and in academic practice. We see an opportunity—define reference architectures for research workflows—PII-aware data lakes for lab data, versioned RAG for literature, signed computational notebooks, and safety-checked LLM interfaces—so that research groups can adopt TRiSM with minimal friction. Including this negative finding here is an invitation for the community to surface patterns that already exist and to study their efficacy.

9.13 | Prescription for Practitioners

If practitioners only implement one change after reading this paper, let it be this: move governance *into* the pipeline. Add one gate, wire one dashboard, run one red-team exercise, and write one model card with subgroup performance. If practitioners can do more, convene a cross-functional council, assign decision rights, and articulate thresholds that bind model behaviour to organisational risk appetite. Build from there. The systems depicted here are not theoretical; they are within reach of typical data and platform teams. Start with a bounded pilot, pick a clear owner, define success and stop criteria, and insist on evidence for every control—then scale what works. A concrete first pass can be operationalised with minimal new tooling:

- Choose the pilot wisely. Select a model with moderate business value, limited blast radius, and real users. Prefer a workload that already has logs and a predictable update cadence.
- Stand up a registry and a card. Register the current model and dataset snapshots; attach a lightweight model card that documents use case, training data lineage, subgroup

metrics, known limitations, and rollback conditions. Treat the card as a living artefact updated at every promotion.

- Insert three pipeline gates. (i) *Data gate*: schema validation, basic PII scanning, and a simple drift statistic on key features; (ii) *Model gate*: reproducibility check (fixed seeds, pinned dependencies), fairness/robustness checks against a held-out stress set, and thresholded acceptance criteria; (iii) *Security gate*: container/image vulnerability scan and a dependency inventory (SBOM) with licence screening.
- Wire one dashboard. Expose a small set of leading indicators: latency and error rates, calibration error, drift score, refusal precision/recall (for LLMs), and alert counts. Tie each metric to an explicit SLO and an on-call escalation route.
- Schedule a red-team drill. Time-box a generative safety exercise that targets jailbreaks, PII leakage, and illicit-assistance prompts. Capture findings as test cases and add them to a recurring evaluation suite.
- Define decision rights. Publish a one-page RACI: who approves promotions, who can hit the kill switch, who owns incident triage, and who accepts residual risk when thresholds are temporarily waived.
- Close the loop. For every failed check, require a tracked remediation with an owner and a due date; for every exception, record the rationale and an expiry. No silent deviations.

9.14 | Why This Matters Now

AI is entering safety- and rights-critical spaces faster than institutions can adapt. The temptation is either to deploy and hope or to delay until “perfect” solutions arrive. AI TRiSM offers a middle path: deploy with eyes open, with controls that fail safe, and with evidence that earns trust. The risk is not merely technical; it is institutional—reputational, legal, ethical. The reward, when done well, is not merely avoidance of harm; it is better systems: more robust under shift, more equitable under scrutiny, and more intelligible to the people who must live with their decisions. A disciplined approach today reduces costly rework later and creates a durable foundation for innovation. Three dynamics make immediacy prudent. First, *governance debt* compounds like technical debt: retrofitting provenance, safety filters, and audit trails after a system scales is slower, riskier, and more expensive than building them in. Second, *supply-chain exposure* is growing as teams depend on third-party models and data services; boundary controls and policy-as-code are the only practical way to manage opaque components without paralysing delivery. Third, *expectations have shifted*: users, regulators, and partners increasingly demand demonstrable safeguards and clear lines of accountability. Organisations that can show calibrated controls, measurable outcomes, and repeatable playbooks earn the practical “license to operate” that enables sustained adoption. TRiSM also protects momentum. Safety-by-construction (gates, guardrails, least privilege) combined with safety-in-operation (monitoring, red-teaming, rollback) reduces incident frequency and severity, freeing teams to focus on product value rather than emergency response. In short: start now, start small, measure everything, and iterate under real constraints.

9.15 | Closing Synthesis

The core message of this paper is simple. Begin with *what can go wrong* (risk taxonomy), make it *measurable and monitorable* (security-management taxonomy and metrics), anticipate *how it will actually show up* in generative systems (toxicity taxonomy), and implement *controls that execute* (policy-as-code, provenance, monitoring, and red-teaming). Do this in a way that is sensitive to sector constraints and human factors, and stage it with an honest view of trade-offs. Our negative result about academic workflow guidance is a reminder that even as AI advances, trustworthy practice lags—and that sharing process knowledge is itself a public good. Practically, the synthesis reduces to an operational loop: map risks to concrete failure modes; attach a minimal set of preventive, detective, and corrective controls; define thresholds and decision rights; observe the system with meaningful metrics; and feed incidents and red-team findings back into data, training, and policy. Repeat on a regular cadence. Organisations that make this loop explicit—in code, in dashboards, and in culture—will move faster with fewer surprises and will be able to explain, with evidence, why their systems are safe enough for the contexts in which they operate.

10 | Conclusion

Looking ahead, the landscape of AI TRiSM will evolve into a living, self-optimising ecosystem—where governance policies are no longer static documents but AI-driven agents that sense regulatory shifts, adapt security controls in real time, and proactively anticipate emerging threats. Imagine ModelOps pipelines that auto-tune privacy budgets according to live risk signals, federated governance networks on blockchain that coordinate cross-industry compliance votes, and causal auditing engines capable of generating “what-if” counterfactual narratives on demand. In this future, every model—whether a mission-critical diagnostic assistant, an autonomous drone, or a consumer-facing recommender—carries its own digital DNA: immutable provenance logs, embedded fairness checks, quantum-safe encryption, and sustainability metrics that report carbon footprints per inference. Continuous monitoring and automated remediation will shrink detection windows from days to seconds, while adversarial simulation sandboxes will stress-test AI systems under ever more sophisticated attack scenarios. By fusing human creativity with AI’s analytical rigour, this next-generation TRiSM paradigm will not simply defend against risks but transform them into strategic insights—enabling organisations to innovate boldly, ethically, and sustainably in an era where intelligence itself becomes a dynamically governed fabric of society.

Disclosure

Author used gpt-40-mini for polishing the language so that readers can perceive an improved readability experience besides the enrichment of the humanized style of paragraphs. However, all the content of this article and its accuracy, correctness and fairness are the sole responsibility of the author.

Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

Endnotes

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